



PFR SPTS No. 26070

## **New Zealand Food Composition Database 2024 Data Manual**

The New Zealand Food Composition Database

August 2024

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## Key to abbreviations

| Abbreviation/Acronym | Meaning                                                       |
|----------------------|---------------------------------------------------------------|
| ABV                  | Alcohol By Volume                                             |
| AFCD                 | Australian Food Composition Database                          |
| AP                   | Applix Presents (file extension)                              |
| ASCII                | American Standard Code for information changes                |
| AOAC                 | Association of Official Analytical Chemists                   |
| BRWN                 | Brown                                                         |
| BLD                  | Boiled                                                        |
| CKD                  | Cooked                                                        |
| CoFID                | Composition of Foods Integrated Dataset                       |
| CSM                  | Common Standard Measure                                       |
| DRIP                 | Dripping                                                      |
| DRND                 | Drained                                                       |
| EP                   | Edible Portion                                                |
| FAO                  | Food and Agriculture Organisation (United Nations)            |
| FIMS                 | Food Information Management System                            |
| FT                   | Full Text index file extension                                |
| FSANZ                | Food Standards Australia and New Zealand                      |
| FoodCASE             | Food Composition And System Environment                       |
| FoodID               | Food Index                                                    |
| FRZN                 | Frozen                                                        |
| HPLC                 | High Performance Liquid Chromatography                        |
| ICP-MS               | Inductivity Coupled Plasma Mass Spectrometry                  |
| ICP-OES              | Inductivity Coupled Plasma Optical Emission Spectroscopy      |
| IU                   | International Unit                                            |
| IUPAC                | International Union of Pure and Applied Chemistry             |
| INFOODS              | International Network of Food Data Systems                    |
| LC                   | Liquid Chromatography                                         |
| NIP                  | Nutrition Information Panel                                   |
| NRF                  | Nutrient Retention Factor                                     |
| NUTTAB               | NUTrition TABLEs                                              |
| NZFCD                | New Zealand Food Composition Database                         |
| PFR                  | The New Zealand Institute for Plant and Food Research Limited |
| RAE                  | Retinol Activity Equivalent                                   |
| RTD                  | Ready To Drink                                                |
| RE                   | Retinol Equivalent                                            |
| SIMMR                | Simmer                                                        |
| TFA                  | Total Fatty Acid                                              |

| Abbreviation/Acronym | Meaning                                      |
|----------------------|----------------------------------------------|
| TMAH                 | Tetramethylammonium hydroxide                |
| UHT                  | Ultra-heat treated<br>Ultra-high-temperature |
| USDA                 | United States Department of Agriculture      |
| UV                   | Ultra-Violet                                 |
| YF                   | Yield Factor                                 |
| W/                   | With                                         |
| W/O                  | Without                                      |
| WHO                  | World Health Organisation                    |
| WTR                  | Water                                        |
| WYF                  | Weight Yield Factor                          |

# New Zealand Food Composition Database 2024 Data Manual

August 2024

The New Zealand Institute for Plant and Food Research Limited and the New Zealand Ministry of Health jointly own the intellectual property in the New Zealand Food Composition Database (NZFCD). The Ministry of Health is the principal funder of the database, with Plant & Food Research investing in the database management system and related infrastructure and hosting the database. The Ministry of Health contracts Plant & Food Research to maintain and develop the NZFCD and to disseminate data, via updated releases of the New Zealand FOODfiles and Online Search.

## Purpose of this manual

This reference manual has been produced by Plant & Food Research to facilitate the use of the New Zealand FOODfiles™ 2024, and Online Search. The manual provides technical information for users, and information for computing personnel involved in setting up database management systems utilising data from New Zealand FOODfiles. It also provides information regarding programmers' application software.

The New Zealand FOODfiles were first released in 1989 by the Biotechnology Division, Department of Scientific and Industrial Research (DSIR), with the majority of the data coming from overseas sources. Since then, 23 upgrades have been released, with the latest being this current edition – New Zealand FOODfiles 2024. The data have been regularly disseminated from the New Zealand Food Composition Database (NZFCD) by producing updated, freely accessible versions of FOODfiles and associated products on the website [www.foodcomposition.co.nz](http://www.foodcomposition.co.nz) since 2011. In June 2018 the website was refreshed to incorporate new online search functions, making data more accessible and user-friendly. Downloads of the New Zealand FOODfiles 2024 remain freely accessible through the website [www.foodcomposition.co.nz](http://www.foodcomposition.co.nz) for all users. In addition, the website provides a 'Search' facility with a number of report options, including a Nutrition Information Panel (NIP) and claimable nutrients, in addition to the full datasets. Previous versions of New Zealand FOODfiles (2010 onwards) and the Concise Tables (8th–14th editions) are archived on the same website.

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# 1 Introduction

The New Zealand Food Composition Database release 2024 is the major source of verified food composition data available in New Zealand. It replaces the data released in March 2022 (those data can still be accessed in the archives on the website).

There are two versions of the data files in the New Zealand FOODfiles 2024:

1. A standard version that contains information for 87 components in 2857 foods. Some foods do not have data for all 87 components because the number of core components has changed over time.
2. An unabridged version that contains information on up to 434 food components of the same 2857 foods listed in the standard version. Most foods do not have data for all 434 components, for a variety of reasons.

The NZFCD data files are intended to be used in database or spreadsheet applications and, depending on their intended use, may require additional computer programming input to be used fully and effectively.

In addition, FOODfiles is the major source of data supporting Online Search and The Concise New Zealand Food Composition Tables. Note, the Concise Tables have not been updated in the 2024 release but the last version (14<sup>th</sup> edition, 2021) is accessible via the website [www.foodcomposition.co.nz](http://www.foodcomposition.co.nz).

## 1.1 Data sources

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Most of the data found in NZFCD release 2024 were derived from analytical laboratory results from appropriately accredited laboratories. Some of the data have been obtained from overseas food composition databases/tables, Nutrition Information Panels on packaged foods, or from the New Zealand Manufactured Food Database; and some data have been imputed from related foods or derived from recipe calculations by applying the nutrition retention factor. The following sources are acknowledged for the contribution of food compositional data:

- Food Composition Table for Use in Africa (Wu Leung 1968)
- Pacific Islands Food Composition Tables (Dignan et al. 2004)
- Canadian Nutrient File (Health Canada 2015)
- Fineli-Finnish Food Composition Database (National Institute for Health and Welfare 2019)
- FoodData Central (U.S. Department of Agriculture 2024)
- Australian Food Composition Database (Food Standards Australia New Zealand 2022)
- Indian Food Composition (Longvah et al. 2017)
- UK/National Nutrient Databank/McCance and Widdowson's The Composition of Foods Integrated Dataset (CoFID) (Finglas et al. 2021)
- Japan Food Composition Tables (Ministry of Education 2017).

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Australian source data are copyright Food Standards Australia New Zealand (2022) and are reproduced by permission.

## 1.2 Updates from 2021 to 2024

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FOODfiles 2024 contains the updates detailed below. Further details can be found in the FOODfiles 2024 Update files (available via [www.foodcomposition.co.nz](http://www.foodcomposition.co.nz)). See Section 3 for more details.

- A total of 191 Food Records (individual food or beverage allocated a unique identifier and Food Name) have been updated (Table 1). The list of updated Food Records can be found in the update files on the website: 'New Food Records replacing the old Food Records in the latest version of FOODfiles.xlsx'.
- In contrast, 97 Food Records reported in FOODfiles 2021 have been archived (this list of food is provided in 'Food Records archived from latest version of FOODfiles.xlsx' on the website). If there is a need to access data for these foods, this can be done by downloading FOODfiles 2021 from the archives on the website.
- Food composition data points have been updated for some foods in the FOODfiles 2021. These updates comparing 2021 and 2024 can be found in the 'Data added to or updated in the Food Records in the latest version of FOODfiles.xlsx' available on the website.
- Two new components, galactose and dietary fibre, measured using the HPLC method (fibre, total dietary (LC method)) added to both standard and unabridged DATA files. In addition to these two components, 72 new components have been added to the unabridged DATA files. Most of the components are individual fatty acids, expressed in two units: g per 100 g edible portion and g per 100 g Total Fatty Acid (TFA). The full list of the added components can be found in the 'List of components updated in current version of FOODfiles.xlsx'.

As raw ingredients can vary slightly within and between seasons, users should note that the data contained in FOODfiles 2024 do not represent absolute values. Rather, the data provided reflect the typical composition of foods as available and/or consumed in New Zealand, and thus actual values may differ at times.

For additional information on how we have prioritised foods for analysing or derived values from recipe methods, or information on images, sampling, analytical and nutrient plans, please contact staff at Plant & Food Research: [foodcomp@plantandfood.co.nz](mailto:foodcomp@plantandfood.co.nz). Some of this information can also be found in the publication Sivakumaran et al. (2018).

Table 1. Summary of the updated Food Records in the 2024 release<sup>a</sup>.

| Derivation          | # Food records                                     | Type of foods                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| New Foods           | By analysis <sup>b</sup>                           | 83<br><b>Bakery products:</b> Rewena (Māori bread); <b>Beverages, non-alcoholic:</b> tomato juice [unsweetened]; <b>Cereals and pseudo-cereals:</b> fresh pasta [uncooked], and instant noodles [uncooked]; <b>Dairy:</b> Double cream, evaporated cream [skim], yoghurt, plain [high fat and fortified]; yoghurt, assorted fruits [lactose-free and high protein], dairy foods [fortified], cheese [paneer], and reduced cream; <b>Fast foods and ready to eat meals:</b> patty, frozen [kumara]; <b>Fruits:</b> apple dried, mixed citrus peel, frozen banana, canned mango [pulp and drained], canned apple in water, canned pear in juice [drained], canned pear in light syrup [drained], canned pear in water [drained and undrained], canned Boysenberry [drained and undrained], canned raspberry [drained], canned blackberry [drained and undrained], canned blueberry [drained and undrained], canned plum [undrained], passionfruit [pulp and pulp with seed] and lime juice; <b>Miscellaneous:</b> dairy-free [sour cream, cream cheese and cheese], and salt [Himalayan pink and lower sodium]; <b>Condiments:</b> tomato sauce [reduced sugar]; <b>Snack foods:</b> snack bar, puffed rice base [reduced sugar, sweetened with intense sweetener and with dark chocolate or white chocolate]; cereal bar [Kellogg's® Nutri Grain® and Nestle® Milo®], cereal soft bar [wheat based and oat based] and seed bar; <b>Soups:</b> onion flavoured dry mix; <b>Vegetable and pulses:</b> celeriac, natto, lentil [brown and French], canned vegetables [black bean, baby corn and jack fruit]. |
|                     | Partial analysis & recipe calculation <sup>c</sup> | 8<br><b>Cereals and pseudo-cereals:</b> instant noodle, boiled [drained]; <b>Dairy:</b> cheese [paneer, cooked]; <b>Fast foods and ready-to-eat meals:</b> patty, cooked [kumara]; <b>Vegetable and pulses:</b> cooked [fennel, okra, butternut, canned jack fruit, and lentil cooked (brown and French)]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | Recipe <sup>d</sup>                                | 10<br><b>Cereals and pseudo-cereals:</b> noodle, boiled [undrained]; <b>Egg:</b> white & yolk [fried without oil], yolk [scrambled without oil] and white [scrambled without oil]; <b>Fruits:</b> tamarillo stewed [sweetened and unsweetened], feijoa stewed [sweetened and unsweetened]; <b>Nuts and seeds:</b> almond dry roasted; <b>Soups:</b> Onion flavoured dip [prepared]; <b>Soups:</b> Onion flavoured dry mix soup [prepared].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     | Borrowed <sup>e</sup>                              | 9<br><b>Cereals and pseudo-cereals:</b> flours from [amaranth, sorghum, buckwheat grains and barley grains], grains [buckwheat and millet]; <b>Miscellaneous:</b> meat alternative [mycoprotein/fungus base]; <b>Vegetable and pulses:</b> black gram dhal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Replacing old foods | By analysis <sup>b</sup>                           | 55<br><b>Beverages, non-alcoholic:</b> tomato juice [sweetened]; <b>Dairy:</b> Sour cream [regular fat and lite], cottage cheese [regular fat and lite], cream [regular fat (refrigerated and UHT), reduced fat [UHT and canned], cream cheese [regular fat and reduced fat], condensed milk [whole and skim], evaporated milk [whole], Instant milk powder [skim and whole], yoghurt, plain [Greek, Greek-style, sweetened, and unsweetened]; yoghurt, assorted fruits [high fat, Greek-style, sugar-sweetened, and fortified], processed cheese [regular fat and reduced fat] and human milk; <b>Fruits:</b> rockmelon, honeydew melon, feijoa, tamarillo, fresh raspberry, fresh cooking apple, fresh mango, canned mango [undrained], canned pear in juice [undrained], canned pear in syrup [drained and undrained], canned pear in light syrup [undrained], canned raspberry [undrained], canned plum [undrained]; <b>Miscellaneous:</b> salt [iodised and non-iodised]; <b>Nuts and seeds:</b> Brazil nut; <b>Condiments:</b> tomato sauce [regular]; <b>Snack foods:</b> snack bar, puffed rice base [regular sugar]; <b>Vegetable and pulses:</b> fresh [fennel, okra and butternut], tempeh [uncooked], red lentil.                                                                                                                                                                                                                                                                                                                                                                            |
|                     | Partial analysis & recipe calculation <sup>c</sup> | 5<br><b>Cereals and pseudo-cereals:</b> fresh pasta [cooked] and noodle boiled [drained]; <b>Vegetable and pulses:</b> celeriac [cooked], red lentil [cooked] and tempeh [cooked].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                     | Recipe <sup>d</sup>                                | 10<br><b>Fruits:</b> cooking apple [stewed (sweetened and unsweetened) and, baked (sweetened and unsweetened)], raspberry stewed [sweetened                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Derivation            | # Food records | Type of foods                                                                                                                                                                                                                                    |
|-----------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                | and unsweetened]; <b>Nuts and seeds:</b> hazel nut, dry roasted; <b>Recipe:</b> guacamole dip, spinach quiche and chilli con carne and corn chip                                                                                                 |
| Borrowed <sup>e</sup> | 11             | <b>Cereals and pseudo-cereals:</b> rye flour; <b>Fin fishes:</b> anchovy [canned]; <b>Fruit:</b> pawpaw or papaya; <b>Meat:</b> pork pate; <b>Miscellaneous:</b> spices [ginger, cinnamon, cardamom, cloves, turmeric, and nutmeg] and gelatine. |
| Total                 | 191            |                                                                                                                                                                                                                                                  |

<sup>a</sup> More details can be found in the Update files 'New Food Records replacing the old Food Records in the latest version of FOODfiles.xlsx'.

<sup>b</sup> By analysis – the majority of the core components values were derived by analysis.

<sup>c</sup> Partial analysis & recipe calculation – some core components' values were derived by analytical means, and the rest from recipe calculation.

<sup>d</sup> Recipe – most of the core components values were derived by recipe calculation

<sup>e</sup> Borrowed – core component values derived from one or more published sources

UHT = ultra-heat treated.

## 2 Notes for users

### 2.1 Limitations in the dataset

FOODfiles 2024 contains a listing of 2857 Food Records from the NZFCD. Although there are 434 individual food components (nutrients and non-nutrients) in the NZFCD, only 87 of these (“core components”, the most commonly referenced food components) are included in the standard version of FOODfiles 2024. Some foods do not have data for all 87 components because the set of core components has changed over time.

The unabridged version of FOODfiles 2024 includes data for up to 434 components. The total data points are approximately 572,000. Where the data are unavailable for particular components, they are recorded as missing values. Food component data are expressed as per 100 g edible portion of food or as specified.

### 2.2 FoodID

Each food is identified by a unique alpha-numeric FoodID: a single letter denoting a food chapter (see Table 2) followed by one to five digits. As new Food Records are added to the NZFCD, they are given a unique FoodID. When Food Records are archived in the previous versions of FOODfiles, their FoodIDs are also archived i.e. never used again. The FoodID is the record key that links data between NAME.FT, DATA.FT, DATA.AP, CSM.FT and INGREDIENT.FT files (these files are described in Section 3).

Table 2. Descriptions of food chapters.

| Chapter | Food chapter                      | Types of products included                                                                                                                                                                                                                                                          |
|---------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A       | Bakery products                   | Bagels, biscuits, breads, buns, cakes, crackers, crumpets, doughnuts, muffins, scones wrap etc.                                                                                                                                                                                     |
| B       | Beverages, alcoholic              | Beers, ciders, liqueurs, mixed drinks (Ready to Drink, RTD), ports, sake, sherries, sprits, vermouths, wines etc.                                                                                                                                                                   |
| C       | Beverages, non-alcoholic          | Beverage bases (e.g. cocoa powder); coffee beverages, energy drinks, energy food drinks, flavoured drinks, juice concentrates, fruit drinks and juices, plant-based milks (e.g. soy, almond, coconut), soft drinks, sports drinks, smoothies, teas (including kombucha), water etc. |
| D       | Breakfast cereals                 | Extruded cereals, mixed grain cereals, mueslis, porridge etc.                                                                                                                                                                                                                       |
| E       | Cereals and pseudo-cereals        | Cereal constituents, flours, noodles, pasta, rice etc.                                                                                                                                                                                                                              |
| F       | Dairy                             | Butters, cheeses, creams, milks, ice creams, protein powder, yoghurt etc.                                                                                                                                                                                                           |
| G       | Eggs                              | Chicken eggs (raw and cooked)                                                                                                                                                                                                                                                       |
| H       | Fast foods and ready to eat meals | Takeaways, including burgers, pizzas, savoury items, and cooked frozen meals                                                                                                                                                                                                        |
| J       | Fats and oils                     | Cooking and salad oils, margarines, shortening etc.                                                                                                                                                                                                                                 |
| K       | Fin fishes                        | Raw, cooked, smoked, and canned fishes                                                                                                                                                                                                                                              |
| L       | Fruits                            | Raw, dried, canned, and cooked fruits                                                                                                                                                                                                                                               |
| M       | Meats                             | Raw and cooked varieties of meat cuts including beef, chicken, lamb, pork, venison, offal items etc.                                                                                                                                                                                |
| N       | Meat products                     | Cured meats include bacon, sausages, salami etc.                                                                                                                                                                                                                                    |

| Chapter | Food chapter                              | Types of products included                                                                                                         |
|---------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| P       | Miscellaneous                             | Salts, stocks, yeast and yeast spreads, spices, herbs, meat alternative, dairy-free etc.                                           |
| Q       | Nuts and seeds                            | Nuts, seeds, and their products                                                                                                    |
| R       | Recipes                                   | Multiple-ingredient recipes, derived from recipe calculations using Weight Yield Factor (WYF) and Nutrient Retention Factors (NRF) |
| S       | Sauces                                    | Dressings, dips, sauces including tomato, simmer, and soy                                                                          |
| T       | Shellfishes                               | Cockles, crab, mussels, prawns, scallops, squid etc.                                                                               |
| U       | Snack foods                               | Chips including potato, soy and corn, and bars including muesli, nuts, and fruits etc.                                             |
| V       | Soups                                     | Dry mixes, prepared and canned                                                                                                     |
| W       | Sugars, confectionaries and sweet spreads | Chocolates, jams, lollies, marmalade, syrups, sugars, honeys etc.                                                                  |
| X       | Vegetables and pulses                     | Raw and cooked vegetables and pulses, tomato puree, tofu, potato fries etc.                                                        |

## 2.3 Value derivation description

Derivation codes are used to explain how the data are sourced for all the component values in the NZFCD. In the FOODfiles 2024, there are two types of value derivation descriptions. In 2021, when NZFCD transitioned from the old Food Information Management System (FIMS) to Food Composition and System Environment (FoodCASE), a new derivation system called 'Data provenance' was adapted and used for 38% (21,500) of data points, which include 708 of the foods in FOODfiles 2024. The remaining 62% (35,700) data points covering 2149 foods, were carried over from FOODfiles 2021 and still use the old description 'source code'. The list of 'Data provenance' including source codes can be found FOODfiles 2024 downloadable files (see Section 3).

### 2.3.1 Data provenance

'Data provenance' is a value derivation description that includes six mandatory properties (Table 3) based on the EuroFIR thesauri (Machackova et al. 2019). The full list of the Data provenance descriptions can be found in the FOODfiles 2024 downloadable files. The 'Data provenance' descriptions applied to each component value can be found in the DATA.FT, only for new and updated data points (214, 650).

Table 3. Mandatory value properties used for 'Data provenance'.

| Value properties | Description                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acquisition Type | Description of the origin of the value reported e.g. independent laboratory and value created within host system.                                                                                    |
| Value Type       | Description of the value, e.g. best estimate, logical zero, below limit of detection, weighted.                                                                                                      |
| Method Type      | A general indication of the type of method used to obtain the associate value, e.g. analytical, estimated as logical reduction, calculated from a recipe, summation from constituent components.     |
| Method Indicator | Details of the method used to obtain a value, including analytical methods, e.g. high performance liquid chromatography, microbiological assay, calculation by formula or recipe method, imputation. |
| Unit             | A measure of the amount of the component, e.g. gram (g), microgram (µg), milligram (mg).                                                                                                             |
| Matrix Unit      | Amount of matrix material that has its quantity reported, usually expressed using the preposition "per" e.g. per 100 g edible portion (W), per 100 g fatty acid (F) and per g Nitrogen (N).          |

Source: Reference: Machackova et al. (2019).

2.3.2 Source code

Data points totalling 357167, covering 2149 foods from the FOODfiles 2021, are assigned with source code(s). The lower-case letter code indicates the source of the data, such as analysis, calculation or borrowing from another source. The full list of the source code and description can be found in the 'Data provenance' downloadable files from the website. The common source codes used are listed in Table 4.

Table 4. Common source codes and descriptions.

| Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| z    | New Zealand analytical data – the majority of the values were derived by analysis.                                                                                                                                                                                                                                                                                                                                                                                  |
| c    | Calculated by the system (FoodCASE) after data entry – component value can be calculated using the recipe calculation method and calculation formula, e.g. energy. The calculated value can have various combinations of source codes, e.g. the source code “cdz” for ‘Protein, total; calculated from total nitrogen’ represent by calculation (c) applying a conversion factor from published source (d) to New Zealand analytical value (z) for nitrogen, total. |
| p    | Presumed or logical zero – if the food is not a source of the particular food component, e.g. for cows’ milk and beef meat, the dietary fibre value is presumed zero, as milk and meat are not source of dietary fibres.                                                                                                                                                                                                                                            |
| r    | Value imputed from a related food – value source from similar or related foods.                                                                                                                                                                                                                                                                                                                                                                                     |
| a    | Australian Food Composition Database (AFCD, previously called NUTTAB) sourced value – borrowed value expected to be representative of the New Zealand food.                                                                                                                                                                                                                                                                                                         |
| d    | Derived from published sources.                                                                                                                                                                                                                                                                                                                                                                                                                                     |

2.4 Food components

Each food component is uniquely identified by an INFOODS (International Network of Food Data Systems) tagname. Tagnames are compiled by an expert committee under the auspices of INFOODS and allocated to unique food components (Klensin et al. 1989; FAO/INFOODS 2012a). The INFOODS tagnames incorporate the component entity; the method of analysis where this is specific to the result (e.g. dietary fibre methodologies); and the default unit of measure. As INFOODS tagnames are not available for some of the food components recorded in the NZFCD, the NZFCD uses ‘Component Identifiers’. The ‘Component Identifier’ includes the available INFOODS tagname or, if there is no tagname, it is generated following a similar naming convention to that used for the INFOODS tagnames. A list of the food components, their ‘Component Identifiers’ INFOODS tagnames and units of measure is given in Appendix 1. All components’ values are expressed in per 100 g of edible portion. Additionally, fatty acids and amino acids values are expressed per 100 g of total fatty acids (TFA) and per g of nitrogen (N). Additional information for each component is also included: availability in data files (Standard and Unabridged), the number of foods having the value, and the percentage of foods having the value (calculated by dividing the number of foods with the value by the total number of foods, 2858). Furthermore, it is indicated which components are analysed (A) and which are calculated (C).

Each component’s values are rounded to the number of significant figures and maximal decimal places described in the Appendix 2.

The chemical methods used for the analyses of the components are summarised in Appendix 3. The components that are calculated from analytical data are described in the following sections.

### 2.4.1 Proximate, total

The proximate total is calculated by summing the following components: Ash (ASH), protein (PROT); fat, total (FAT); water (WATER); dietary fibre (FIBTG); alcohol (ALC); and available carbohydrate by weight (CHOAVL). In the FOODfiles 2024, sorbitol (SORTL\_G) and organic acid (OA\_G) are included in the proximate total formula for the first time. If dietary fibre values are obtained by methods, gravimetric (FIBTG) and/or high performance liquid chromatography (FIBTLC), the proximate total using the value from the identified dietary fibre method, indicated by the method indicators MINZ1902 and MINZ1906 respectively, is reported in the DATA.FT files (see Section 3.5).

The sum of proximates should fall within a range of 95–105 g per 100 g edible portion. A margin of plus or minus 5% is considered acceptable (FAO/INFOODS 2012a) because many of the food components are determined independently on different samples in different laboratories. In FOODfiles 2024, we accept that 8% of the Food Records have a sum outside this acceptable range. Explanations for this include the presence of constituents other than the major proximates listed above, and other miscellaneous organic compounds (e.g. resistant starch, oligosaccharides and polydextrose), which are not measured independently but which may contribute to the values. Where proximate values of entries are outside the  $\pm 5\%$  allowance, the typical food component values are still reliable.

### 2.4.2 Energy

The energy values are calculated in four different ways according to the FAO/INFOODS guidelines (FAO/INFOODS 2012b) and Standard 1.2.8 (Food Standards Australia New Zealand 2021). In addition, they are expressed both in kilocalories (kcal) and kilojoules (kJ), thus providing eight calculated energy values for each food:

1. Energy, total metabolisable (kJ, ENERC)
2. Energy, total metabolisable (kcal, ENERC\_KCAL)
3. Energy, total metabolisable (kJ, including dietary fibre, ENERC1)
4. Energy, total metabolisable (kcal, including dietary fibre, ENERC1\_KCAL)
5. Energy, total metabolisable, carbohydrate by difference, FSANZ (kJ, ENERC\_FSANZ1)
6. Energy, total metabolisable, carbohydrate by difference, FSANZ (kcal, ENERC\_FSANZ1\_KCAL)
7. Energy, total metabolisable, available carbohydrates, FSANZ (kJ, ENERC\_FSANZ2)
8. Energy, total metabolisable, available carbohydrates, FSANZ (kcal, ENERC\_FSANZ2\_KCAL).

Note: Total dietary fibre values were obtained through analysis by two methods: Enzymatic–gravimetric [FIBTG, AOAC 985.29 and 991.43 (Official Methods of Analysis of AOAC International 2023)] and Enzymatic–HPLC (FIBTLC, AOAC 2017.16 (Official Methods of Analysis of AOAC International 2023)) for a few foods in FOODfiles 2024. If the FIBTLC value is greater than the FIBTG value, FIBTG is replaced with FIBTLC in equations 3, 4, 5 and 7 as indicated by the method indicators in the DATA.FT files.

The values are calculated from the energy-producing food components (carbohydrate, protein, fat, total and alcohol), with and without dietary fibre and other energy-producing food components, using the appropriate energy conversion factors (Table 5) as described below.



Table 5. Conversion factors used for calculation of energy.

| Component identifier for conversion factor | Component (Component Identifier)                                         | kJ/g | kcal/g |
|--------------------------------------------|--------------------------------------------------------------------------|------|--------|
| XPROT                                      | Protein, total (PROT)                                                    | 17   |        |
| XPROT_KCAL                                 |                                                                          |      | 4      |
| XFAT                                       | Fat, total (FAT)                                                         | 37   |        |
| XFAT_KCAL                                  |                                                                          |      | 9      |
| XCHOAVL                                    | Available carbohydrate [by the weight, difference (CHOAVL)] <sup>a</sup> | 17   |        |
| XCHOAVL_KCAL                               |                                                                          |      | 4      |
| XCHOCDF                                    | Total carbohydrate by difference (CHOCDF)                                | 17   |        |
| XCHOCDF_KCAL                               |                                                                          |      | 4      |
| XALC                                       | Alcohol (ALC)                                                            | 29   |        |
| XALC_KCAL                                  |                                                                          |      | 7      |
| XFIBTG                                     | Dietary fibre (FIBTG)                                                    | 8    |        |
| XFIBTG_KCAL                                |                                                                          |      | 2      |
| XOA                                        | Organic acid (OA_G)                                                      | 13   |        |
| XOA_KCAL                                   |                                                                          |      |        |
| XOA_KCAL                                   |                                                                          |      | 3      |

<sup>a</sup>The conversion factors for available carbohydrates by weight (XCHOAVL and XCHOAVL\_KCAL) are used as conversion factors for all other carbohydrates when converting to energy.  
Source:(Food and Agriculture Organisation of the United Nations and the World Health Organisation 2003; Food Standards Australia New Zealand 2021).

FAO/INFOODS

This set of energy calculations meets international conventions as given in FAO/INFOODS (2012b).

1. **Energy, total metabolisable (kJ, ENERC)** is calculated from the values of energy-producing food components and appropriate conversion factors listed in Table 5. The energy-producing food components taken into account are protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), and alcohol (ALC),

Equation 1:

ENERC (kJ/100 g)

$$\begin{aligned} &= [\text{PROT (g/100 g)} \times \text{XPROT(kJ/g)}] \\ &+ [\text{CHOAVL(g/100 g)} \times \text{XCHOAVL(kJ/g)}] \\ &+ [\text{FAT(g/100 g)} \times \text{XFAT(kJ/g)}] \\ &+ [\text{ALC(g/100 g)} \times \text{XALC (kJ/g)}] \end{aligned}$$

where the ‘Component Identifier’ XPROT, XCHOAVL, XFAT and XALC are the energy conversion factors expressed in kJ/g for protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), and alcohol (ALC) as listed in Table 5.

2. **Energy, total metabolisable (kcal, ENERC\_KCAL)** is calculated from the energy-producing food components: protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), and alcohol (ALC),

Equation 2:

$$\begin{aligned} \text{ENERC\_KCAL (kcal/100 g)} &= [\text{PROT (g/100 g)} \times \text{XPROT\_KCAL (kcal/g)}] \\ &+ [\text{CHOAVL(g/100 g)} \times \text{XCHOAVL\_KCAL (kcal/g)}] \\ &+ [\text{FAT(g/100 g)} \times \text{XFAT\_KCAL (kcal/g)}] \\ &+ [\text{ALC(g/100 g)} \times \text{XALC\_KCAL (kcal/g)}] \end{aligned}$$

where XPROT\_KCAL, XCHOAVL\_KCAL, XFAT\_KCAL and XALC\_KCAL are the energy conversion factors expressed in kcal/g for protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), and alcohol (ALC) as listed in Table 5.

3. **Energy, total metabolisable (kJ, including dietary fibre, ENERC1)** is calculated from the energy-producing food components: protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), alcohol (ALC) and fibre, total dietary (FIBTG),

Equation 3:

$$\begin{aligned} \text{ENERC1 (kJ/100 g)} &= [\text{PROT (g/100 g)} \times \text{XPROT(kJ/g)}] \\ &+ [\text{CHOAVL(g/100 g)} \times \text{XCHOAVL(kJ/g)}] \\ &+ [\text{FAT(g/100 g)} \times \text{XFAT(kJ/g)}] \\ &+ [\text{ALC(g/100 g)} \times \text{XALC (kJ/g)}] \\ &+ [\text{FIBTG(g/100 g)} \times \text{XFIBTG(kJ/g)}] \end{aligned}$$

where the 'Component Identifier' XFIBTG is the energy conversion factor expressed in kJ/g for fibre. Other energy conversion factors are explained above in Equation 1.

**Energy, total metabolisable (kcal, including dietary fibre, ENERC1\_KCAL)** is calculated from the energy-producing food components: protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), alcohol (ALC) and dietary fibre (FIBTG),

Equation 4:

$$\begin{aligned} \text{ENERC1\_KCAL (kcal/100 g)} &= [\text{PROT (g/100 g)} \times \text{XPROT\_KCAL (kcal/g)}] \\ &+ [\text{CHOAVL(g/100 g)} \times \text{XCHOAVL\_KCAL + (kcal/g)}] \\ &+ [\text{FAT(g/100 g)} \times \text{XFAT\_KCAL (kcal/g)}] \\ &+ [\text{ALC(g/100 g)} \times \text{XALC\_KCAL (kcal/g)}] \\ &+ [\text{FIBTG(g/100 g)} \times \text{XFIBTG\_KCAL (kcal/g)}] \end{aligned}$$

where the 'Component Identifier' XFIBTG\_KCAL is the energy conversion factor expressed in kcal/g for fibre. Other energy conversion factors are explained above in Equation 2.

## FSANZ Standard 1.2.8

This set of energy calculations follows the FSANZ guidelines (Food Standards Australia New Zealand 2021) and is used for food labelling in New Zealand.

4. **Energy, total metabolisable, carbohydrate by difference, FSANZ (kJ, ENERC\_FSANZ1)** is calculated according to Standard 1.2.8. The energy-producing food components accounted for are protein (PROT), carbohydrates by difference (CHOAVLDF\_FSANZ), fat, total (FAT), alcohol (ALC), dietary fibre (FIBTG) and organic acids (OA\_G). The calculation of carbohydrate by difference (CHOAVLDF\_FSANZ) value is explained below in Section 2.4.5.

Equation 5:

$$\begin{aligned} \text{ENERC\_FSANZ1 (kJ/100 g)} &= [\text{PROT (g/100 g)} \times \text{XPROT(kJ/g)}] \\ &+ [\text{CHOAVLDF\_FSANZ (g/100 g)} \times \text{XCHOAVL(kJ/g)}] \\ &+ [\text{FAT(g/100 g)} \times \text{XFAT(kJ/g)}] \\ &+ [\text{ALC(g/100 g)} \times \text{XALC (kJ/g)}] \\ &+ [\text{FIBTG(g/100 g)} \times \text{XFIBTG(kJ/g)}] \\ &+ [\text{OA\_G(g/100 g)} \times \text{XOA(kJ/g)}] \end{aligned}$$

Note: Food components 'unavailable carbohydrates', 'polyols' and 'polydextrose' are not recorded in FOODfiles 2024. Therefore, they are not included when calculating ENERC\_FSANZ1.

5. **Energy, total metabolisable, carbohydrate by difference, FSANZ (kcal, ENERC\_FSANZ1\_KCAL)** is converted from ENERC\_FSANZ1 by using the unit conversion factor 4.18.

Equation 6:

$$\text{ENERC\_FSANZ1\_KCAL(kcal/100 g)} = \text{ENERC\_FSANZ1 (kJ/100 g)}/4.18$$

6. **Energy, total metabolisable, available carbohydrates, FSANZ (kJ, ENERC\_FSANZ2)** is calculated according to Standard 1.2.8. The energy-producing food components accounted are protein (PROT), available carbohydrates (CHOAVL\_FSANZ), fat, total (FAT), alcohol (ALC), dietary fibre (FIBTG) and organic acids (OA\_G). The calculation of available carbohydrates (CHOAVL\_FSANZ) value is explained in Section 2.4.5.

Equation 7:

$$\begin{aligned} \text{ENERC\_FSANZ2 (kJ/100 g)} &= [\text{PROT (g/100 g)} \times \text{XPROT(kJ/g)}] \\ &+ [\text{CHOAVL\_FSANZ (g/100 g)} \times \text{XCHOAVL(kJ/g)}] \\ &+ [\text{FAT(g/100 g)} \times \text{XFAT(kJ/g)}] \\ &+ [\text{ALC(g/100 g)} \times \text{XALC (kJ/g)}] \\ &+ [\text{FIBTG(g/100 g)} \times \text{XFIBTG(kJ/g)}] \\ &+ [\text{OA\_G(g/100 g)} \times \text{XOA(kJ/g)}] \end{aligned}$$

Note: Food components 'unavailable carbohydrates', 'polyols' and 'polydextrose' are not recorded in FOODfiles 2024. Therefore, they are not included when calculating ENERC\_FSANZ2.

7. **Energy, total metabolisable, available carbohydrates, FSANZ (kcal, ENERC\_FSANZ2\_KCAL)** is converted from ENERC\_FSANZ2 by using the unit conversion factor 4.18,

Equation 8:

$$\text{ENERC\_FSANZ2\_KCAL (kcal/100 g)} = \text{ENERC\_FSANZ2 (kJ/100 g)}/4.18$$

Note: Total dietary fibre values were obtained through analysis by two methods: Enzymatic–gravimetric [FIBTG, AOAC 985.29 and 991.43 (Official Methods of Analysis of AOAC International 2023)] and Enzymatic–HPLC (FIBTLC, AOAC 2017.16 (Official Methods of Analysis of AOAC International 2023)) for a few foods in FOODfiles 2024. If the FIBTLC value is greater than the FIBTG value, FIBTG is replaced with FIBTLC in equations 3, 4, 5 and 7 indicated by the method indicators in the DATA.FT files (see section 3.5).

Some examples showing the calculation using the different methods of determining energy can be found in Appendix 4.

2.4.3 Protein, total

For all foods, protein, total; calculated from total nitrogen (PROT) is calculated from the measured total nitrogen (NT) multiplied by a specific nitrogen-to-protein conversion factor (XN) (Jones 1941; Greenfield & Southgate 2003). The conversion factor that is used for the calculation of the PROT for each food is tabulated in FOODfiles 2024, CONVERSION FACTOR.FT (see Section 3).

Equation 9:

$$\text{PROT (g / 100 g)} = \text{NT (g / 100 g)} \times \text{XN (g / g Nitrogen)}$$

2.4.4 Sugars

Sugars total (SUGAR) is the sum of the free individual monosaccharides, glucose (GLUS), fructose (FRUS), and galactose (GALS) and the disaccharides lactose (LACS), maltose (MALS) and sucrose (SUCS),

Equation 10:

$$\begin{aligned} \text{SUGAR (g / 100 g food)} \\ &= \text{GLUS (g/100 g)} + \text{FRUS (g/100 g)} + \text{LACS (g/100 g)} + \text{GALS (g/100 g)} \\ &+ \text{MALS (g/100 g)} + \text{SUCS (g/100 g)} \end{aligned}$$

Glycogen (GLYC), lactose (LACS), maltose (MALS), sucrose (SUCS), starch, total (STARCH), and oligosaccharides, total available (OLSAC) are expressed as monosaccharide equivalents (GLYCM, LACSM, MALSM, SUCSM, STARCHM and OLSACM respectively) calculated using the conversion factors shown in Table 6.

Table 6. Conversion factors used for the calculation of monosaccharide equivalents.

| Food component         | Conversion to monosaccharide equivalents (g/g) |
|------------------------|------------------------------------------------|
| Glycogen (GLYC)        | x 1.10                                         |
| Lactose (LACS)         | x 1.05                                         |
| Maltose (MALS)         | x 1.05                                         |
| Sucrose (SUCR)         | x 1.05                                         |
| Starch, total (STARCH) | x 1.10                                         |

Source: Finglas et al. (2015).

**Disaccharide, total; expressed in monosaccharide equivalents (DISACM)** is calculated by summation of lactose, maltose and sucrose expressed in monosaccharide equivalents (LACSM, MALSM and SUCSM respectively),

Equation 11:

$$\text{DISACM (g/100 g)} = \text{LACSM (g/100 g)} + \text{MALSM (g/100 g)} + \text{SUCSM (g/100 g)}$$

**Sugar, total; expressed in monosaccharide equivalents (SUGARM)** is calculated by summation of the free monosaccharide (GLUS and FRUS) and disaccharides expressed in monosaccharide equivalents (DISACM),

Equation 12:

$$\text{SUGARM (g/100 g)} = \text{GLUSM (g/100 g)} + \text{FRUSM (g/100 g)} + \text{DISCM (g/100 g)}$$

**Added sugars (SUGAD)** are defined according to the United States Food and Drug Administration (US FDA) as: “sugars that are either added during the processing of foods, or are packaged as such, and include sugars (free, mono- and disaccharides), sugars from syrups and honey, and sugars from concentrated fruit or vegetable juices that are in excess of what would be expected from the same volume of 100 percent fruit or vegetable juices of the same type. The definition excludes fruit or vegetable juice concentrated from 100 percent fruit juice that is sold to consumers (e.g. frozen 100 percent fruit juice concentrate) as well as some sugars found in fruit and vegetable juices, jellies, jams, preserves, and fruit spreads” (Erickson & Slavin 2015).

**Free sugars (SUGFR)** are defined according to the World Health Organisation (WHO) as: “free sugars include monosaccharides and disaccharides added to foods and beverages by the manufacturer, cook or consumer, and sugars naturally present in honey, syrups, fruit juices, and fruit juice concentrates” (World Health Organization 2015).

Added and free sugars are estimated using the 10-step method described by Louie et al. (2015) and Louie et al. (2016). Free sugars are estimated adapting the method of Louie et al. (2015) with some modifications (e.g. to include fruit juice) as described by Kibblewhite et al. (2017). Appendix 5 describes the 10-step process, steps 1–6 being objective and steps 7–10 being subjective.

## 2.4.5 Carbohydrate totals

Carbohydrate values are expressed in different ways in FOODfiles 2024 to meet the definitions for INFOODS (FAO/INFOODS 2012a) and for FSANZ Standard 1.2.8 (Food Standards Australia New Zealand 2021).

Note: Total dietary fibre values were obtained through analysis by two methods: Enzymatic–gravimetric [FIBTG, AOAC 985.29 and 991.43 (Official Methods of Analysis of AOAC International 2023)] and Enzymatic–HPLC (FIBTLC, AOAC 2017.16 (Official Methods of Analysis of AOAC International 2023)) for a few foods in FOODfiles 2024. If the FIBTLC value is greater than the FIBTG value, FIBTG is replaced with FIBTLC in equations 15, 17 and 18, as indicated by the method indicators in the DATA.FT files (see section 3.5).

## INFOODS

The carbohydrate contents in foods are expressed in FOODfiles 2024 in five different ways, based on the FAO/INFOODS (2012b) guidelines. Total carbohydrates value includes fibre, total dietary (FIBTG or FIBTLC) whereas available carbohydrates are expressed 'by weight' or as a 'monosaccharide equivalent', excluding the FIBTG or FIBTLC value. Carbohydrate values are calculated either as the sum of the analytically analysed carbohydrate components or as the difference of 100 minus the sum of the other proximate components.

1. **Available carbohydrates by weight (CHOAVL)** is the sum of the analytical values of sugar, total (SUGAR); starch (STARCH) and glycogen (GLYC),

Equation 13:

$$\text{CHOAVL (g/100 g)} = \text{SUGAR (g/100 g)} + \text{STARCH (g/100 g)} + \text{GLYC (g/100 g)}$$

2. **Available carbohydrates in monosaccharide equivalent (CHOAVLM)** is the sum of the analytical values of sugar, total (SUGAR), starch (STARCH) and glycogen (GLYC). It includes the residual water from hydrolysis around each monosaccharide (monosaccharide equivalents SUGARM, STARCHM and GLYCM),

Equation 14:

$$\text{CHOAVLM (g/100 g)} = \text{SUGARM (g/100 g)} + \text{STARCHM (g/100 g)} + \text{GLYCM (g/100 g)}$$

3. **Available carbohydrate by difference (CHOAVLDF)** is calculated by deducting the sum of the percentage of water (WATER), protein, total (PROT), fat, total (FAT), alcohol (ALC), ash (ASH) and fibre, total dietary (FIBTG) from 100,

Equation 15:

$$\text{CHOAVLDF (g/100 g)} = 100 - [\text{WATER(g/100 g)} + \text{PROT(g/100 g)} + \text{FAT(g/100 g)} + \text{ALC(g/100 g)} + \text{ASH(g/100 g)} + \text{FIBTG(g/100 g)}]$$

Note: Total dietary fibre values were obtained through analysis by two methods: Enzymatic–gravimetric (FIBTG) and Enzymatic–HPLC (FIBTLC) for a few foods in FOODfiles 2024. If the FIBTLC value is greater than the FIBTG value, FIBTG is replaced with FIBTLC in the formula, indicated by the unique Method Indicator MINZ1108 in the DATA.FT files (see section 3.5).

4. **Total carbohydrate by difference (CHOCDF)** is calculated by deducting the sum of the percentage of water (WATER), protein, total (PROT), fat (FAT), alcohol (ALC) and ash (ASH) from 100,

Equation 16:

$$\text{CHOCDF (g/100 g)} = 100 - [\text{WATER(g/100 g)} + \text{PROT(g/100 g)} + \text{FAT(g/100 g)} + \text{ALC(g/100 g)} + \text{ASH(g/100 g)}]$$

5. **Total carbohydrate by summation (CHOCSM)** is the sum of analytical values of sugar, total (SUGAR), starch (STARCH) oligosaccharides (OLSAC) and fibre, total dietary (FIBTG),

Equation 17:

$$\text{CHOCSM (g/100 g)} = \text{SUGAR (g/100 g)} + \text{STARCH (g/100 g)} + \text{OLSAC (g/100 g)} + \text{FIBTG (g/100 g)}$$

Note: Total dietary fibre values were obtained through analysis by two methods: Enzymatic–gravimetric (FIBTG) and Enzymatic–HPLC (FIBTLC) for a few foods in FOODfiles 2024. If the FIBTLC value is greater than the FIBTG value, FIBTG is replaced with FIBTLC in the formula, indicated by the unique Method Indicator MINZ1109 in the DATA.FT files (see section 3.5).

### FSANZ Standard 1.2.8

For nutrient labelling, two definitions of carbohydrate are found in Standard 1.2.8 of the FSANZ Food Standards Code (Food Standards Australia New Zealand 2021) either Carbohydrate by difference or Available carbohydrate. Neither definition includes dietary fibre (fibre, total dietary (FIBTG)).

6. **Carbohydrate by difference, FSANZ (CHOAVLDF\_FSANZ)** is calculated by deducting the sum of the percentage of water (WATER), protein (protein, total; PROT), fat (fat, total; FAT), dietary fibre (fibre, total dietary (FIBTG)), ash (ASH), alcohol (ALC) and, if qualified or added to the food (any amount as additive or ingredients to the final food), any other unavailable carbohydrates and substances: sugar alcohols [erythritol (ERYTHL\_G), glycerol (GLYRL\_G), isomalt (ISOMAL\_G), lactitol (LACTL\_G), maltitol (MALTL\_G), mannitol (MANTL\_G), sorbitol (SORTL\_G), xylitol (XYLTL\_G)], D-tagatose, polydextrose and/or organic acids, total (OA\_G) from 100,

Equation 18:

$$\begin{aligned} \text{CHOAVLDF\_FSANZ (g/100 g)} \\ = 100 - [\text{WATER (g/100 g)} + \text{PROT (g/100 g)} + \text{FAT (g/100 g)} \\ + \text{FIBTG (g/100 g)} + \text{ASH (g/100 g)} + \text{ALC (g/100 g)} + \text{OA\_G (g/100 g)}] \end{aligned}$$

Note: **Carbohydrate by difference (CHOAVLDF\_FSANZ)** is equal to the **Available carbohydrate; by difference (CHOAVLDF)** value if other additives or ingredients and unavailable carbohydrates and substances are not present or accounted for. The food components: erythritol, glycerol, isomalt, lactitol, maltitol, mannitol, xylitol, D-tagatose, and polydextrose are not recorded and sorbitol (SORTL\_G) is recorded for only a few foods in FOODfiles 2024 and therefore are not included in this equation. A few foods do have values for organic acids, total (OA\_G).

7. **Available carbohydrate (CHOAVL\_FSANZ)** is the sum of analytical values of total available sugars (sugar, total; SUGAR) and starch (STARCH) and, if qualified or added to the food (any amount as additive or ingredients to the final food), any available oligosaccharides (oligosaccharides, total available, OLSAC), glycogen (GLYC) and/or maltodextrin (MALTDEX),

Equation 19:

$$\begin{aligned} \text{CHOAVL\_FSANZ (g/100 g)} \\ = \text{SUGAR (g/100 g)} + \text{STARCH (g/100 g)} + \text{GLYC (g/100 g)} \\ + \text{MALTDEX (g/100 g)} \end{aligned}$$

Note: **Available carbohydrate (CHOAVL\_FSANZ)** is equal to the **Available carbohydrates by weight (CHOAVL)** value if values for oligosaccharides and maltodextrin are not present or accounted for. The food components oligosaccharides, total available (OLSAC) are not recorded

in FOODfiles 2021. A few foods do have values for glycogen (GLYC) and maltodextrin (MALTDEX).

## 2.4.6 Fatty acids, total (FACID)<sup>1</sup>

**Fatty acid, total (FACID)** is determined in two different ways for FOODfiles 2024:

1. It is calculated by multiplying the fat, total (FAT) with fat-to-fatty-acid conversion factor (XFA),  
Equation 20:

$$\text{FACID (g / 100 g)} = \text{FAT (g / 100 g)} \times \text{XFA (g / g)}$$

2. It is determined from the analytical chromatography method by Sukhija & Palmquist (1988), so a conversion factor is not required.

Note: The method for determining the total fatty acids is not overtly specified in FOODfiles 2024; either method can be used, and they are all tabulated under the **Fatty acid, total (FACID)**. However, if XFA is listed for a food, the FACID has been calculated using Equation 20 above, and if the XFA value is not listed (left as blank) then the FACID has been determined analytically.

## 2.4.7 Individual fatty acids

Fatty acids data are expressed as both g/100 g of food and as a percentage of the total fatty acid content. Individual fatty acid (g/100 g TFA<sup>2</sup>) values in food are derived from the value of each individual fatty acid (g/100 g Food) using the following equation.

Equation 21:

$$\begin{aligned} \text{Individual fatty acid (g / 100 g)} \\ = (\text{Individual fatty acid (g/100 g TFA)} \times \text{FACID (g / 100 g)}) / 100 \end{aligned}$$

The individual fatty acids in FOODfiles 2024 are listed in Tables 7–9 with the 'Component Identifier', component names systematic names, and trivial names (Klensin et al. 1989; AOCS Lipid Library 2018).

**Fatty acids, total saturated (FASAT)** is the sum of all the individual saturated fatty acids listed in Table 7.

<sup>1</sup> FACID is the INFOOD tagname for Fatty acid, total; determined by calculation using Equation 21.

<sup>2</sup> TFA is a commonly used abbreviation for the total fatty acid, which is equivalent to the 'Component Identifier' fatty acid, total (FACID); Note, TFA is not an INFOODS tagname or 'Component Identifier'. This abbreviation TFA is used for expressing the fatty acid content (g) as per 100 g of total fatty acid (g/100 g TFA).



Table 7. Details of the saturated fatty acids.

| Component Identifier | Fatty acid name | Systematic name | Trivial/common name |
|----------------------|-----------------|-----------------|---------------------|
| F4D0                 | Fatty acid 4:0  | butanoic        | butyric             |
| F6D0                 | Fatty acid 6:0  | hexanoic        | caproic             |
| F8D0                 | Fatty acid 8:0  | octanoic        | caprylic            |
| F10D0                | Fatty acid 10:0 | decanoic        | capric              |
| F11D0                | Fatty acid 11:0 | undecanoic      | undecylic           |
| F12D0                | Fatty acid 12:0 | dodecanoic      | lauric              |
| F13D0                | Fatty acid 13:0 | tridecanoic     | tridecylic          |
| F14D0                | Fatty acid 14:0 | tetradecanoic   | myristic            |
| F15D0                | Fatty acid 15:0 | pentadecanoic   | pentadecylic acid   |
| F16D0                | Fatty acid 16:0 | hexadecanoic    | palmitic            |
| F17D0                | Fatty acid 17:0 | heptadecanoic   | margaric            |
| F18D0                | Fatty acid 18:0 | octadecanoic    | stearic             |
| F19D0                | Fatty acid 19:0 | nonadecanoic    | nonadecylic         |
| F20D0                | Fatty acid 20:0 | eicosanoic      | arachidic           |
| F21D0                | Fatty acid 21:0 | heneicosanoic   | heneicosylic        |
| F22D0                | Fatty acid 22:0 | docosanoic      | behenic             |
| F23D0                | Fatty acid 23:0 | tricosanoic     | tricosylic          |
| F24D0                | Fatty acid 24:0 | tetracosanoic   | lignoceric          |

**Fatty acids, total monounsaturated (FAMU)** is the sum of undifferentiated individual fatty acids listed in Table 8. The undifferentiated fatty acid, e.g. fatty acid 18:1 omega-9 (F18D1N9), is calculated from the sum of its differentiated isomers, i.e. *cis* (F18D1CN9) and *trans* (F18D1TN9).

Table 8. Details of the monounsaturated fatty acids.

| Component Identifier | Fatty acid name                          | Systematic name                | Trivial/common name       |
|----------------------|------------------------------------------|--------------------------------|---------------------------|
| F10D1                | Fatty acid 10:1                          | decenoic                       |                           |
| F12D1                | Fatty acid 12:1                          | dodecenoic                     | lauroleic                 |
| F14D1                | Fatty acid 14:1 undifferentiated         | tetradecenoic                  | myristoleic               |
| F14D1C               | Fatty acid <i>cis</i> 14:1               |                                |                           |
| F14D1CN5             | Fatty acid <i>cis</i> 14:1 omega-5       |                                |                           |
| F14D1N5              | Fatty acid 14:1 omega-5                  | tetradec-9-enoic               | myristoleic               |
| F15D1                | Fatty acid 15:1 undifferentiated         | pentadecenoic                  | oncobic                   |
| F15D1C               | Fatty acid <i>cis</i> 15:1               |                                |                           |
| F15D1CN5             | Fatty acid <i>cis</i> 15:1 omega-5       |                                |                           |
| F15D1N5              | Fatty acid 15:1 omega-5                  |                                |                           |
| F16D1                | Fatty acid 16:1 undifferentiated         | hexadecenoic                   | palmitoleic               |
| F16D1C               | Fatty acid <i>cis</i> 16:1               | <i>cis</i> -hexadec -9-enoic   | <i>cis</i> -palmitoleic   |
| F16D1CN7             | Fatty acid <i>cis</i> 16:1 omega-7       |                                |                           |
| F16D1N7              | Fatty acid 16:1 omega-7                  | hexadec-9-enoic                | palmitoleic               |
| F16D1T               | Fatty acid <i>trans</i> 16:1             | <i>trans</i> -hexadecenoic     | <i>trans</i> -palmitoleic |
| F17D1                | Fatty acid 17:1                          | heptadenoic                    | civetic                   |
| F17D1C               | Fatty acid <i>cis</i> 17:1               |                                |                           |
| F17D1CN7             | Fatty acid <i>cis</i> 17:1 omega-7       |                                |                           |
| F17D1N7              | Fatty acid 17:1 omega-7                  | heptadec-8-enoic               | civetic                   |
| F18D1                | Fatty acid 18:1 undifferentiated         | octadecenoic                   |                           |
| F18D1C               | Fatty acid <i>cis</i> 18:1               | <i>cis</i> -octadecenoic       |                           |
| F18D1CN7             | Fatty acid <i>cis</i> 18:1 omega-7       | <i>cis</i> -octadec-11-enoic   | <i>cis</i> -vaccenic      |
| F18D1CN9             | Fatty acid <i>cis</i> 18:1 omega-9       | <i>cis</i> -octadec-9-enoic    | oleic                     |
| F18D1T               | Fatty acid <i>trans</i> 18:1             | <i>trans</i> -octadecenoic     |                           |
| F18D1TN7             | Fatty acid <i>trans</i> 18:1 omega-7     | <i>trans</i> -octadec-11-enoic | <i>trans</i> -vaccenic    |
| F18D1TN9             | Fatty acid <i>trans</i> 18:1 omega-9     | <i>trans</i> -octadec-9-enoic  | elaidic                   |
| F18D1N7              | Fatty acid 18:1 omega-7 undifferentiated | octadec-11-enoic               | vaccenic                  |
| F18D1N9              | Fatty acid 18:1 omega-9 undifferentiated | octadec-9-enoic                |                           |
| F20D1                | Fatty acid 20:1 undifferentiated         | eicosenoic                     |                           |
| F20D1C               | Fatty acid <i>cis</i> 20:1               |                                |                           |
| F20D1CN9             | Fatty acid <i>cis</i> 20:1 omega-9       | <i>cis</i> -eicos-11-enoic     | gondoic                   |
| F20D1N9              | Fatty acid 20:1 omega-9 undifferentiated | eicos-11-enoic                 | gondoic                   |
| F20D1N11             | Fatty acid 20:1 omega-11                 | eicos-9-enoic                  | gadoleic                  |
| F22D1                | Fatty acid 22:1 undifferentiated         | docosenoic                     |                           |
| F22D1N9              | Fatty acid <i>cis</i> 22:1 omega-9       | <i>cis</i> -docos-13-enoic     |                           |
| F22D1N9              | Fatty acid 22:1 omega-9 undifferentiated | docos-13-enoic                 | erucic                    |
| F22D1N11             | Fatty acid 22:1 omega-11                 | docos-11-enoic                 | catelaidic                |
| F24D1                | Fatty acid 24:1                          | tetracosenoic                  | nervonic                  |
| F20D4C               | Fatty acid <i>cis</i> 24:1               |                                |                           |
| F24D1CN9             | Fatty acid <i>cis</i> 24:1 omega-9       | <i>cis</i> -tetracos-15-enoic  | nervonic                  |
| F24D1N9              | Fatty acid 24:1 omega-9 undifferentiated | tetracos-15-enoic              | nervonic                  |

**Fatty acids, total polyunsaturated (FAPU)** is the sum of undifferentiated individual fatty acids listed in Table 9.

Table 9. Details of the polyunsaturated fatty acids.

| Component Identifier | Fatty acid name                               | Systematic name                                 | Trivial/common name         | Abbreviation |
|----------------------|-----------------------------------------------|-------------------------------------------------|-----------------------------|--------------|
| F18D2                | Fatty acid 18:2 undifferentiated              | octadecadienoic                                 |                             |              |
| F18D2C               | Fatty acid <i>cis</i> 18:2                    | <i>cis</i> -octadecadienoic                     |                             |              |
| F18D2CN6             | Fatty acid <i>cis,cis</i> 18:2 omega-6        | <i>cis,cis</i> -octadeca-9,12-dienoic           | linoleic                    |              |
| F18D2T               | Fatty acid <i>trans</i> 18:2                  | <i>trans</i> -octadecadienoic                   |                             |              |
| F18D2TN6             | Fatty acid <i>trans</i> 18:2 omega-6          | <i>trans</i> -octadeca-9,12-dienoic             | linoelaidic                 |              |
| F18D2N6              | Fatty acid 18:2 omega-6                       | octadeca-9,12-dienoic                           |                             |              |
| F18D2CN9TN11         | Fatty acids <i>cis,trans</i> 18:2 omega-9, 11 | <i>cis,trans</i> -octadeca-9,11-dienoic         | rumenic                     | CLA          |
| F18D3                | Fatty acid 18:3 undifferentiated              | octadecatrienoic                                |                             |              |
| F18D3C               | Fatty acid <i>cis</i> 18:3                    | <i>cis</i> -octadecatrienoic                    |                             |              |
| F18D3CN3             | Fatty acid <i>cis,cis,cis</i> 18:3 omega-3    | all- <i>cis</i> -octadeca-9,12,15-trienoic      | $\alpha$ -linolenic         | ALA          |
| F18D3CN6             | Fatty acid <i>cis, cis, cis</i> 18:3 omega-6  | all- <i>cis</i> -octadeca-6,9,12-trienoic       | $\gamma$ -linolenic         | GLA          |
| F18D3N3              | Fatty acid 18:3 omega-3 undifferentiated      | octadeca-9,12,15-trienoic                       |                             |              |
| F18D3N6              | Fatty acid 18:3 omega-6 undifferentiated      | octadeca-6,9,12-trienoic                        |                             |              |
| F18D3TN3             | Fatty acid <i>trans</i> 18:3 omega-3          | <i>trans</i> -octadeca-9,12,15-trienoic         |                             |              |
| F18D4                | Fatty acid 18:4 undifferentiated              | octadecatetraenoic                              | parinaric                   |              |
| F18D4N3              | Fatty acid 18:4 omega-3                       | all- <i>cis</i> -octadeca-6,9,12,15-tetraenoic  | stearidonic                 |              |
| F20D2                | Fatty acid 20:2 undifferentiated              | eicosadienoic                                   |                             |              |
| F20D2CN6             | Fatty acid <i>cis,cis</i> 20:2 omega-6        | all- <i>cis</i> -eicosa-11,14-dienoic           | dihomolinsoleic             |              |
| F20D2N6              | Fatty acid 20:2 omega-6                       | eicosa-11,14-dienoic                            |                             |              |
| F20D3                | Fatty acid 20:3 undifferentiated              | eicosatrienoic                                  |                             |              |
| F20D3C               | Fatty acid <i>cis</i> 20:3                    | <i>cis</i> -eicosatrienoic                      |                             |              |
| F20D3CN3             | Fatty acid <i>cis,cis,cis</i> 20:3 omega-3    | all- <i>cis</i> -eicosa-11,14,17-trienoic       | dihomo- $\alpha$ -linolenic | ETE          |
| F20D3CN6             | Fatty acid <i>cis,cis,cis</i> 20:3 omega-6    | all- <i>cis</i> -eicosa-8,11,14-trienoic        | dihomo- $\gamma$ -linolenic | DHLA         |
| F20D3N3              | Fatty acid 20:3 omega-3                       | eicosa-11,14,17-trienoic                        |                             |              |
| F20D3N6              | Fatty acid 20:3 omega-6                       | eicosa-8,11,14-trienoic                         |                             |              |
| F20D4                | Fatty acid 20:4 undifferentiated              | eicosatetraenoic                                |                             |              |
| F20D4CN6             | Fatty acid 20:4 omega-6                       | all- <i>cis</i> -eicosa-5,8,11,14-tetraenoic    | arachidonic                 | AA           |
| F20D4N3              | Fatty acid 20:4 omega-3                       | eicosa-8,11,14,17-tetraenoic                    |                             |              |
| F20D4N6              | Fatty acid 20:4 omega-6                       | eicosa-5,8,11,14-tetraenoic                     | arachidonic                 |              |
| F20D5                | Fatty acid 20:5 undifferentiated              | eicosapentaenoic                                |                             |              |
| F20D5C               | Fatty acid <i>cis</i> 20:5                    | <i>cis</i> -eicosapentaenoic                    |                             |              |
| F20D5CN3             | Fatty acid <i>cis,cis,cis</i> 20:5 omega-3    | all- <i>cis</i> -eicosa-5,8,11,14,17-pentaenoic | timnodonic                  | EPA          |
| F20D5N3              | Fatty acid 20:5 omega-3                       | eicosa-5,8,11,14,17-pentaenoic                  |                             |              |
| F22D2                | Fatty acid 22:2                               | docosadienoic                                   |                             |              |

| Component Identifier | Fatty acid name                        | Systematic name                                   | Trivial/common name | Abbreviation |
|----------------------|----------------------------------------|---------------------------------------------------|---------------------|--------------|
| F22D2C               | Fatty acid <i>cis</i> 22:2             | <i>cis</i> -docosadienoic                         |                     |              |
| F22D2CN6             | Fatty acid <i>cis,cis</i> 22:2 omega 6 | all- <i>cis</i> -docosa-13,22-dienoic             | docosadienoic       |              |
| F22D2N6              | Fatty acid 22:2 omega-6                | docosa-13,22-dienoic                              | docosadienoic       |              |
| F22D4                | Fatty acid 22:4 undifferentiated       | docosatetraenoic                                  |                     |              |
| F22D4N6              | Fatty acid 22:4 omega-6                | all- <i>cis</i> -docosa-7,10,13,22-tetraenoic     | adrenic             |              |
| F22D5                | Fatty acid 22:5 undifferentiated       | docosapentaenoic                                  |                     |              |
| F22D5C               | Fatty acid <i>cis</i> 22:5             | <i>cis</i> -docosapentaenoic                      |                     |              |
| F22D5CN3             | Fatty acid <i>cis</i> 22:5 omega 3     | all- <i>cis</i> -docosa-7,10,13,22,19-pentaenoic  | clupanodonic        | DPA          |
| F22D5N3              | Fatty acid 22:5 omega-3                | docosa-7,10,13,22,19-pentaenoic                   |                     |              |
| F22D5N6              | Fatty acid <i>cis</i> 22:5 omega-6     | all- <i>cis</i> -docosa-4,7,10,13,22-pentaenoic   | osbond              |              |
| F22D5N6              | Fatty acid 22:5 omega-6                | docosa-4,7,10,13,22-pentaenoic                    |                     |              |
| F22D6                | Fatty acid 22:6 undifferentiated       | docosahexaenoic                                   |                     |              |
| F22D6C               | Fatty acid <i>cis</i> 22:6             | <i>cis</i> -docosahexaenoic                       |                     |              |
| F22D6CN3             | Fatty acid <i>cis</i> 22:6 omega-3     | all- <i>cis</i> -docosa-4,7,10,13,22,19-hexaenoic | cervonic            | DHA          |
| F22D6N3              | Fatty acid 22:6 omega-3                | docosa-4,7,10,13,22,19-hexaenoic                  |                     |              |

Fatty acids, total omega-3 polyunsaturated (FAPUN3) and Fatty acids, total omega-6 polyunsaturated (FAPUN6) are the sums of the undifferentiated omega-3 fatty acid and undifferentiated omega-6 fatty acid in Table 9.

**Fatty acids, total *trans* (FATRN)** is the sum of the percentage of individual *trans* fatty acids in Tables 8 and 9. See Appendix 6 for the list of the fatty acids that contribute to the FATRN.

## 2.4.8 Vitamins

The following vitamin data are expressed in equivalents in FOODfiles 2024 and were calculated using the appropriate conversion factors given in Table 10.

Table 10. Conversion factors used to determine the equivalents of selected vitamins.

| Compound Identifier       | Description                                                                    | Conversion factor (g/g) <sup>a</sup> |
|---------------------------|--------------------------------------------------------------------------------|--------------------------------------|
| <b>Vitamin A-related</b>  |                                                                                |                                      |
| XBCARTA                   | Beta-carotene equivalent of alpha-carotene                                     | 0.5                                  |
| XRECARTB<br>XRECARTBEQ    | Retinol equivalent (RE) of beta-carotene or beta-carotene equivalent           | 0.167 (1/6)                          |
| XRAECARTB<br>XRAECARTBEQ  | Retinol activity equivalent (RAE) of beta-carotene or beta-carotene equivalent | 0.083 (1/12)                         |
| <b>Vitamin B3-related</b> |                                                                                |                                      |
| XTRYP                     | Niacin equivalents of tryptophan                                               | 0.017                                |
| <b>Vitamin E-related</b>  |                                                                                |                                      |
| XTOCOPHA                  | Alpha-tocopherol                                                               | 1                                    |
| XTOCOPHB                  | Alpha-tocopherol equivalent of beta tocopherol                                 | 0.4                                  |
| XTOCOPHD                  | Delta-tocopherol                                                               | 0.01                                 |
| XTOCOPHG                  | Gamma-tocopherol                                                               | 0.1                                  |
| <b>Folate-related</b>     |                                                                                |                                      |
| XFOLAC                    | Dietary folate equivalents of folic acid                                       | 1.67                                 |

<sup>a</sup> Sources: Finglas et al. (2015) and National Health and Medical Research Council (NHMRC). (2006).

**Beta-carotene equivalents (CARTBEQ)** are calculated by multiplying alpha-carotene (CARTA) by the conversion factor for beta-carotene equivalent of alpha-carotene (XCARTA) and adding beta-carotene (CARTB),

Equation 22:

$$\text{CARTBEQ } (\mu\text{g} / 100 \text{ g}) = \text{CARTB } (\mu\text{g} / 100 \text{ g}) + [\text{CARTA}(\mu\text{g} / 100 \text{ g}) \times \text{XBCARTA } (\text{g} / \text{g})]$$

Note: Other provitamin A carotenoids are not reported in FOODfiles 2024, and therefore are not included in beta-carotene equivalents calculation. However, for a few foods where data are “borrowed” from other sources, the food component cryptoxanthin might be reported as half the activity of beta-carotene, where that is the standard practice of the original source.

**Vitamin A, expressed as retinol equivalents (VITA)**, is calculated by multiplying beta-carotene equivalents (CARTBEQ) by the conversion factor (1/6, i.e. 0.167) and adding retinol (RETOL),

Equation 23:

$$\text{VITA } (\mu\text{g} / 100 \text{ g}) = \text{RETOL } (\mu\text{g} / 100 \text{ g}) + [(\text{CARTBEQ } (\mu\text{g} / (100 \text{ g}))/6)]$$

**Vitamin A, expressed as retinol activity equivalents (VITA\_RAE)**, is calculated by multiplying beta-carotene equivalents (CARTBEQ) by the conversion factor (1/12, i.e. 0.083) and adding retinol (RETOL),

Equation 24:

$$\text{VITA\_RAE } (\mu\text{g} / 100 \text{ g}) = \text{RETOL } (\mu\text{g} / 100 \text{ g}) + [(\text{CARTBEQ } (\mu\text{g} / (100 \text{ g}))/12)]$$

**Niacin equivalents, from tryptophan (NIATRP)**, are calculated by multiplying tryptophan (TRYP) by the conversion factor for niacin equivalents of tryptophan (XTRYP),

Equation 25:

$$\text{NIATRP (mg/100 g)} = \text{TRYP (mg/100 g)} \times \text{XTRYP (g/g)}$$

where the tryptophan value was not available, it has been imputed as 1.1% of total protein (Health Canada 2015).

**Niacin equivalents total (NIAEQ)** is the sum of the percentage of niacin, preformed (NIA) and niacin equivalent from tryptophan (NIATRP),

Equation 26:

$$\text{NIAEQ (mg/100 g)} = \text{NIA (mg/100 g)} + \text{NIATRP (mg/100 g)}$$

**Vitamin E, expressed as alpha-tocopherol equivalents (VITE)**, is calculated by multiplying alpha-beta-, gamma- and delta-tocopherols (TOCPHA, TOCPHB, TOCPHG, and TOCPHD) by their corresponding conversion factors for alpha-tocopherol equivalents of XTOCPHA, XTOCPHB, XTOCPHG, and XTOCPHD respectively and summing them,

Equation 27:

$$\begin{aligned} \text{VITE (mg/100 g)} &= [\text{TOCPHA (mg/100 g)} \times \text{XTOCPHA (g/g)}] \\ &+ [\text{TOCPHB (mg/100 g)} \times \text{XTOCPHB (g/g)}] \\ &+ [\text{TOCPHG (mg/100 g)} \times \text{XTOCPHG (g/g)}] \\ &+ [\text{TOCPHD (mg/100 g)} \times \text{XTOCPHD (g/g)}] \end{aligned}$$

Note: Where the tocopherol profile is incomplete, it has been assumed that all activity is alpha-tocopherol. There is an exception for seeds and seed oils where the gamma-tocopherol and other forms may be present in significant amounts and part of the tocopherol calculation for VITE. Tocotrienols are not recorded in FOODfiles 2024; therefore, they are not included in the VITE calculation.

**Dietary folate equivalents (FOLDFE)** are calculated by multiplying folic acid (FOLAC) by the conversion factor for dietary folate equivalents of folic acid (XFOLAC) and adding food folate, naturally occurring folates (FOLFD),

Equation 28:

$$\text{FOLDFE } (\mu\text{g}/(100 \text{ g})) = [\text{FOLAC } (\mu\text{g}/100 \text{ g}) \times \text{XFOLAC } (\text{g}/\text{g})] + \text{FOLFD } (\mu\text{g}/100 \text{ g})$$

**Folate food, naturally occurring food folate (FOLFD)** is calculated by subtracting folic acid (FOLAC) from folate, total (FOL),

Equation 29:

$$\text{FOLFD } (\mu\text{g}/100 \text{ g}) = \text{FOL } (\mu\text{g}/100 \text{ g}) - \text{FOLAC } (\mu\text{g}/100 \text{ g})$$

**Vitamin D (VITD)** is the sum of cholecalciferol or vitamin D<sub>3</sub> (CHOCAL) and ergocalciferol or vitamin D<sub>2</sub> (ERGCAL),

Equation 30:

$$\text{VITD } (\mu\text{g}/100 \text{ g}) = \text{CHOCAL } (\mu\text{g}/100 \text{ g}) + \text{ERGCAL } (\mu\text{g}/100 \text{ g})$$

#### 2.4.9 Amino acids

Amino acid profiles are expressed in three ways: mg/g of total nitrogen (mg/g N), mg/100 g of food, and g/100 g of food. The amino acids have been determined by analysis as mg/100 g of food. The values can be expressed as mg/g N using the total nitrogen (NT) value of the food as follows:

Equation 31:

$$\text{Amino acid (mg/g N)} = \text{Amino acid (mg/100 g)} / \text{NT (g/100 g)}$$

The amino acid (mg/g N) values can be used to calculate the amounts of amino acids in 100 g food by using the total nitrogen values for foods containing proteins with the same amino acid profiles (Orr & Watt 1957; Paul & Southgate 1978).

#### 2.4.10 Organic acids, total

**Organic acids, total (grams, OA\_G)** are the sum of the grams per 100 g of acetic acid (ACEAC\_G), citric acid (CITAC\_G), lactic acid (LACAC\_G), malic acid (MALAC\_G), oxalic acid (OXALAC\_G) and quinic acid (QUINAC\_G),

Equation 32:

$$\begin{aligned} \text{OA\_G (g/100 g)} &= \text{ACEAC\_G (g/100 g)} + \text{CITAC\_G (g/100 g)} + \text{LACAC\_G (g/100 g)} \\ &+ \text{MALAC\_G (g/100 g)} + \text{OXALAC\_G (g/100 g)} + \text{QUINAC (g/100 g)} \end{aligned}$$

Note: Only a few foods have values for organic acids.

## 2.5 Recipes

Recipe Food Records are generated by calculation following a method known as the 'Mixed method' (FAO/INFOODS 2012a) using the food composition data management system, FoodCASE. In this recipe method, nutrient gains or losses during recipe preparation are estimated by applying Nutrient Retention Factors (NRFs) at the individual ingredient level and Weight Yield Factor (WYF) at the recipe level for total weight gain or loss.

### 2.5.1 Nutrient Retention Factor (NRF)

Nutrient retention is defined as the measure of the proportion of the nutrient remaining in the cooked dish in relation to the nutrient originally present in the raw ingredient.

The equation used for calculation of NRF (%) is:

Equation 33:

$$\text{NRF}(\%) = (\text{Nutrient content in cooked dish (g/100 g)}) / (\text{Nutrient content in raw ingredients (g/100 g)}) \times \text{WYF}(\%)$$

The NRFs have been derived largely from the U.S. Department of Agriculture (2007) document for vitamins and minerals. Iodine retention factors have been derived from Food Standards Australia New Zealand (2021b). A list of NRFs linked to specific NRF classification and preparation methods can be downloaded from the NUTRIENT RETENTION FACTOR.FT file. NRF classification and preparation method linked to each recipe Food Record can be found in the INGREDIENTS.FT file (Section 3).

The equation used for estimating the nutrient content of a recipe (cooked dish) is:

Equation 34:

$$\begin{aligned} &\text{Nutrient content of cooked dish (g/100 g)} \\ &= (\text{Sum of nutrient content of each ingredient (g)} \times \text{NRF}(\%)) \\ &\quad / (\text{Total cooked dish weight (g)}) \times 100 \end{aligned}$$

Note: Where the analytical values were available for vitamins and minerals in the cooked food, these were used instead of values derived (recipe calculation) using NRFs.

### 2.5.2 Weight Yield Factor (WYF)

The WYFs have been obtained largely during preparation of recipes by measuring the weight of the ingredients before and after cooking, using the formula below. Where the measure of weight change was not available, the WYF values were obtained from published sources (Matthews et al. 1975; Food Standards Australia New Zealand 2014a; Finglas et al. 2015). WYFs that are used for each recipe can be downloaded from the WEIGHT YIELD FACTOR.FT (Section 3).

Equation 35:

$$\text{WYF}(\%) = (\text{Total cooked dish weight (g)}) / (\text{Total weight of raw ingredients(g)}) \times 100$$

\*All ingredient weights are based on edible portions.



### 2.5.3 Calculation of component values in recipes

The food component values for each recipe can be derived by the mixture of methods: recipe calculation (using WYF and NRFs) and direct chemical analysis. The values derived by these methods (calculation and analysis) are identified with codes in the DATA.FT file for each component.

If an analytical value was unavailable for water, the value was calculated by difference (subtracting other proximate components from 100). The proximate total (sum of the macro-nutrient set of ash, protein, fat (total), water, dietary fibre, alcohol and available carbohydrate by weight) per ingredient was also checked that it was ~100 g before performing the calculation.

An example of calculation of the values of the selected components for a recipe can be found in Appendix 7.

### 2.5.4 Recipe Food Records

Approximately 14% (385) of the Food Records in FOODfiles 2024 are derived by recipe calculation. Of these, 45 are complex recipes derived from multiple ingredients and are listed only in Chapter R (Recipes). The rest of the 340 recipe Food Records are single ingredient recipes or simple recipes (e.g. N1021 – Sausage, beef, pan fried, with no added fat, composite) listed in the associated food chapters (Table 2). All recipe Food Records are listed in the INGREDIENT.FT.

## 3 Downloadable files

FOODfiles 2024 contains six principal data files and six supporting files. Four of these are update files, which include foods that have been archived, replaced and new; and component values updated since the release of FOODfiles 2021 are also available.

The 11 principal and supporting files listed below are available in two formats: as DSV (tilde delimited UTF8 encoded) and as Microsoft® Excel® files. These files occupy approximately 25 MB and 36 MB, respectively. The Excel files are in .xlsx format.

### Principal files

- DATA.AP (Standard and unabridged data files) – contains data in a table format.
- DATA.FT (Standard and unabridged data files) – contains data in a list format along with Data provenance (source code).
- CSM.FT – contains common standard measures along with density.
- INGREDIENT.FT – contains data on ingredients used to create a recipe Food Record and where relevant the NRF Classification and preparation method.

### Supporting files

- NAME.FT – provides full food details including more edible portion, detailed description and sampling details.
- CODE.FT – provides a list of the components included in the standard and unabridged dataset, their identifier, Unit Code and Matrix Unit Code.
- NUTRIENT RETENTION FACTOR.FT – contains the value for NRF used where foods have been prepared (e.g. boiled).
- WEIGHT YIELD FACTOR.FT – gives the value for WYF for foods that have been cooked.
- CONVERSION FACTOR.FT – gives the general conversion factors from the published sources for nitrogen to protein (Jones 1941; Greenfield & Southgate 2003) and fat to fatty acid (Exler & Weihrauch 1977; Greenfield & Southgate 2003).
- Data Provenance Description.xlsx – lists data derivation and other details: Unit Code, Matrix Unit Code, Data Source Code, Value Type Code, Acquisition Type Code, Method Type Code and Method Indicator based on Machackova et al. (2019). The short descriptions are listed in Table 3.

### Update files

These four files provide the added, changed, or deleted foods, nutrient values and major changes made to the database since the last release of FOODfiles, in 2021.

- New Food Records replacing the old Food Records in latest version of FOODfiles.xlsx.
- Food Records archived from latest version of FOODfiles.xlsx.
- Data added to or updated in the Food Records in the latest version of FOODfiles.xlsx.

- List of components updated in current version of FOODfiles.xlsx.

## 3.1 Instructions for downloading

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All files can be found at [www.foodcomposition.co.nz](http://www.foodcomposition.co.nz) and are provided as a Microsoft Windows® installer file (.msi). Use “This PC” or “Windows Explorer” to browse to the folder (where you saved the installation program when downloading it) and double click on the installer FOODfilesSetup.msi (foodfiles-2024-v1.msi normally in your “Downloads” directory) to begin the installation. (NOTE: you may not be able to see the .msi extension, depending on your computer settings). You will need local administrative rights on your computer to complete this. If you do not specify otherwise during the installation, the New Zealand FOODfiles will be installed in the folder **New Zealand FOODfiles 2024** on your C: drive. If you wish to remove the New Zealand FOODfiles from your computer, you can do so through the Control Panel | Programs and Features. |Uninstall or change a program.

## 3.2 ASCII text files

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ASCII text files are delimited and text fields are separated by tildes (~). The tilde characters can be replaced with other field separators. To open these files in Microsoft Excel, go to the “Open file” of the program and find the folder. Make sure you select “Files of Type”. Choose the appropriate file and open it. The “Text Import Wizard” will be activated. In the first screen choose “Delimited” and in the “File Origin” drop-down box, ensure that the entry “Unicode (UTF8)” is selected and press “Next”. In the second screen choose “Other” and type “~” in the text box and select “[none]” for “Text Qualifier”. Press “Finish” button.

Note that all text fields are separated by a tilde (“~”) character. When a field is null or blank, two tilde characters will be adjacent to each other. The first line of all text files contains “© Copyright The New Zealand Institute for Plant and Food Research Limited and the Ministry of Health (New Zealand) 2024 All rights reserved”. The field structure starts from the second line in all the text files.

## 3.3 NAME.FT file

---

The NAME.FT file contains the FoodID, Food Name, Short Food Name, and descriptions of the foods in FOODfiles 2024. The Food Name includes the multi-faceted descriptors according to the INFOODS Food Description System (Truswell et al. 1991). The faceted descriptions are also reported in separate fields. Each Short Food Name is unique, represents more common words for general public use, and does not typically follow the multi-faceted descriptors from INFOODS. The details of the fields included in the NAME.FT file can be found in Table 11.

Table 11. Details of the fields used in the NAME.FT file.

| Field | Name              | Field type alphanumeric | Notes (Required (R) or Optional (O))                                                                                                                                                                                                                                                                         |
|-------|-------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | FoodID            | Identifier              | (R) The unique code; first character alphabetic, representing food group (Table 2); remaining characters sequential numbers.                                                                                                                                                                                 |
| 2     | Food Name         | Text                    | (R) Includes the multi-faceted descriptors according to the INFOODS Food Description System (Truswell et al. 1991). Use this field as the 'name' of the food in printed tables and computer packages. Each Food Name is unique.                                                                              |
| 3     | Short Food Name   | Text                    | (R) The Short Food Name has fewer words than the 'Food Name'. The Short Food Name is used in The Concise New Zealand Food Composition Tables. Each Short Food Name is unique, represents more common words for general public use, and does not typically follow the multi-faceted descriptors from INFOODS. |
| 4     | Alternative Name  | Text                    | (O) Other common names to be used for cross-referencing indices.                                                                                                                                                                                                                                             |
| 5     | Food Description  | Text                    | (O) Detailed information of the food, including ingredients, fortification, appearance, production, preparation and whether the Food Records generated by borrowing from other databases or recipe calculated.                                                                                               |
| 6     | Edible portion    | Numeric                 | (R) Percentage of the edible part of the food and all data refer to the denominator 100% of edible portion unless specified with a different denominator.                                                                                                                                                    |
| 7     | Generic Name      | Text                    | (R) Food descriptor facet, the most general of food descriptors.                                                                                                                                                                                                                                             |
| 8     | Kind              | Text                    | (O) More specific descriptor.                                                                                                                                                                                                                                                                                |
| 9     | Part              | Text                    | (O) Portion of the food represented by the data.                                                                                                                                                                                                                                                             |
| 10    | State             | Text                    | (O) State, condition, treatment, cooking etc.                                                                                                                                                                                                                                                                |
| 11    | Grade             | Text                    | (O) Usually defined by food standards.                                                                                                                                                                                                                                                                       |
| 12    | Maturity          | Text                    | (O) Age or stage of development.                                                                                                                                                                                                                                                                             |
| 13    | Scientific Name   | Text                    | (O) Latin name; binary nomenclature; the first part of the name identifies the genus, and second part identifies the species within the genus, followed by variety (cultivar or sub-species name); for animals and plants.                                                                                   |
| 14    | Sampling details  | Text                    | (O) Detailed information on samples such as numbers of samples, product names including brands and flavours, place of purchase, date of purchase and analysis. If the Food Records are "borrowed" from other databases, the source of the Food Records information is included.                              |
| 15    | Component message | Text                    | (O) Includes the additional information on Food Composition Data derivation.                                                                                                                                                                                                                                 |

Each food is uniquely described using the following naming structure in the second line of the text file and in tabular form (Table 12):

Format:

FoodID~Food Name~Short Food Name~Alternative Names~Food Description~Edible portion (%)~Generic Name~Kind~Part~State~Grade~Maturity~Scientific Name~Sampling Details~Component Message

Example:

Q1042~Seed, hemp, hulled, dried, composite~Hemp seed, hulled, dried~Hemp hearts~Hulled hemp seed commercially made by removing crunchy outer shell of the whole hemp seed. Product of New Zealand and imported.~100~SEED~HEMP ~Seed~Dried, Hulled~~~Cannabis sativa~A composite sample of the brands: Hemp Farm (x 2), new hemisphere (x 1), Good Pantry (x 1), Bin Inn (x 1), Organic buzz (x 1), Love Organics (x 1), Floating leaf (x 1), and Organic Mechanic (x 1) were sampled from retail outlets in April 2019.~Majority of food components data were derived by analysis. Alcohol, caffeine, cholesterol, retinol, folic acid, and vitamins (B12 and D) data were presumed zero.

Table 12. Example of the naming structure used in the NAME.FT file in tabular form.

| Field | Name               | Entry                                                                                                                                                                                                                                               |
|-------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | FoodID             | Q1042                                                                                                                                                                                                                                               |
| 2     | Food Name          | Seed, hemp, hulled, dried, composite                                                                                                                                                                                                                |
| 3     | Short Food Name    | Hemp seed, hulled, dried                                                                                                                                                                                                                            |
| 4     | Alternative Names  | Hemp hearts                                                                                                                                                                                                                                         |
| 5     | Food Description   | Hulled hemp seed commercially made by removing crunchy outer shell of the whole hemp seed. Product of New Zealand and imported.                                                                                                                     |
| 6     | Edible portion (%) | 100                                                                                                                                                                                                                                                 |
| 7     | Generic Name       | SEED                                                                                                                                                                                                                                                |
| 8     | Kind               | HEMP                                                                                                                                                                                                                                                |
| 9     | Part               | Seed                                                                                                                                                                                                                                                |
| 10    | State              | Dried, Hulled                                                                                                                                                                                                                                       |
| 11    | Grade              |                                                                                                                                                                                                                                                     |
| 12    | Maturity           |                                                                                                                                                                                                                                                     |
| 13    | Scientific Name    | <i>Cannabis sativa</i>                                                                                                                                                                                                                              |
| 14    | Sampling Details   | A composite sample of the brands: Hemp Farm (x 2), new hemisphere (x 1), Good Pantry (x 1), Bin Inn (x 1), Organic buzz (x 1), Love Organics (x 1), Floating leaf (x 1), and Organic Mechanic (x 1) were sampled from retail outlets in April 2019. |
| 15    | Component Message  | Majority of food components data were derived by analysis. Alcohol, caffeine, cholesterol, retinol, folic acid, and vitamins (B12 and D) data were presumed zero.                                                                                   |

### 3.4 CODE.FT file

The CODE.FT file contains the nutritional component name, with its Component Identifier being used in the data files to define the component. Each line is one record and holds information on one component. The unit is a measure of the amount of the component values followed by a code for Matrix unit,

Format:

Code~Description~Unit Code~Matrix Unit Code

Example: RETOL~Retinol~µg~W

where:

- RETOL is the ‘Component Identifier’
- Retinol is the component name of the ‘Component Identifier’
- Unit of measure is µg
- W is the code for the Matrix Unit Code, ‘per 100 grams edible portion’.

Appendix 1 contains a full list of ‘Component Identifier’ and food component names. The details of the fields that are included in the CODE.FT file can be found in Table 13.

Table 13. Details of the fields used in the CODE.FT file.

| Field | Name                 | Field type | Notes                                                                                                                                                                                                     |
|-------|----------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Component Identifier | Identifier | Klensin et al. (1989) and FAO/INFOODS 2012a)                                                                                                                                                              |
| 2     | Component            | Text       | The component name                                                                                                                                                                                        |
| 3     | Unit                 | Identifier | The measure of the amount of the component values, a dimensionless number                                                                                                                                 |
| 4     | Matrix Unit Code     | Identifier | The amount of matrix material that has a quantity reported, usually expressed using the preposition “per”. Codes: W – per 100 g of edible portion; N – per g of nitrogen; F- per 100 g Total fatty acids. |

### 3.5 DATA.FT file

The DATA.FT file contains the data for each component for all foods. Each component forms a record for each food. The format as defined on the second line of the file contains the header:

FoodID~Component Identifier~Value~Unit Code~Matrix Unit Code~Source Code~Value Type Code~Acquisition Type Code~Method Type Code~Method Indicator Code

Each record in the file is composed of a FoodID followed by a field for each of the Component Identifier, the component, value, unit code, matrix unit code and data provenance descriptions, Example:

K1034~WATER~78.5~g~W~~BE~D~A~MI1212

where:

- K1034 is the FoodID for Snapper, fillet, flesh, fresh, raw, related in the file NAME.FT
- WATER is the ‘Component Identifier’ for WATER, related in the file CODE.FT file
- 78.5 is the value of water in this food
- ‘g’ is the measurement unit code for ‘gram’
- ‘W’ is the Matrix Unit Code “per 100 g edible portion”.

The rest of the field’s description codes are related to data derivation (refer to Data provenance.xlsx): ‘BE’ is the Value Type Code “Best Estimation”; D is the Acquisition Type Code data derived from “Independent laboratory”; A is the Method Type Code “Analytical” and MI1212 is the Method Indicator Code for “Vacuum drying”.

The details of the fields that are included in the DATA.FT file can be found in Table 14.

Table 14. Details of the fields used in the DATA.FT file.

| Field | Name                  | Field type | Notes                                                                                                    |
|-------|-----------------------|------------|----------------------------------------------------------------------------------------------------------|
| 1     | FoodID                | Identifier | NAME.FT                                                                                                  |
| 2     | Component Identifier  | Identifier | CODE.FT                                                                                                  |
| 2     | Value                 | Numeric    | Significant figures and maximal decimal places for the food composition values are listed in Appendix 2. |
| 3     | Unit Code             | Identifier | Data provenance.xlsx (see Tables 3 and 4)                                                                |
| 4     | Matrix Unit Code      | Identifier |                                                                                                          |
| 5     | Source Code           | Identifier |                                                                                                          |
| 6     | Value Type Code       | Identifier |                                                                                                          |
| 7     | Acquisition Type Code | Identifier |                                                                                                          |
| 8     | Method Type Code      | Identifier |                                                                                                          |
| 9     | Method Indicator Code | Identifier |                                                                                                          |

### 3.6 DATA.AP file

The DATA.AP file contains some of the information already present in DATA.FT but in an alternative format. It features only the values of the components in each food within rows and columns like a spreadsheet. The second line of the file contains the name of each of the components, and the third line contains the units of measure. The subsequent lines contain a field for each component. These fields are filled by values:

Second line:

FoodID~Food Name~Alcohol~Alpha-carotene~Alpha-tocopherol~Ash~Available carbohydrate by difference~Available carbohydrate, FSANZ~Available carbohydrates by weight~Available carbohydrates in monosaccharide equivalent~Beta-carotene~Beta-carotene equivalents~Beta-tocopherol~Caffeine~Calcium~Carbohydrate by difference, FSANZ~Cholesterol~Copper~etc.

Third line:

FoodID~Food Name~g/100 g~µg/100 g~mg/100 g~g/100 g~g/100 g~g/100 g~g/100 g~g/100 g~µg/100 g~µg/100 g~mg/100 g~mg/100 g~mg/100 g~g/100 g~mg/100 g~mg/100 g~etc.

Subsequent lines, an example (37<sup>th</sup> line):

A1119~Crumpet, white, as purchased, commercial~0~0~0.07~2.4~36.8~31.6~31.6~34.7~0~0~0.02~0~71~36.8~0~0.104~etc,

The details of the fields that are included in the DATA.AP file can be found in Table 15.

Table 15. Details of the fields used in the DATA.FT file.

| Field | Name      | Field type | Notes                                                                                                    |
|-------|-----------|------------|----------------------------------------------------------------------------------------------------------|
| 1     | FoodID    | Identifier | NAME.FT                                                                                                  |
| 2     | Food Name | Identifier | NAME.FT                                                                                                  |
| 3     | Unit      | Text       | CODE.FT                                                                                                  |
| 4     | Value     | Numeric    | Significant figures and maximal decimal places for the food composition values are listed in Appendix 2. |

Table 16 provides an example of the information provided in the DATA.AP data file in tabular form for an individual Food Record.

Table 16. Excerpt of the details in the DATA.AP for Food Record A1119 provided in tabular form (note only the first 19 lines are shown here).

| Field | Second line                                          | Third line | 37 <sup>th</sup> line (example)          |
|-------|------------------------------------------------------|------------|------------------------------------------|
| 1     | FoodID                                               | FoodID     | A1119                                    |
| 3     | Food Name                                            | Food name  | Crumpet, white, as purchased, commercial |
| 4     | Alcohol                                              | g/100 g    | 0                                        |
| 5     | Alpha-carotene                                       | µg/100 g   | 0                                        |
| 6     | Alpha-tocopherol                                     | mg/100 g   | 0.07                                     |
| 7     | Ash                                                  | g/100 g    | 2.4                                      |
| 8     | Available carbohydrate by difference                 | g/100 g    | 36.8                                     |
| 9     | Available carbohydrate, FSANZ                        | g/100 g    | 31.6                                     |
| 10    | Available carbohydrates by weight                    | g/100 g    | 31.6                                     |
| 11    | Available carbohydrates in monosaccharide equivalent | g/100 g    | 34.7                                     |
| 12    | Beta-carotene                                        | µg/100 g   | 0                                        |
| 13    | Beta-carotene equivalents                            | µg/100 g   | 0                                        |
| 14    | Beta-tocopherol                                      | mg/100 g   | 0.02                                     |
| 15    | Caffeine                                             | mg/100 g   | 0                                        |
| 16    | Calcium                                              | mg/100 g   | 71                                       |
| 17    | Carbohydrate by difference, FSANZ                    | g/100 g    | 36.8                                     |
| 18    | Cholesterol                                          | mg/100 g   | 0.00                                     |
| 19    | Copper                                               | mg/100 g   | 0.104                                    |

FSANZ = Food Standards Australia New Zealand.

### 3.7 CSM.FT file

The CSM.FT file contains the data for the common standard measures (CSM) and density data for most foods. The details of the fields included in the CSM.FT file can be found in Table 16. The CSM is either expressed as New Zealand metric standards (Appendix 8) or as the amount commonly purchased or eaten. All these description measures are listed under the CSM column, noting that there may be more than one CSM for any particular food. All the measures (g) were made on an edible portion, so no adjustments are necessary to account for the inedible portion. For the alcoholic



beverages in New Zealand, one standard drink containing 10 g of pure alcohol is used (Health Promotion Agency. April 2016).

Density is measured as specific gravity, mass density, bulk density and/or use mass and volume by water or rapeseed displacement method. The Mass density of a food is its mass per unit volume. The term Specific gravity is used for liquid foods (e.g. milk, fruit juices), whereas the term Mass density is typically used for semi-solids and solids with regular shapes (e.g. bread, biscuits). Bulk density is the term typically used in measurement for powders, grains, or other particulate solids that contain occluded air. The bulk density total volume includes particle volume, inter-particle void volume and internal pore volume (e.g. almonds ground and slivered). Water displacement method is used for foods with irregular shapes (e.g. eggplant, meat cuts) to establish the volume that relates to an item of known mass/weight to allow density to be calculated. Similarly, a baked product can be measured using rapeseed instead of water.

The format, as defined on the second line of the CSM.FT file is:

FoodID~Food Name~CSM~Measure~Density (g/cm3)

For example:

A1169~Bread, from white wheat flour with multigrain, light, sliced, prepacked, as purchased, commercial, composite~1 slice sandwich (11.4 x 10.3 x 1.1 cm)~29.8~0.25

In the example above:

A1069 is the FoodID for a Food Name, Bread, from white wheat flour with multigrain, light, sliced, prepacked, as purchased, commercial, composite

1 slice sandwich (11.4 x 10.3 x 1.1 cm) weighs 29.8 g and has a density of 0.25 g/cm3

The details of the fields that are included in the CSM.FT file can be found in Table 17.

Table 17. Details of the fields used in the CSM.FT file.

| Field | Name        | Field type | Notes                                                                                                                                           |
|-------|-------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Food ID     | Identifier | NAME.FT file                                                                                                                                    |
| 2     | Food Name   | Text       | NAME.FT file                                                                                                                                    |
| 3     | CSM         | Text       | The description of the Common Standard Measure (CSM) or amount commonly purchased or eaten                                                      |
| 4     | Measure (g) | Numeric    | The weight of the CSM amount commonly purchased or eaten in grams                                                                               |
| 5     | Density     | Numeric    | g/cm <sup>3</sup> ; Food Records may have multiple densities based on the physical state e.g. 1 cup (250 mL) almond: ground, slivered and whole |

### 3.8 INGREDIENT.FT file

The INGREDIENT.FT file contains the FoodID of a recipe, FoodIDs of the ingredients and their Food Names, Weight Fractions used in that particular recipe, Nutrition Retention Factor ID and USDA Retention Factor description. The Nutrition Retention Factor ID is specific to the cooking method (USDA Retention Factor description). The details of the fields included in the INGREDIENT.FT file can be found in Table 18.

Table 18. Details of the fields used in the INGREDIENT.FT file.

| Field | Name                                     | Field type | Notes                                                    |
|-------|------------------------------------------|------------|----------------------------------------------------------|
| 1     | Recipe FoodID                            | Identifier | NAME.FT file                                             |
| 2     | Ingredient FoodID                        | Identifier | NAME.FT file                                             |
| 3     | Ingredient Name                          | Text       | NAME.FT file                                             |
| 4     | Weight Fraction (%)                      | Numeric    | The weight fraction of the food ingredient in the recipe |
| 5     | Nutrient Retention Factor Classification | Text       | U.S. Department of Agriculture (2007)                    |
| 6     | Preparation Method                       | Text       |                                                          |

The naming structure for each record in this file is:

Recipe FoodID~Ingredient Food Name~Weight Fraction(%)~Nutrient Retention Factor Classification~Preparation Method

For example, the recipe “Cake, sponge, fat added, baked” (R10034) contains the following:

- R10034~F1046~Butter, salted, composite~25.0~~
- R10034~G1016~Egg, chicken, white & yolk (whole), raw, fresh, composite~24.4~EGGS~BAKED
- R10034~W20~Sugar, caster~25.0~~
- R10034~E1088~Flour, wheat, white, standard, plain, composite~25.0~FLOUR/MEAL~BAKED
- R10034~P62~Baking soda~0.6~~

The tabulated form of the above example Food Record R10028 can be found in Table 19.

Table 19. Example of the details in the INGREDIENT.FT file for Food Record R5442 provided in tabular form.

| Recipe FoodID | Ingredient FoodID | Ingredient name                                           | Weight fraction (g) <sup>a</sup> | Nutrient Retention Factor Classification <sup>b</sup> | Preparation Method <sup>b</sup> |
|---------------|-------------------|-----------------------------------------------------------|----------------------------------|-------------------------------------------------------|---------------------------------|
| R10034        | F1046             | Butter, salted, composite                                 | 25.0                             |                                                       |                                 |
| R10034        | G1016             | Egg, chicken, white & yolk (whole), raw, fresh, composite | 24.4                             | EGGS                                                  | BAKED                           |
| R10034        | W20               | Sugar, caster                                             | 25.0                             |                                                       |                                 |
| R10034        | E1088             | Flour, wheat, white, standard, plain, composite           | 25.0                             | FLOUR/MEAL                                            | BAKED                           |
| R10034        | P62               | Baking soda                                               | 0.6                              |                                                       |                                 |

<sup>a</sup>Weight fraction is the percentage of each ingredient used in the recipe; Source: <sup>b</sup>U.S. Department of Agriculture (2007).

### 3.9 NUTRIENT RETENTION FACTOR.FT file

The NUTRIENT RETENTION FACTOR.FT contains the NRF for selected nutrients and ID. The details of the fields included in the NUTRIENT RETENTION FACTOR.FT file can be found in Table 20.

The naming structure in this file is:

Nutrient Retention Factor Classification ~ Preparation Method ~Component Identifier~Nutrient Retention Factor

Example: CHICKEN~ROASTED~FOL~0.6

where:

- CHICKEN is the 'Nutrient Retention Factor Classification' and 'ROASTED' is the Preparation Method from U.S. Department of Agriculture (2007) related to the INGREDIENT.FT
- FOL is the 'Component Identifier' for 'Folate, total', related in the file CODE.FT file'
- 0.6 is 'Nutrient Retention Factor'.

Table 20. Details of the fields used in the NUTRIENT FACTOR.FT file.

| Field | Name                                     | Field type | Notes                                                                                                                                                           |
|-------|------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Nutrient Retention Factor Classification | Text       | Related to the INGREDIENT.FT                                                                                                                                    |
| 2     | Preparation Method                       | Text       | U.S. Department of Agriculture (2007)                                                                                                                           |
| 3     | Component Identifier                     | Identifier | CODE.FT file                                                                                                                                                    |
| 3     | Nutrient Retention Factor                | Numeric    | Ratio of the nutrient retention between cooked and raw ingredients (g/g) U.S. Department of Agriculture (2007) and Food Standards Australia New Zealand (2014b) |

### 3.10 WEIGHT YIELD FACTOR.FT file

The WEIGHT YIELD FACTOR.FT contains the Weight YF for each recipe food. The details of the fields included in the WEIGHT YIELD FACTOR.FT file can be found in Table 21.

The naming structure in this file is:

Recipe Food ID~Recipe Food Name~Weight Yield Factor

The details of the fields included in the WEIGHT YIELD.FACTOR.FT file can be found in Table 21.

Table 21. Details of the fields used in the WEIGHT YIELD FACTOR.FT file.

| Field | Name                | Field type | Notes                                                                                                                      |
|-------|---------------------|------------|----------------------------------------------------------------------------------------------------------------------------|
| 1     | Recipe Food ID      | Identifier | INGREDIENT.FT file (NAME.FT)                                                                                               |
| 2     | Recipe Food Name    | Text       | INGREDIENT.FT file (NAME.FT)                                                                                               |
| 3     | Weight Yield Factor | Numeric    | Measure of the percentage of the weight of the cooked dish in relation to the weight of the raw ingredient/s (Equation 35) |

### 3.11 CONVERSION FACTOR.FT file

The CONVERSION FACTOR.FT contains the general factor conversions for fat to fatty acids for some of the Food Records, and for nitrogen to protein for all the Food Records from the published sources. The details of the fields included in the CONVERSION FACTOR.FT file can be found in Table 22.

The naming structure in this file is:

FoodID~XFA~XN

Table 22. Details of the fields used in the CONVERSION.FACTOR.FT file.

| Field | Name   | Field type | Notes                                                                                                          |
|-------|--------|------------|----------------------------------------------------------------------------------------------------------------|
| 1     | FoodID | Identifier | INGREDIENT.FT file (NAME.FT)                                                                                   |
| 2     | XFA    | Identifier | INFOODS tagname for fat to fatty acids conversion factor (Exler & Weihrauch 1977; Greenfield & Southgate 2003) |
| 3     | XN     | Identifier | INFOODS tagname for nitrogen to protein conversion factor (Jones 1941; Greenfield & Southgate 2003)            |

INFOODS - International Network of Food Data Systems.

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## Appendices

### Appendix 1. List of components with supporting details

This table provides details of the components included in the database, including the number of foods and the percentage of the total number of foods (2857) that contain the indicated nutrient for which values are available.

|                      |                           |                              |          |                                | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|---------------------------|------------------------------|----------|--------------------------------|----------------------------|------------|-----------------|------------|
| Component identifier | Component name            | INFOODS Tagname <sup>a</sup> | Unit     | Analysed (A) or Calculated (C) | Standard                   | Unabridged | Number of foods | Percentage |
| ACEAC_G              | Acetic acid (g)           | ACEAC_G                      | g/100 g  | A                              |                            | ✓          | 66              | 2.31       |
| AL                   | Aluminium                 | AL                           | µg/100 g | A                              |                            | ✓          | 478             | 16.7       |
| ALA                  | Alanine                   | ALA                          | mg/100 g | A                              |                            | ✓          | 472             | 16.5       |
| ALA_G                | Alanine (g)               | ALA_G                        | g/100 g  | A                              |                            | ✓          | 460             | 16.1       |
| ALAN                 | Alanine (/g N)            | ALAN                         | mg/g N   | C                              |                            | ✓          | 459             | 16.1       |
| ALC                  | Alcohol                   | ALC                          | g/100 g  | A                              | ✓                          | ✓          | 2857            | 100        |
| ARG                  | Arginine                  | ARG                          | mg/100 g | A                              |                            | ✓          | 474             | 16.6       |
| ARG_G                | Arginine (g)              | ARG_G                        | g/100 g  | A                              |                            | ✓          | 463             | 16.2       |
| ARGN                 | Arginine (/g N)           | ARGN                         | mg/g N   | C                              |                            | ✓          | 461             | 16.1       |
| AS                   | Arsenic                   | AS                           | µg/100 g | A                              |                            | ✓          | 652             | 22.8       |
| ASH                  | Ash                       | ASH                          | g/100 g  | A                              | ✓                          | ✓          | 2857            | 100        |
| ASN                  | Asparagine                | ASN                          | mg/100 g | A                              |                            | ✓          | 178             | 6.23       |
| ASN_G                | Asparagine (g)            | ASN_G                        | g/100 g  | A                              |                            | ✓          | 178             | 6.23       |
| ASNN                 | Asparagine (/g N)         | ASNN                         | mg/g N   | C                              |                            | ✓          | 178             | 6.23       |
| ASP                  | Aspartic acid             | ASP                          | mg/100 g | A                              |                            | ✓          | 294             | 10.3       |
| ASP_G                | Aspartic acid (g)         | ASP_G                        | g/100 g  | A                              |                            | ✓          | 282             | 9.87       |
| ASPN                 | Aspartic acid (/g N)      | ASPN                         | mg/g N   | C                              |                            | ✓          | 281             | 9.83       |
| B                    | Boron                     | B                            | µg/100 g | A                              |                            | ✓          | 462             | 16.2       |
| BIOT                 | Biotin                    | BIOT                         | µg/100 g | A                              |                            | ✓          | 1060            | 37.1       |
| CA                   | Calcium                   | CA                           | mg/100 g | A                              | ✓                          | ✓          | 2857            | 100        |
| CAFFN                | Caffeine                  | CAFFN                        | mg/100 g | A                              | ✓                          | ✓          | 2857            | 100        |
| CARTA                | Alpha-carotene            | CARTA                        | µg/100 g | A                              | ✓                          | ✓          | 2395            | 83.8       |
| CARTB                | Beta-carotene             | CARTB                        | µg/100 g | A                              | ✓                          | ✓          | 2572            | 90.0       |
| CARTBEQ              | Beta-carotene equivalents | CARTBEQ                      | mg/100 g | C                              | ✓                          | ✓          | 2857            | 100        |
| CD                   | Cadmium                   | CD                           | µg/100 g | A                              |                            | ✓          | 565             | 19.8       |

| Component identifier | Component name                                            | INFOODS Tagname <sup>a</sup> | Unit     | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|-----------------------------------------------------------|------------------------------|----------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                           |                              |          |                                | Standard                   | Unabridged | Number of foods | Percentage |
| CHOAVL               | Available carbohydrates by weight                         | CHOAVL                       | g/100 g  | C                              | ✓                          | ✓          | 2857            | 100        |
| CHOAVL_FSANZ         | Available carbohydrate, FSANZ                             |                              | g/100 g  | C                              | ✓                          | ✓          | 2857            | 100        |
| CHOAVLDF             | Available carbohydrate by difference                      | CHOAVLDF                     | g/100 g  | C                              | ✓                          | ✓          | 2857            | 100        |
| CHOAVLDF_FSANZ       | Carbohydrate by difference, FSANZ                         |                              | g/100 g  | C                              | ✓                          | ✓          | 2857            | 100        |
| CHOAVLM              | Available carbohydrates in monosaccharide equivalent      | CHOAVLM                      | g/100 g  | C                              | ✓                          | ✓          | 2857            | 100        |
| CHOCAL               | Cholecalciferol (Vitamin D <sub>3</sub> )                 | CHOCAL                       | µg/100 g | A                              |                            | ✓          | 2857            | 100        |
| CHOCALOH             | 25-hydroxyvitamin D <sub>3</sub>                          | CHOCALOH                     | µg/100 g | A                              |                            | ✓          | 198             | 6.9        |
| CHOCDF               | Total carbohydrate by difference                          | CHOCDF                       | g/100 g  | C                              | ✓                          | ✓          | 2857            | 100        |
| CHOCSM               | Total carbohydrates by summation                          | CHOCSM                       | g/100 g  | C                              | ✓                          | ✓          | 2857            | 100        |
| CHOLE                | Cholesterol                                               | CHOLE                        | mg/100 g | A                              | ✓                          | ✓          | 2857            | 100        |
| CHOLN                | Choline                                                   | CHOLN                        | mg/100 g | A                              |                            | ✓          | 16              | 0.56       |
| CITAC_G              | Citric acid (g)                                           | CITAC_G                      | g/100 g  | A                              |                            | ✓          | 51              | 1.78       |
| CLD                  | Chloride                                                  | CLD                          | mg/100 g | A                              |                            | ✓          | 911             | 31.9       |
| CO                   | Cobalt                                                    | CO                           | µg/100 g | A                              |                            | ✓          | 407             | 14.2       |
| CR                   | Chromium                                                  | CR                           | µg/100 g | A                              |                            | ✓          | 499             | 17.5       |
| CS                   | Cesium                                                    | CS                           | µg/100 g | A                              |                            | ✓          | 298             | 10.4       |
| CU                   | Copper                                                    | CU                           | mg/100 g | A                              | ✓                          | ✓          | 2857            | 100        |
| CYS                  | Cystine                                                   | CYS                          | mg/100 g | A                              |                            | ✓          | 456             | 16.0       |
| CYS_G                | Cystine (g)                                               | CYS_G                        | g/100 g  | A                              |                            | ✓          | 445             | 15.7       |
| CYSN                 | Cystine (/g N)                                            | CYSN                         | mg/g N   | C                              |                            | ✓          | 444             | 15.5       |
| DISAC                | Disaccharides, total                                      | DISAC                        | g/100 g  | C                              |                            | ✓          | 2857            | 100        |
| DISACM               | Disaccharides, total (monosaccharide equivalents)         | DISACM                       | g/100 g  | C                              |                            | ✓          | 2857            | 100        |
| DM                   | Dry matter                                                | DM                           | g/100 g  | C                              | ✓                          | ✓          | 2857            | 100        |
| ENERC                | Energy, total metabolisable (kJ)                          | ENERC                        | kJ/100 g | C                              | ✓                          | ✓          | 2857            | 100        |
| ENERC_FSANZ1         | Energy, total metabolisable (kJ, including dietary fibre) |                              | kJ/100 g | C                              | ✓                          | ✓          | 2857            | 100        |

| Component identifier | Component name                                                        | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |           |                 |            |
|----------------------|-----------------------------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|-----------|-----------------|------------|
|                      |                                                                       |                              |             |                                | Standard                   | Unbridged | Number of foods | Percentage |
| ENERC_FSANZ1_KCAL    | Energy, total metabolisable (kcal, including dietary fibre)           |                              | kcal/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| ENERC_FSANZ2         | Energy, total metabolisable, carbohydrate by difference, FSANZ (kJ)   |                              | kJ/100 g    | C                              | ✓                          | ✓         | 2857            | 100        |
| ENERC_FSANZ2_KCAL    | Energy, total metabolisable, carbohydrate by difference, FSANZ (kcal) |                              | kcal/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| ENERC_KCAL           | Energy, total metabolisable, available carbohydrate, FSANZ (kJ)       | ENERC_KCAL                   | kcal/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| ENERC1               | Energy, total metabolisable, available carbohydrate, FSANZ (kcal)     |                              | kcal/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| ENERC1_KCAL          | Energy, total metabolisable (kcal)                                    |                              | kcal/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| ERGCAL               | Ergocalciferol (Vitamin D <sub>2</sub> )                              | ERGCAL                       | µg/100 g    | A                              |                            | ✓         | 2857            | 100        |
| ERGCALOH             | 25-hydroxyvitamin D <sub>2</sub>                                      | ERGCALOH                     | µg/100 g    | A                              |                            | ✓         | 1               | 0.03       |
| F10D0                | Fatty acid 10:0                                                       | F10D0                        | g/100 g     | A                              |                            | ✓         | 1714            | 60.0       |
| F10D0F               | Fatty acid 10:0 (/100 g TFA)                                          | F10D0F                       | g/100 g TFA | C                              |                            | ✓         | 1714            | 60.0       |
| F10D1                | Fatty acid 10:1                                                       | F10D1                        | g/100 g     | A                              |                            | ✓         | 350             | 12.3       |
| F10D1F               | Fatty acid 10:1 (/100 g TFA)                                          | F10D1F                       | g/100 g TFA | C                              |                            | ✓         | 350             | 12.3       |
| F11D0                | Fatty acid 11:0                                                       | F11D0                        | g/100 g     | A                              |                            | ✓         | 909             | 31.8       |
| F11D0F               | Fatty acid 11:0 (/100 g TFA)                                          | F11D0F                       | g/100 g TFA | C                              |                            | ✓         | 909             | 31.8       |
| F12D0                | Fatty acid 12:0                                                       | F12D0                        | g/100 g     | A                              |                            | ✓         | 1786            | 62.5       |
| F12D0F               | Fatty acid 12:0 (/100 g TFA)                                          | F12D0F                       | g/100 g TFA | C                              |                            | ✓         | 1770            | 62.0       |
| F12D1                | Fatty acid 12:1                                                       | F12D1                        | g/100 g     | A                              |                            | ✓         | 134             | 4.69       |
| F12D1F               | Fatty acid 12:1 (/100 g TFA)                                          | F12D1F                       | g/100 g TFA | C                              |                            | ✓         | 134             | 4.69       |
| F13D0                | Fatty acid 13:0                                                       | F13D0                        | g/100 g     | A                              |                            | ✓         | 1456            | 51.0       |
| F13D0F               | Fatty acid 13:0 (/100 g TFA)                                          | F13D0F                       | g/100 g TFA | C                              |                            | ✓         | 1456            | 51.0       |

| Component identifier | Component name                                  | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|-------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                 |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F14D0                | Fatty acid 14:0                                 | F14D0                        | g/100 g     | A                              |                            | ✓          | 2104            | 73.7       |
| F14D0F               | Fatty acid 14:0 (/100 g TFA)                    | F14D0F                       | g/100 g TFA | C                              |                            | ✓          | 2088            | 73.1       |
| F14D1                | Fatty acid 14:1                                 | F14D1                        | g/100 g     | C                              |                            | ✓          | 1655            | 57.9       |
| F14D1C               | Fatty acid <i>cis</i> 14:1                      | F14D1C                       | g/100 g     | C                              |                            | ✓          | 901             | 31.6       |
| F14D1CF              | Fatty acid <i>cis</i> 14:1 (/100 g TFA)         | F14D1CF                      | g/100 g TFA | C                              |                            | ✓          | 901             | 31.6       |
| F14D1CN5             | Fatty acid <i>cis</i> 14:1 omega-5              | F14D1CN5                     | g/100 g     | A                              |                            | ✓          | 882             | 30.9       |
| F14D1CN5F            | Fatty acid <i>cis</i> 14:1 omega-5 (/100 g TFA) | F14D1CN5F                    | g/100 g TFA | C                              |                            | ✓          | 882             | 30.9       |
| F14D1F               | Fatty acid 14:1 (/100 g TFA)                    | F14D1F                       | g/100 g TFA | C                              |                            | ✓          | 1642            | 57.5       |
| F14D1N5              | Fatty acid 14:1 omega-5                         | F14D1N5                      | g/100 g     | C                              |                            | ✓          | 1256            | 44.0       |
| F14D1N5F             | Fatty acid 14:1 omega-5 (/100 g TFA)            | F14D1N5F                     | g/100 g TFA | C                              |                            | ✓          | 1256            | 44.0       |
| F15D0                | Fatty acid 15:0                                 | F15D0                        | g/100 g     | A                              |                            | ✓          | 784             | 27.4       |
| F15D0F               | Fatty acid 15:0 (/100 g TFA)                    | F15D0F                       | g/100 g TFA | C                              |                            | ✓          | 775             | 27.1       |
| F15D1                | Fatty acid 15:1                                 | F15D1                        | g/100 g     | C                              |                            | ✓          | 1581            | 55.3       |
| F15D1C               | Fatty acid <i>cis</i> 15:1                      | F15D1C                       | g/100 g     | C                              |                            | ✓          | 886             | 31.1       |
| F15D1CF              | Fatty acid <i>cis</i> 15:1 (/100 g TFA)         | F15D1CF                      | g/100 g TFA | C                              |                            | ✓          | 886             | 31.1       |
| F15D1CN5             | Fatty acid <i>cis</i> 15:1 omega-5              | F15D1CN5                     | g/100 g     | A                              |                            | ✓          | 881             | 30.9       |
| F15D1CN5F            | Fatty acid <i>cis</i> 15:1 omega-5 (/100 g TFA) | F15D1CN5F                    | g/100 g TFA | C                              |                            | ✓          | 881             | 30.9       |
| F15D1F               | Fatty acid 15:1 (/100 g TFA)                    | F15D1F                       | g/100 g TFA | C                              |                            | ✓          | 1581            | 55.4       |
| F15D1N5              | Fatty acid 15:1 omega-5                         | F15D1N5                      | g/100 g     | C                              |                            | ✓          | 889             | 31.1       |
| F15D1N5F             | Fatty acid 15:1 omega-5 (/100 g TFA)            | F15D1N5F                     | g/100 g TFA | C                              |                            | ✓          | 889             | 31.1       |
| F16D0                | Fatty acid 16:0                                 | F16D0                        | g/100 g     | A                              |                            | ✓          | 2169            | 75.9       |
| F16D0F               | Fatty acid 16:0 (/100 g TFA)                    | F16D0F                       | g/100 g TFA | C                              |                            | ✓          | 2153            | 75.4       |
| F16D1                | Fatty acid 16:1                                 | F16D1                        | g/100 g     | C                              |                            | ✓          | 2085            | 73.0       |
| F16D1C               | Fatty acid <i>cis</i> 16:1                      | F16D1C                       | g/100 g     | C                              |                            | ✓          | 1552            | 54.3       |
| F16D1CF              | Fatty acid <i>cis</i> 16:1 (/100 g TFA)         | F16D1CF                      | g/100 g TFA | C                              |                            | ✓          | 1536            | 53.8       |

| Component identifier | Component name                                  | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|-------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                 |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F16D1CN7             | Fatty acid <i>cis</i> 16:1 omega-7              | F16D1CN7                     | g/100 g     | A                              |                            | ✓          | 921             | 32.3       |
| F16D1CN7F            | Fatty acid <i>cis</i> 16:1 omega-7 (/100 g TFA) | F16D1CN7F                    | g/100 g TFA | C                              |                            | ✓          | 921             | 32.3       |
| F16D1F               | Fatty acid 16:1 (/100 g TFA)                    | F16D1F                       | g/100 g TFA | C                              |                            | ✓          | 2068            | 72.4       |
| F16D1N7              | Fatty acid 16:1 omega-7                         | F16D1N7                      | g/100 g     | C                              |                            | ✓          | 968             | 33.9       |
| F16D1N7F             | Fatty acid 16:1 omega-7 (/100 g TFA)            | F16D1N7F                     | g/100 g TFA | C                              |                            | ✓          | 968             | 33.9       |
| F16D1T               | Fatty acid <i>trans</i> 16:1                    | F16D1T                       | g/100 g     | A                              |                            | ✓          | 582             | 20.4       |
| F16D1TF              | Fatty acid <i>trans</i> 16:1 (/100 g TFA)       | F16D1TF                      | g/100 g TFA | C                              |                            | ✓          | 566             | 19.8       |
| F17D0                | Fatty acid 17:0                                 | F17D0                        | g/100 g     | A                              |                            | ✓          | 1911            | 66.9       |
| F17D0F               | Fatty acid 17:0 (/100 g TFA)                    | F17D0F                       | g/100 g TFA | C                              |                            | ✓          | 1895            | 66.3       |
| F17D1                | Fatty acid 17:1                                 | F17D1                        | g/100 g     | C                              |                            | ✓          | 1865            | 65.3       |
| F17D1C               | Fatty acid <i>cis</i> 17:1                      | F17D1C                       | g/100 g     | C                              |                            | ✓          | 894             | 31.3       |
| F17D1CF              | Fatty acid <i>cis</i> 17:1 (/100 g TFA)         | F17D1CF                      | g/100 g TFA | C                              |                            | ✓          | 894             | 31.3       |
| F17D1CN7             | Fatty acid <i>cis</i> 17:1 omega-7              | F17D1CN7                     | g/100 g     | A                              |                            | ✓          | 883             | 31.3       |
| F17D1CN7F            | Fatty acid <i>cis</i> 17:1 omega-7 (/100 g TFA) | F17D1CN7F                    | g/100 g TFA | C                              |                            | ✓          | 883             | 31.3       |
| F17D1F               | Fatty acid 17:1 (/100 g TFA)                    | F17D1F                       | g/100 g TFA | C                              |                            | ✓          | 1855            | 64.9       |
| F17D1N7              | Fatty acid 17:1 omega-7                         | F17D1N7                      | g/100 g     | C                              |                            | ✓          | 887             | 31.1       |
| F17D1N7F             | Fatty acid 17:1 omega-7 (/100 g TFA)            | F17D1N7F                     | g/100 g TFA | C                              |                            | ✓          | 887             | 31.1       |
| F18D0                | Fatty acid 18:0                                 | F18D0                        | g/100 g     | A                              |                            | ✓          | 2168            | 75.9       |
| F18D0F               | Fatty acid 18:0 (/100 g TFA)                    | F18D0F                       | g/100 g TFA | C                              |                            | ✓          | 2152            | 75.3       |
| F18D1                | Fatty acid 18:1                                 | F18D1                        | g/100 g     | C                              |                            | ✓          | 2209            | 77.3       |
| F18D1C               | Fatty acid <i>cis</i> 18:1                      | F18D1C                       | g/100 g     | C                              |                            | ✓          | 1585            | 55.5       |
| F18D1CF              | Fatty acid <i>cis</i> 18:1 (/100 g TFA)         | F18D1CF                      | g/100 g TFA | C                              |                            | ✓          | 1569            | 54.9       |
| F18D1CN7             | Fatty acid <i>cis</i> 18:1 omega-7              | F18D1CN7                     | g/100 g     | A                              |                            | ✓          | 1397            | 48.9       |
| F18D1CN7F            | Fatty acid <i>cis</i> 18:1 omega-7 (/100 g TFA) | F18D1CN7F                    | g/100 g TFA | C                              |                            | ✓          | 1397            | 48.9       |

| Component identifier | Component name                                             | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|------------------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                            |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F18D1CN9             | Fatty acid <i>cis</i> 18:1 omega-9                         | F18D1CN9                     | g/100 g     | A                              |                            | ✓          | 1510            | 52.9       |
| F18D1CN9F            | Fatty acid <i>cis</i> 18:1 omega-9 (/100 g TFA)            | F18D1CN9F                    | g/100 g TFA | C                              |                            | ✓          | 1510            | 52.9       |
| F18D1F               | Fatty acid 18:1 (/100 g TFA)                               | F18D1F                       | g/100 g TFA | C                              |                            | ✓          | 2192            | 76.7       |
| F18D1N7              | Fatty acid 18:1 omega-7                                    | F18D1N7                      | g/100 g     | C                              |                            | ✓          | 1505            | 52.7       |
| F18D1N7F             | Fatty acid 18:1 omega-7 (/100 g TFA)                       | F18D1N7F                     | g/100 g TFA | C                              |                            | ✓          | 1505            | 52.7       |
| F18D1N9              | Fatty acid 18:1 omega-9                                    | F18D1N9                      | g/100 g     | C                              |                            | ✓          | 1554            | 54.4       |
| F18D1N9F             | Fatty acid 18:1 omega-9 (/100 g TFA)                       | F18D1N9F                     | g/100 g TFA | C                              |                            | ✓          | 1554            | 54.4       |
| F18D1T               | Fatty acid <i>trans</i> 18:1                               | F18D1T                       | g/100 g     | C                              |                            | ✓          | 1481            | 51.9       |
| F18D1TF              | Fatty acid <i>trans</i> 18:1 (/100 g TFA)                  | F18D1TF                      | g/100 g TFA | C                              |                            | ✓          | 1465            | 51.3       |
| F18D1TN7             | Fatty acid <i>trans</i> 18:1 omega-7                       | F18D1TN7                     | g/100 g     | A                              |                            | ✓          | 1126            | 39.4       |
| F18D1TN7F            | Fatty acid <i>trans</i> 18:1 omega-7 (/100 g TFA)          | F18D1TN7F                    | g/100 g TFA | C                              |                            | ✓          | 1126            | 39.4       |
| F18D1TN9             | Fatty acid <i>trans</i> 18:1 omega-9                       | F18D1TN9                     | g/100 g     | A                              |                            | ✓          | 1386            | 48.5       |
| F18D1TN9F            | Fatty acid <i>trans</i> 18:1 omega-9 (/100 g TFA)          | F18D1TN9F                    | g/100 g TFA | C                              |                            | ✓          | 1386            | 48.5       |
| F18D2                | Fatty acid 18:2                                            | F18D2                        | g/100 g     | A                              |                            | ✓          | 2167            | 75.9       |
| F18D2C               | Fatty acid <i>cis</i> 18:2                                 | F18D2C                       | g/100 g     | C                              |                            | ✓          | 1740            | 60.9       |
| F18D2CF              | Fatty acid <i>cis</i> 18:2 (/100 g TFA)                    | F18D2CF                      | g/100 g TFA | C                              |                            | ✓          | 1724            | 60.4       |
| F18D2CN6             | Fatty acid <i>cis,cis</i> 18:2 omega-6                     | F18D2CN6                     | g/100 g     | A                              | ✓                          | ✓          | 1797            | 62.9       |
| F18D2CN6F            | Fatty acid <i>cis,cis</i> 18:2 omega-6 (/100 g TFA)        | F18D2CN6F                    | g/100 g TFA | C                              |                            | ✓          | 1781            | 62.4       |
| F18D2CN9TN11         | Fatty acid <i>cis, trans</i> 18:2 omega-9, 11              |                              | g/100 g     | A                              | ✓                          | ✓          | 1381            | 48.4       |
| F18D2CN9TN11F        | Fatty acid <i>cis, trans</i> 18:2 omega-9, 11 (/100 g TFA) |                              | g/100 g TFA | C                              |                            | ✓          | 1381            | 48.4       |
| F18D2F               | Fatty acid 18:2 (/100 g TFA)                               | F18D2F                       | g/100 g TFA | C                              |                            | ✓          | 2150            | 75.3       |
| F18D2N6              | Fatty acid 18:2 omega-6                                    | F18D2N6                      | g/100 g     | C                              |                            | ✓          | 1893            | 66.3       |

| Component identifier | Component name                                          | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|---------------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                         |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F18D2N6F             | Fatty acid 18:2 omega-6 (/100 g TFA)                    | F18D2N6F                     | g/100 g TFA | C                              |                            | ✓          | 1876            | 65.7       |
| F18D2T               | Fatty acid <i>trans</i> 18:2                            | F18D2T                       | g/100 g     | C                              |                            | ✓          | 1522            | 53.3       |
| F18D2TF              | Fatty acid <i>trans</i> 18:2 (/100 g TFA)               | F18D2TF                      | g/100 g TFA | C                              |                            | ✓          | 1505            | 52.7       |
| F18D2TN6             | Fatty acid <i>trans</i> 18:2 omega-6                    | F18D2TN6                     | g/100 g     | A                              |                            | ✓          | 1314            | 46.0       |
| F18D2TN6F            | Fatty acid <i>trans</i> 18:2 omega-6 (/100 g TFA)       | F18D2TN6F                    | g/100 g TFA | C                              |                            | ✓          | 1298            | 46.0       |
| F18D3                | Fatty acid 18:3                                         | F18D3                        | g/100 g     | C                              |                            | ✓          | 2011            | 70.4       |
| F18D3C               | Fatty acid <i>cis</i> 18:3                              | F18D3C                       | g/100 g     | C                              |                            | ✓          | 1251            | 43.8       |
| F18D3CF              | Fatty acid <i>cis</i> 18:3 (/100 g TFA)                 | F18D3CF                      | g/100 g TFA | C                              |                            | ✓          | 1251            | 43.8       |
| F18D3CN3             | Fatty acid <i>cis,cis,cis</i> 18:3 omega-3              | F18D3CN3                     | g/100 g     | A                              |                            | ✓          | 1267            | 44.4       |
| F18D3CN3F            | Fatty acid <i>cis,cis,cis</i> 18:3 omega-3 (/100 g TFA) | F18D3CN3F                    | g/100 g TFA | C                              |                            | ✓          | 1267            | 44.4       |
| F18D3CN6             | Fatty acid <i>cis</i> 18:3 omega-6                      | F18D3CN6                     | g/100 g     | A                              |                            | ✓          | 880             | 30.8       |
| F18D3CN6F            | Fatty acid <i>cis</i> 18:3 omega-6 (/100 g TFA)         | F18D3CN6F                    | g/100 g TFA | C                              |                            | ✓          | 880             | 30.8       |
| F18D3F               | Fatty acid 18:3 (/100 g TFA)                            | F18D3F                       | g/100 g     | C                              |                            | ✓          | 1997            | 69.9       |
| F18D3N3              | Fatty acid 18:3 omega-3                                 | F18D3N3                      | g/100 g     | C                              | ✓                          | ✓          | 1745            | 61.1       |
| F18D3N3F             | Fatty acid 18:3 omega-3 (/100 g TFA)                    | F18D3N3F                     | g/100 g TFA | C                              |                            | ✓          | 1745            | 61.1       |
| F18D3N6              | Fatty acid 18:3 omega-6                                 | F18D3N6                      | g/100 g     | C                              |                            | ✓          | 1475            | 51.6       |
| F18D3N6F             | Fatty acid 18:3 omega-6 (/100 g TFA)                    | F18D3N6F                     | g/100 g TFA | C                              |                            | ✓          | 1459            | 51.1       |
| F18D3TN3             | Fatty acid <i>trans</i> 18:3 omega-3                    | F18D3TN3                     | g/100 g     | A                              |                            | ✓          | 7               | 0.20       |
| F18D3TN3F            | Fatty acid <i>trans</i> 18:3 omega-3 (/100 g TFA)       | F18D3TN3F                    | g/100 g TFA | C                              |                            | ✓          | 7               | 0.20       |
| F18D4                | Fatty acid 18:4                                         | F18D4                        | g/100 g     | C                              |                            | ✓          | 287             | 10         |
| F18D4F               | Fatty acid 18:4 (/100 g TFA)                            | F18D4F                       | g/100 g TFA | C                              |                            | ✓          | 287             | 10         |
| F18D4N3              | Fatty acid 18:4 omega-3                                 | F18D4N3                      | g/100 g     | A                              |                            | ✓          | 230             | 8.0        |
| F18D4N3F             | Fatty acid 18:4 omega-3 (/100 g TFA)                    | F18D4N3F                     | g/100 g TFA | C                              |                            | ✓          | 230             | 8.0        |



| Component identifier | Component name                                      | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|-----------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                     |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F19D0                | Fatty acid 19:0                                     | F19D0                        | g/100 g     | A                              |                            | ✓          | 31              | 1.1        |
| F19D0F               | Fatty acid 19:0 (/100 g TFA)                        | F19D0F                       | g/100 g TFA | C                              |                            | ✓          | 31              | 1.1        |
| F20D0                | Fatty acid 20:0                                     | F20D0                        | g/100 g     | A                              |                            | ✓          | 1809            | 63.3       |
| F20D0F               | Fatty acid 20:0 (/100 g TFA)                        | F20D0F                       | g/100 g TFA | C                              |                            | ✓          | 1809            | 63.2       |
| F20D1                | Fatty acid 20:1                                     | F20D1                        | g/100 g     | C                              |                            | ✓          | 1879            | 65.8       |
| F20D1C               | Fatty acid <i>cis</i> 20:1                          | F20D1C                       | g/100 g     | C                              |                            | ✓          | 892             | 31.2       |
| F20D1CF              | Fatty acid <i>cis</i> 20:1 (/100 g TFA)             | F20D1CF                      | g/100 g TFA | C                              |                            | ✓          | 892             | 31.2       |
| F20D1CN9             | Fatty acid <i>cis</i> 20:1 omega-9                  | F20D1CN9                     | g/100 g     | A                              |                            | ✓          | 882             | 30.9       |
| F20D1CN9F            | Fatty acid <i>cis</i> 20:1 omega-9 (/100 g TFA)     | F20D1CN9F                    | g/100 g TFA | C                              |                            | ✓          | 882             | 30.9       |
| F20D1F               | Fatty acid 20:1 (/100 g TFA)                        | F20D1F                       | g/100 g TFA | C                              |                            | ✓          | 1878            | 65.7       |
| F20D1N11             | Fatty acid 20:1 omega-11                            | F20D1N11                     | g/100 g TFA | A                              |                            | ✓          | 236             | 8.30       |
| F20D1N11F            | Fatty acid 20:1 omega-11 (/100 g TFA)               | F20D1N11F                    | g/100 g TFA | C                              |                            | ✓          | 236             | 8.30       |
| F20D1N9              | Fatty acid 20:1 omega-9                             | F20D1N9                      | g/100 g     | C                              |                            | ✓          | 1467            | 51.4       |
| F20D1N9F             | Fatty acid 20:1 omega-9 (/100 g TFA)                | F20D1N9F                     | g/100 g TFA | C                              |                            | ✓          | 1467            | 51.4       |
| F20D2                | Fatty acid 20:2                                     | F20D2                        | g/100 g     | C                              |                            | ✓          | 1525            | 53.4       |
| F20D2CN6             | Fatty acid <i>cis,cis</i> 20:2 omega-6              | F20D2CN6                     | g/100 g     | A                              |                            | ✓          | 877             | 30.7       |
| F20D2CN6F            | Fatty acid <i>cis,cis</i> 20:2 omega-6 (/100 g TFA) | F20D2CN6F                    | g/100 g TFA | C                              |                            | ✓          | 877             | 30.7       |
| F20D2F               | Fatty acid 20:2 (/100 g TFA)                        | F20D2F                       | g/100 g TFA | C                              |                            | ✓          | 1525            | 53.4       |
| F20D2N6              | Fatty acid 20:2 omega-6                             | F20D2N6                      | g/100 g     | C                              |                            | ✓          | 1358            | 47.6       |
| F20D2N6F             | Fatty acid 20:2 omega-6 (/100 g TFA)                | F20D2N6F                     | g/100 g TFA | C                              |                            | ✓          | 1358            | 47.6       |
| F20D3                | Fatty acid 20:3                                     | F20D3                        | g/100 g     | C                              |                            | ✓          | 1633            | 57.2       |
| F20D3C               | Fatty acid <i>cis</i> 20:3                          | F20D3C                       | g/100 g     | C                              |                            | ✓          | 886             | 31.0       |
| F20D3CF              | Fatty acid <i>cis</i> 20:3 (/100 g TFA)             | F20D3CF                      | g/100 g TFA | C                              |                            | ✓          | 886             | 31.0       |
| F20D3CN3             | Fatty acid <i>cis</i> 20:3 omega-3                  | F20D3CN3                     | g/100 g     | A                              |                            | ✓          | 873             | 30.6       |

| Component identifier | Component name                                  | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|-------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                 |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F20D3CN3F            | Fatty acid <i>cis</i> 20:3 omega-3 (/100 g TFA) | F20D3CN3F                    | g/100 g TFA | C                              |                            | ✓          | 873             | 30.6       |
| F20D3CN6             | Fatty acid <i>cis</i> 20:3 omega-6              | F20D3CN6                     | g/100 g     | A                              |                            | ✓          | 873             | 30.6       |
| F20D3CN6F            | Fatty acid <i>cis</i> 20:3 omega-6 (/100 g TFA) | F20D3CN6F                    | g/100 g TFA | C                              |                            | ✓          | 873             | 30.6       |
| F20D3F               | Fatty acid 20:3 (/100 g TFA)                    | F20D3F                       | g/100 g     | C                              |                            | ✓          | 1633            | 57.2       |
| F20D3N3              | Fatty acid 20:3 omega-3                         | F20D3N3                      | g/100 g     | C                              |                            | ✓          | 1356            | 47.5       |
| F20D3N3F             | Fatty acid 20:3 omega-3 (/100 g TFA)            | F20D3N3F                     | g/100 g TFA | C                              |                            | ✓          | 1356            | 47.5       |
| F20D3N6              | Fatty acid 20:3 omega-6                         | F20D3N6                      | g/100 g     | C                              |                            | ✓          | 1452            | 50.8       |
| F20D3N6F             | Fatty acid 20:3 omega-6 (/100 g TFA)            | F20D3N6F                     | g/100 g TFA | C                              |                            | ✓          | 1452            | 50.8       |
| F20D4                | Fatty acid 20:4                                 | F20D4                        | g/100 g     | C                              |                            | ✓          | 1712            | 59.9       |
| F20D4CN6             | Fatty acid <i>cis</i> 20:4 omega-6              | F20D4CN6                     | g/100 g     | A                              |                            | ✓          | 858             | 30.1       |
| F20D4CN6F            | Fatty acid <i>cis</i> 20:4 omega-6 (/100 g TFA) | F20D4CN6F                    | g/100 g TFA | C                              |                            | ✓          | 858             | 30.1       |
| F20D4F               | Fatty acid 20:4 (/100 g TFA)                    | F20D4F                       | g/100 g TFA | C                              |                            | ✓          | 1712            | 59.9       |
| F20D4N3              | Fatty acid 20:4 omega-3                         | F20D4N3                      | g/100 g     | C                              |                            | ✓          | 254             | 8.9        |
| F20D4N3F             | Fatty acid 20:4 omega-3 (/100 g TFA)            | F20D4N3F                     | g/100 g TFA | C                              |                            | ✓          | 254             | 8.9        |
| F20D4N6              | Fatty acid 20:4 omega-6                         | F20D4N6                      | g/100 g     | C                              |                            | ✓          | 1447            | 50.7       |
| F20D4N6F             | Fatty acid 20:4 omega-6 (/100 g TFA)            | F20D4N6F                     | g/100 g TFA | C                              |                            | ✓          | 1447            | 50.7       |
| F20D5                | Fatty acid 20:5                                 | F20D5                        | g/100 g     | C                              |                            | ✓          | 1686            | 59.0       |
| F20D5C               | Fatty acid <i>cis</i> 20:5                      | F20D5C                       | g/100 g     | C                              |                            | ✓          | 888             | 31.1       |
| F20D5CF              | Fatty acid <i>cis</i> 20:5 (/100 g TFA)         | F20D5CF                      | g/100 g TFA | C                              |                            | ✓          | 888             | 31.1       |
| F20D5CN3             | Fatty acid <i>cis</i> 20:5 omega-3              | F20D5CN3                     | g/100 g     | A                              |                            | ✓          | 885             | 31.0       |
| F20D5CN3F            | Fatty acid <i>cis</i> 20:5 omega-3 (/100 g TFA) | F20D5CN3F                    | g/100 g TFA | C                              |                            | ✓          | 885             | 31.0       |
| F20D5F               | Fatty acid 20:5 (/100 g TFA)                    | F20D5F                       | g/100 g TFA | C                              |                            | ✓          | 1686            | 59.0       |
| F20D5N3              | Fatty acid 20:5 omega-3                         | F20D5N3                      | g/100 g     | C                              | ✓                          | ✓          | 1561            | 54.7       |

| Component identifier | Component name                                  | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|-------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                 |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F20D5N3F             | Fatty acid 20:5 omega-3 (/100 g TFA)            | F20D5N3F                     | g/100 g TFA | C                              |                            | ✓          | 1561            | 54.7       |
| F21D0                | Fatty acid 21:0                                 | F21D0                        | g/100 g     | C                              |                            | ✓          | 1122            | 39.3       |
| F21D0F               | Fatty acid 21:0 (/100 g TFA)                    | F21D0F                       | g/100 g TFA | C                              |                            | ✓          | 1122            | 39.3       |
| F21D5                | Fatty acid 21:5                                 | F21D5                        | g/100 g     | C                              |                            | ✓          | 236             | 8.30       |
| F21D5F               | Fatty acid 21:5 (/100 g TFA)                    | F21D5F                       | g/100 g TFA | C                              |                            | ✓          | 236             | 8.30       |
| F21D5N3              | Fatty acid 21:5 omega-3                         | F21D5N3                      | g/100 g     | A                              |                            | ✓          | 167             | 5.80       |
| F21D5N3F             | Fatty acid 21:5 omega-3 (/100 g TFA)            | F21D5N3F                     | g/100 g TFA | C                              |                            | ✓          | 167             | 5.80       |
| F22D0                | Fatty acid 22:0                                 | F22D0                        | g/100 g     | A                              |                            | ✓          | 1599            | 56.0       |
| F22D0F               | Fatty acid 22:0 (/100 g TFA)                    | F22D0F                       | g/100 g TFA | C                              |                            | ✓          | 1599            | 56.0       |
| F22D1                | Fatty acid 22:1                                 | F22D1                        | g/100 g     | C                              |                            | ✓          | 1610            | 56.4       |
| F22D1C               | Fatty acid <i>cis</i> 22:1                      | F22D1C                       | g/100 g     | C                              |                            | ✓          | 877             | 30.7       |
| F22D1CF              | Fatty acid <i>cis</i> 22:1 (/100 g TFA)         | F22D1CF                      | g/100 g TFA | C                              |                            | ✓          | 877             | 30.7       |
| F22D1CN9             | Fatty acid <i>cis</i> 22:1 omega-9              | F22D1CN9                     | g/100 g     | A                              |                            | ✓          | 872             | 30.5       |
| F22D1CN9F            | Fatty acid <i>cis</i> 22:1 omega-9 (/100 g TFA) | F22D1CN9F                    | g/100 g TFA | C                              |                            | ✓          | 872             | 30.5       |
| F22D1F               | Fatty acid 22:1 (/100 g TFA)                    | F22D1F                       | g/100 g TFA | C                              |                            | ✓          | 1610            | 56.4       |
| F22D1N11             | Fatty acid 22:1 omega-11                        | F22D1N11                     | g/100 g     | C                              |                            | ✓          | 312             | 10.9       |
| F22D1N11F            | Fatty acid 22:1 omega-11 (/100 g TFA)           | F22D1N11F                    | g/100 g TFA | C                              |                            | ✓          | 312             | 10.9       |
| F22D1N9              | Fatty acid 22:1 omega-9                         | F22D1N9                      | g/100 g     | C                              |                            | ✓          | 1247            | 43.6       |
| F22D1N9F             | Fatty acid 22:1 omega-9 (/100 g TFA)            | F22D1N9F                     | g/100 g TFA | C                              |                            | ✓          | 1246            | 43.6       |
| F22D2                | Fatty acid 22:2                                 | F22D2                        | g/100 g     | C                              |                            | ✓          | 1206            | 42.2       |
| F22D2C               | Fatty acid <i>cis</i> 22:2                      | F22D2C                       | g/100 g     | C                              |                            | ✓          | 822             | 28.8       |
| F22D2CF              | Fatty acid <i>cis</i> 22:2 (/100 g TFA)         | F22D2CF                      | g/100 g TFA | C                              |                            | ✓          | 822             | 28.8       |
| F22D2CN6             | Fatty acid <i>cis</i> 22:2 omega-6              | F22D2CN6                     | g/100 g     | A                              |                            | ✓          | 830             | 29.1       |
| F22D2CN6F            | Fatty acid <i>cis</i> 22:2 omega-6 (/100 g TFA) | F22D2CN6F                    | g/100 g TFA | C                              |                            | ✓          | 830             | 29.1       |

| Component identifier | Component name                                  | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|-------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                 |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F22D2F               | Fatty acid 22:2 (/100 g TFA)                    | F22D2F                       | g/100 g TFA | C                              |                            | ✓          | 1206            | 42.2       |
| F22D2N6              | Fatty acid 22:2 omega-6                         | F22D2N6                      | g/100 g     | C                              |                            | ✓          | 1169            | 40.9       |
| F22D2N6F             | Fatty acid 22:2 omega-6 (/100 g TFA)            | F22D2N6F                     | g/100 g TFA | C                              |                            | ✓          | 1169            | 40.9       |
| F22D4                | Fatty acid 22:4                                 | F22D4                        | g/100 g     | C                              |                            | ✓          | 377             | 13.2       |
| F22D4F               | Fatty acid 22:4 (/100 g TFA)                    | F22D4F                       | g/100 g TFA | C                              |                            | ✓          | 377             | 13.2       |
| F22D4N6              | Fatty acid 22:4 omega-6                         | F22D4N6                      | g/100 g     | C                              |                            | ✓          | 206             | 7.2        |
| F22D4N6F             | Fatty acid 22:4 omega-6 (/100 g TFA)            | F22D4N6F                     | g/100 g TFA | C                              |                            | ✓          | 206             | 7.2        |
| F22D5                | Fatty acid 22:5                                 | F22D5                        | g/100 g     | C                              |                            | ✓          | 1636            | 57.3       |
| F22D5C               | Fatty acid <i>cis</i> 22:5                      | F22D5C                       | g/100 g     | C                              |                            | ✓          | 886             | 31.0       |
| F22D5CF              | Fatty acid <i>cis</i> 22:5 (/100 g TFA)         | F22D5CF                      | g/100 g TFA | C                              |                            | ✓          | 886             | 31.0       |
| F22D5CN3             | Fatty acid <i>cis</i> 22:5 omega-3              | F22D5CN3                     | g/100 g     | A                              |                            | ✓          | 887             | 31.0       |
| F22D5CN3F            | Fatty acid <i>cis</i> 22:5 omega-3 (/100 g TFA) | F22D5CN3F                    | g/100 g TFA | C                              |                            | ✓          | 887             | 31.0       |
| F22D5F               | Fatty acid 22:5 (/100 g TFA)                    | F22D5F                       | g/100 g TFA | C                              |                            | ✓          | 1636            | 57.3       |
| F22D5N3              | Fatty acid 22:5 omega-3                         | F22D5N3                      | g/100 g     | C                              | ✓                          | ✓          | 1538            | 53.8       |
| F22D5N3F             | Fatty acid 22:5 omega-3 (/100 g TFA)            | F22D5N3F                     | g/100 g TFA | C                              |                            | ✓          | 1538            | 53.8       |
| F22D5N6              | Fatty acid 22:5 omega-6                         | F22D5N6                      | g/100 g     | C                              |                            | ✓          | 260             | 9.1        |
| F22D5N6F             | Fatty acid 22:5 omega-6 (/100 g TFA)            | F22D5N6F                     | g/100 g TFA | C                              |                            | ✓          | 260             | 9.1        |
| F22D6                | Fatty acid 22:6                                 | F22D6                        | g/100 g     | C                              |                            | ✓          | 1737            | 60.8       |
| F22D6C               | Fatty acid <i>cis</i> 22:6                      | F22D6C                       | g/100 g     | C                              |                            | ✓          | 889             | 31.1       |
| F22D6CF              | Fatty acid <i>cis</i> 22:6 (/100 g TFA)         | F22D6CF                      | g/100 g TFA | C                              |                            | ✓          | 889             | 31.1       |
| F22D6CN3             | Fatty acid <i>cis</i> 22:6 omega-3              | F22D6CN3                     | g/100 g     | A                              |                            | ✓          | 881             | 30.8       |
| F22D6CN3F            | Fatty acid <i>cis</i> 22:6 omega-3 (/100 g TFA) | F22D6CN3F                    | g/100 g TFA | C                              |                            | ✓          | 881             | 30.8       |
| F22D6F               | Fatty acid 22:6 (/100 g TFA)                    | F22D6F                       | g/100 g TFA | C                              |                            | ✓          | 1737            | 60.8       |
| F22D6N3              | Fatty acid 22:6 omega-3                         | F22D6N3                      | g/100 g     | C                              | ✓                          | ✓          | 1551            | 54.3       |

| Component identifier | Component name                                                     | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|--------------------------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                                    |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| F22D6N3F             | Fatty acid 22:6 omega-3 (/100 g TFA)                               | F22D6N3F                     | g/100 g TFA | C                              |                            | ✓          | 1551            | 54.3       |
| F23D0                | Fatty acid 23:0                                                    | F23D0                        | g/100 g     | A                              |                            | ✓          | 1106            | 38.7       |
| F23D0F               | Fatty acid 23:0 (/100 g TFA)                                       | F23D0F                       | g/100 g TFA | C                              |                            | ✓          | 1106            | 38.7       |
| F24D0                | Fatty acid 24:0                                                    | F24D0                        | g/100 g     | A                              |                            | ✓          | 1558            | 54.5       |
| F24D0F               | Fatty acid 24:0 (/100 g TFA)                                       | F24D0F                       | g/100 g TFA | C                              |                            | ✓          | 1558            | 54.5       |
| F24D1                | Fatty acid 24:1                                                    | F24D1                        | g/100 g     | C                              |                            | ✓          | 1444            | 50.6       |
| F24D1C               | Fatty acid <i>cis</i> 24:1                                         | F24D1C                       | g/100 g     | C                              |                            | ✓          | 893             | 31.3       |
| F24D1CF              | Fatty acid <i>cis</i> 24:1 (/100 g TFA)                            | F24D1CF                      | g/100 g TFA | C                              |                            | ✓          | 893             | 31.3       |
| F24D1CN9             | Fatty acid <i>cis</i> 24:1 omega-9                                 | F24D1CN9                     | g/100 g     | A                              |                            | ✓          | 883             | 30.9       |
| F24D1CN9F            | Fatty acid <i>cis</i> 24:1 omega-9 (/100 g TFA)                    | F24D1CN9F                    | g/100 g TFA | C                              |                            | ✓          | 883             | 30.9       |
| F24D1F               | Fatty acid 24:1 (/100 g TFA)                                       | F24D1F                       | g/100 g TFA | C                              |                            | ✓          | 1444            | 50.6       |
| F24D1N9              | Fatty acid 24:1 omega-9                                            | F24D1N9                      | g/100 g     | C                              |                            | ✓          | 889             | 31.1       |
| F24D1N9F             | Fatty acid 24:1 omega-9 (/100 g TFA)                               | F24D1N9F                     | g/100 g TFA | C                              |                            | ✓          | 889             | 31.1       |
| F4D0                 | Fatty acid 4:0                                                     | F4D0                         | g/100 g     | A                              |                            | ✓          | 479             | 16.8       |
| F4D0F                | Fatty acid 4:0 (/100 g TFA)                                        | F4D0F                        | g/100 g TFA | C                              |                            | ✓          | 479             | 16.8       |
| F6D0                 | Fatty acid 6:0                                                     | F6D0                         | g/100 g     | A                              |                            | ✓          | 1384            | 48.5       |
| F6D0F                | Fatty acid 6:0 (/100 g TFA)                                        | F6D0F                        | g/100 g TFA | C                              |                            | ✓          | 1384            | 48.5       |
| F8D0                 | Fatty acid 8:0                                                     | F8D0                         | g/100 g     | A                              |                            | ✓          | 1593            | 55.8       |
| F8D0F                | Fatty acid 8:0 (/100 g TFA)                                        | F8D0F                        | g/100 g TFA | C                              |                            | ✓          | 1593            | 55.8       |
| FACID                | Fatty acids, total                                                 | FACID                        | g/100 g     | C                              |                            | ✓          | 2840            | 99.4       |
| FALCPUN3             | Fatty acids, total long chain polyunsaturated omega-3              |                              | g/100 g     | C                              | ✓                          | ✓          | 1552            | 54.3       |
| FALCPUN3F            | Fatty acids, total long chain polyunsaturated omega-3 (/100 g TFA) |                              | g/100 g TFA | C                              |                            | ✓          | 1552            | 54.3       |
| FAMS                 | Fatty acids, total monounsaturated                                 | FAMS                         | g/100 g     | C                              | ✓                          | ✓          | 2857            | 100        |

| Component identifier | Component name                                          | INFOODS Tagname <sup>a</sup> | Unit        | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|---------------------------------------------------------|------------------------------|-------------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                                         |                              |             |                                | Standard                   | Unabridged | Number of foods | Percentage |
| FAMSF                | Fatty acids, total monounsaturated (/100 g TFA)         | FAMSF                        | g/100 g TFA | C                              |                            | ✓          | 2840            | 99.4       |
| FAMST                | Fatty acids, total monounsaturated trans                | FAMST                        | g/100 g     | C                              |                            | ✓          | 1548            | 54.2       |
| FAMSTF               | Fatty acids, total monounsaturated trans (/100 g TFA)   | FAMSTF                       | g/100 g TFA | C                              |                            | ✓          | 1531            | 53.6       |
| FAPU                 | Fatty acids, total polyunsaturated                      | FAPU                         | g/100 g     | C                              | ✓                          | ✓          | 2857            | 100        |
| FAPUF                | Fatty acids, total polyunsaturated (/100 g TFA)         | FAPUF                        | g/100 g TFA | C                              |                            | ✓          | 2840            | 99.4       |
| FAPUN3               | Fatty acids, total polyunsaturated omega-3              | FAPUN3                       | g/100 g     | C                              | ✓                          | ✓          | 1721            | 60.3       |
| FAPUN3F              | Fatty acids, total polyunsaturated omega-3 (/100 g TFA) | FAPUN3F                      | g/100 g TFA | C                              |                            | ✓          | 1721            | 60.3       |
| FAPUN6               | Fatty acids, total polyunsaturated omega-6              | FAPUN6                       | g/100 g     | C                              | ✓                          | ✓          | 1831            | 64.1       |
| FAPUN6F              | Fatty acids, total polyunsaturated omega-6 (/100 g TFA) | FAPUN6F                      | g/100 g TFA | C                              |                            | ✓          | 1815            | 63.5       |
| FAPUT                | Fatty acids, total polyunsaturated trans                | FAPUT                        | g/100 g     | C                              |                            | ✓          | 1563            | 54.7       |
| FAPUTF               | Fatty acids, total polyunsaturated trans (/100 g TFA)   | FAPUTF                       | g/100 g TFA | C                              |                            | ✓          | 1546            | 54.1       |
| FASAT                | Fatty acids, total saturated                            | FASAT                        | g/100 g     | C                              | ✓                          | ✓          | 2857            | 100        |
| FASATF               | Fatty acids, total saturated (/100 g TFA)               | FASATF                       | g/100 g TFA | C                              |                            | ✓          | 2840            | 99.4       |
| FAT                  | Fat, total                                              | FAT                          | g/100 g     | A                              | ✓                          | ✓          | 2857            | 100        |
| FATRAN               | Fatty acids, total trans                                | FATRAN                       | g/100 g     | C                              | ✓                          | ✓          | 1660            | 58.1       |
| FATRNF               | Fatty acids, total trans (/100 g TFA)                   | FATRNF                       | g/100 g TFA | C                              |                            | ✓          | 1643            | 57.5       |
| FD                   | Fluoride                                                | FD                           | µg/100 g    | A                              |                            | ✓          | 32              | 1.10       |
| FE                   | Iron                                                    | FE                           | mg/100 g    | A                              | ✓                          | ✓          | 2857            | 100        |
| FIBHWW               | Fibre, high molecular weight (LC)                       |                              | g/100 g     | A                              |                            | ✓          | 7               | 0.2        |

| Component identifier | Component name                                | INFOODS Tagname <sup>a</sup> | Unit     | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |           |                 |            |
|----------------------|-----------------------------------------------|------------------------------|----------|--------------------------------|----------------------------|-----------|-----------------|------------|
|                      |                                               |                              |          |                                | Standard                   | Unbridged | Number of foods | Percentage |
| FIBINS               | Fibre, water-insoluble                        | FIBINS                       | g/100 g  | A                              | ✓                          | ✓         | 1989            | 69.7       |
| FIBLMW               | Fibre, low molecular weight (LC)              |                              | g/100 g  | A                              |                            | ✓         | 7               | 0.2        |
| FIBSOL               | Fibre, water-soluble                          | FIBSOL                       | g/100 g  | A                              | ✓                          | ✓         | 1996            | 69.9       |
| FIBTG                | Fibre, total dietary                          | FIBTG                        | g/100 g  | A                              | ✓                          | ✓         | 2855            | 99.9       |
| FIBTLC               | Fibre, total dietary (LC method)              |                              | g/100 g  | A                              | ✓                          | ✓         | 30              | 1.00       |
| FOL                  | Folate, total                                 | FOL                          | µg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| FOLAC                | Folic acid, synthetic folic acid              | FOLAC                        | µg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| FOLDFE               | Dietary folate equivalents                    | FOLDFE                       | µg/100 g | C                              | ✓                          | ✓         | 2857            | 100        |
| FOLFD                | Folate food, naturally occurring food folates | FOLFD                        | µg/100 g | C                              | ✓                          | ✓         | 2857            | 100        |
| FRUS                 | Fructose                                      | FRUS                         | g/100 g  | A                              | ✓                          | ✓         | 2857            | 100        |
| GALS                 | Galactose                                     | GALS                         | g/100 g  | A                              | ✓                          | ✓         | 482             | 16.9       |
| GLU                  | Glutamic acid                                 | GLU                          | mg/100 g | A                              | ✓                          | ✓         | 472             | 16.5       |
| GLU_G                | Glutamic acid (g)                             | GLU_G                        | g/100 g  | C                              |                            | ✓         | 460             | 16.1       |
| GLUN                 | Glutamic acid (/g N)                          | GLUN                         | mg/g N   | C                              |                            | ✓         | 459             | 16.1       |
| GLUS                 | Glucose                                       | GLUS                         | g/100 g  | A                              | ✓                          | ✓         | 2857            | 100        |
| GLY                  | Glycine                                       | GLY                          | mg/100 g | A                              |                            | ✓         | 472             | 16.5       |
| GLY_G                | Glycogen                                      | GLY_G                        | g/100 g  | C                              |                            | ✓         | 460             | 16.1       |
| GLYC                 | Glycogen (monosaccharide equivalents)         | GLYC                         | g/100 g  | A                              |                            | ✓         | 29              | 1.00       |
| GLYCM                | Glycine (g)                                   | GLYCM                        | g/100 g  | C                              |                            | ✓         | 29              | 1.00       |
| GLYN                 | Glycine (/g N)                                | GLYN                         | mg/g N   | C                              |                            | ✓         | 459             | 16.1       |
| HG                   | Mercury                                       | HG                           | µg/100 g | A                              |                            | ✓         | 400             | 14.0       |
| HIS                  | Histidine                                     | HIS                          | mg/100 g | A                              |                            | ✓         | 470             | 16.4       |
| HIS_G                | Histidine (g)                                 | HIS_G                        | g/100 g  | C                              |                            | ✓         | 458             | 16.0       |
| HISN                 | Histidine (/g N)                              | HISN                         | mg/g N   | C                              |                            | ✓         | 457             | 16.0       |
| HYP                  | Hydroxyproline                                | HYP                          | mg/100 g | A                              |                            | ✓         | 39              | 1.4        |
| HYP_G                | Hydroxyproline (g)                            | HYP_G                        | g/100 g  | C                              |                            | ✓         | 38              | 1.3        |
| HYPN                 | Hydroxyproline (/g N)                         | HYPN                         | mg/g N   | C                              |                            | ✓         | 38              | 1.3        |
| ID                   | Iodide                                        | ID                           | µg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| ILE                  | Isoleucine                                    | ILE                          | mg/100 g | A                              |                            | ✓         | 474             | 16.6       |

| Component identifier | Component name                       | INFOODS Tagname <sup>a</sup> | Unit     | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |           |                 |            |
|----------------------|--------------------------------------|------------------------------|----------|--------------------------------|----------------------------|-----------|-----------------|------------|
|                      |                                      |                              |          |                                | Standard                   | Unbridged | Number of foods | Percentage |
| ILE_G                | Isoleucine (g)                       | ILE_G                        | g/100 g  | C                              |                            | ✓         | 462             | 16.2       |
| ILEN                 | Isoleucine (/g N)                    | ILEN                         | mg/g N   | C                              |                            | ✓         | 461             | 16.1       |
| K                    | Potassium                            | K                            | mg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| LACAC_G              | Lactic acid (g)                      | LACAC_G                      | g/100 g  | A                              |                            | ✓         | 46              | 1.60       |
| LACS                 | Lactose                              | LACS                         | g/100 g  | A                              | ✓                          | ✓         | 2857            | 100        |
| LACSM                | Lactose (monosaccharide equivalents) | LACSM                        | g/100 g  | C                              |                            | ✓         | 2857            | 100        |
| LEU                  | Leucine                              | LEU                          | mg/100 g | A                              |                            | ✓         | 474             | 16.6       |
| LEU_G                | Leucine (g)                          | LEU_G                        | g/100 g  | C                              |                            | ✓         | 463             | 16.2       |
| LEUN                 | Leucine (/g N)                       | LEUN                         | mg/g N   | C                              |                            | ✓         | 461             | 16.1       |
| LI                   | Lithium                              | LI                           | µg/100 g | A                              |                            | ✓         | 335             | 11.7       |
| LUTN                 | Lutein                               | LUTN                         | µg/100 g | A                              |                            | ✓         | 43              | 1.50       |
| LYCPN                | Lycopene                             | LYCPN                        | µg/100 g | A                              |                            | ✓         | 95              | 3.30       |
| LYS                  | Lysine                               | LYS                          | mg/100 g | A                              |                            | ✓         | 478             | 16.7       |
| LYS_G                | Lysine (g)                           | LYS_G                        | g/100 g  | C                              |                            | ✓         | 466             | 16.3       |
| LYSN                 | Lysine (/g N)                        | LYSN                         | mg/g N   | C                              |                            | ✓         | 465             | 16.3       |
| MALAC_G              | Malic acid (g)                       | MALAC_G                      | g/100 g  | A                              |                            | ✓         | 48              | 1.70       |
| MALS                 | Maltose                              | MALS                         | g/100 g  | A                              | ✓                          | ✓         | 2857            | 100        |
| MALSM                | Maltose (monosaccharide equivalents) | MALSM                        | g/100 g  | C                              |                            | ✓         | 2857            | 100        |
| MALTDEX              | Maltodextrin                         |                              | g/100 g  | A                              |                            | ✓         | 26              | 0.9        |
| MET                  | Methionine                           | MET                          | mg/100 g | A                              |                            | ✓         | 473             | 16.6       |
| MET_G                | Methionine (g)                       | MET_G                        | g/100 g  | C                              |                            | ✓         | 461             | 16.1       |
| METN                 | Methionine (/g N)                    | METN                         | mg/g N   | C                              |                            | ✓         | 460             | 16.1       |
| MG                   | Magnesium                            | MG                           | mg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| MN                   | Manganese                            | MN                           | µg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| MNSAC                | Monosaccharides, total               | MNSAC                        | g/100 g  | C                              |                            | ✓         | 2857            | 100        |
| MO                   | Molybdenum                           | MO                           | µg/100 g | A                              |                            | ✓         | 432             | 15.1       |
| NA                   | Sodium                               | NA                           | mg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| NI                   | Nickel                               | NI                           | µg/100 g | A                              |                            | ✓         | 419             | 14.7       |
| NIA                  | Niacin, preformed                    | NIA                          | mg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| NIAEQ                | Niacin equivalents, total            | NIAEQ                        | mg/100 g | C                              | ✓                          | ✓         | 2857            | 100        |



| Component identifier | Component name                                 | INFOODS Tagname <sup>a</sup> | Unit     | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |           |                 |            |
|----------------------|------------------------------------------------|------------------------------|----------|--------------------------------|----------------------------|-----------|-----------------|------------|
|                      |                                                |                              |          |                                | Standard                   | Unbridged | Number of foods | Percentage |
| NIATRP               | Niacin equivalents from tryptophan             | NIATRP                       | mg/100 g | C                              | ✓                          | ✓         | 2857            | 100        |
| NT                   | Nitrogen, total                                | NT                           | g/100 g  | N                              | ✓                          | ✓         | 2857            | 100        |
| OA_G                 | Organic acids, total (g)                       | OA_G                         | g/100 g  | C                              |                            | ✓         | 100             | 3.50       |
| OXALAC_G             | Oxalic acid (g)                                | OXALAC_G                     | g/100 g  | A                              |                            | ✓         | 1               | 0.03       |
| P                    | Phosphorus                                     | P                            | mg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| PANTAC               | Pantothenic acid                               | PANTAC                       | mg/100 g | A                              |                            | ✓         | 1364            | 47.7       |
| PB                   | Lead                                           | PB                           | µg/100 g | A                              |                            | ✓         | 496             | 17.4       |
| PHE                  | Phenylalanine                                  | PHE                          | g/100 g  | A                              |                            | ✓         | 476             | 16.7       |
| PHE_G                | Phenylalanine (g)                              | PHE_G                        | g/100 g  | C                              |                            | ✓         | 465             | 16.3       |
| PHEN                 | Phenylalanine (/g N)                           | PHEN                         | mg/g N   | C                              |                            | ✓         | 465             | 16.3       |
| PHYSTR               | Phytosterols, total                            | PHYSTR                       | g/100 g  | A                              |                            | ✓         | 28              | 0.98       |
| PRO                  | Proline                                        | PRO                          | mg/100 g | A                              |                            | ✓         | 476             | 16.7       |
| PRO_G                | Proline (g)                                    | PRO_G                        | g/100 g  | C                              |                            | ✓         | 465             | 16.3       |
| PRON                 | Proline (/g N)                                 | PRON                         | mg/g N   | C                              |                            | ✓         | 465             | 16.3       |
| PROT                 | Protein, total; calculated from total nitrogen | PROT                         | g/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| PROXTOT              | Proximates, total                              |                              | g/100 g  | C                              |                            | ✓         | 2857            | 100        |
| PSACNS               | Polysaccharides, non-starch                    | PSACNS                       | g/100 g  | A                              |                            | ✓         | 1706            | 59.7       |
| PSACNSI              | Polysaccharides, non-starch, water-insoluble   | PSACNSI                      | g/100 g  | A                              |                            | ✓         | 1609            | 56.3       |
| PSACNSS              | Polysaccharides, non-starch, water-soluble     | PSACNSS                      | g/100 g  | C                              |                            | ✓         | 1582            | 55.4       |
| QUINAC_G             | Quinic acid (g)                                | QUINAC_G                     | g/100 g  | A                              |                            | ✓         | 4               | 0.14       |
| RB                   | Rubidium                                       | RB                           | µg/100 g | A                              |                            | ✓         | 334             | 11.7       |
| RETOL                | Retinol                                        | RETOL                        | µg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| RIBF                 | Riboflavin                                     | RIBF                         | mg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| S                    | Sulphur                                        | S                            | µg/100 g | A                              |                            | ✓         | 1203            | 42.1       |
| SE                   | Selenium                                       | SE                           | µg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| SER                  | Serine                                         | SER                          | mg/100 g | A                              |                            | ✓         | 472             | 16.5       |
| SER_G                | Serine (g)                                     | SER_G                        | g/100 g  | C                              |                            | ✓         | 460             | 16.1       |
| SERN                 | Serine (/g N)                                  | SERN                         | mg/g N   | C                              |                            | ✓         | 460             | 16.1       |
| SISOL                | Silicon (acid soluble)                         | SISOL                        | µg/100 g | A                              |                            | ✓         | 253             | 8.86       |
| SN                   | Tin                                            | SN                           | µg/100 g | A                              |                            | ✓         | 459             | 16.1       |

| Component identifier | Component name                             | INFOODS Tagname <sup>a</sup> | Unit     | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |           |                 |            |
|----------------------|--------------------------------------------|------------------------------|----------|--------------------------------|----------------------------|-----------|-----------------|------------|
|                      |                                            |                              |          |                                | Standard                   | Unbridged | Number of foods | Percentage |
| SORTL_G              | Sorbitol (g)                               | SORTL_G                      | g/100 g  | A                              |                            | ✓         | 33              | 1.2        |
| STARCH               | Starch, total                              | STARCH                       | g/100 g  | A                              | ✓                          | ✓         | 2857            | 100        |
| STARCHM              | Starch, total (monosaccharide equivalents) | STARCHM                      | g/100 g  | A                              |                            | ✓         | 2857            | 100        |
| STARES               | Starch, resistant                          | STARES                       | g/100 g  | A                              |                            | ✓         | 336             | 11.8       |
| SUCAC_G              | Succinic acid (g)                          | SUCAC_G                      | g/100 g  | A                              |                            | ✓         | 17              | 0.59       |
| SUCS                 | Sucrose                                    | SUCS                         | g/100 g  | A                              | ✓                          | ✓         | 2857            | 100        |
| SUCSM                | Sucrose (monosaccharide equivalents)       | SUCSM                        | g/100 g  | C                              |                            | ✓         | 2857            | 100        |
| SUGAD                | Sugar, added                               | SUGAD                        | g/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| SUGAR                | Sugars, total                              | SUGAR                        | g/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| SUGARM               | Sugars, total (monosaccharide equivalents) | SUGARM                       | g/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| SUGFR                | Sugar, free                                |                              | g/100 g  | C                              | ✓                          | ✓         | 2857            | 100        |
| TAU                  | Taurine                                    | TAU                          | mg/100 g | A                              |                            | ✓         | 162             | 5.67       |
| TAU_G                | Taurine (g)                                | TAU_G                        | g/100 g  | C                              |                            | ✓         | 160             | 5.60       |
| TAUN                 | Taurine (/g N)                             | TAUN                         | mg/g N   | C                              |                            | ✓         | 160             | 5.60       |
| THIA                 | Thiamin                                    | THIA                         | mg/100 g | A                              | ✓                          | ✓         | 2857            | 100        |
| THR                  | Threonine                                  | THR                          | mg/100 g | A                              |                            | ✓         | 472             | 16.5       |
| THR_G                | Threonine (g)                              | THR_G                        | g/100 g  | C                              |                            | ✓         | 462             | 16.2       |
| THRN                 | Threonine (/g N)                           | THRN                         | mg/g N   | C                              |                            | ✓         | 461             | 16.1       |
| TOCPHA               | Alpha-tocopherol                           | TOCPHA                       | mg/100 g | A                              | ✓                          | ✓         | 2149            | 75.2       |
| TOCPHB               | Beta-tocopherol                            | TOCPHB                       | mg/100 g | A                              | ✓                          | ✓         | 1149            | 40.2       |
| TOCPHBG              | Beta-tocopherol + Gamma-tocopherol         |                              | mg/100 g | A                              | ✓                          | ✓         | 4               | 0.14       |
| TOCPHD               | Delta-tocopherol                           | TOCPHD                       | mg/100 g | A                              | ✓                          | ✓         | 1446            | 50.6       |
| TOCPHG               | Gamma-tocopherol                           | TOCPHG                       | mg/100 g | A                              | ✓                          | ✓         | 1471            | 51.5       |
| TRP                  | Tryptophan                                 | TRP                          | mg/100 g | A                              | ✓                          | ✓         | 1753            | 61.4       |
| TRP_G                | Tryptophan (g)                             | TRP_G                        | g/100 g  | C                              |                            | ✓         | 515             | 18.0       |
| TRPN                 | Tryptophan (/g N)                          | TRPN                         | mg/g N   | C                              |                            | ✓         | 514             | 18.0       |
| TYR                  | Tyrosine                                   | TYR                          | mg/100 g | A                              |                            | ✓         | 477             | 16.7       |
| TYR_G                | Tyrosine (g)                               | TYR_G                        | g/100 g  | C                              |                            | ✓         | 466             | 16.3       |
| TYRN                 | Tyrosine (/g N)                            | TYRN                         | mg/g N   | C                              |                            | ✓         | 465             | 16.3       |

| Component identifier | Component name                          | INFOODS Tagname <sup>a</sup> | Unit     | Analysed (A) or Calculated (C) | FOODfiles™ 2024 Version 01 |            |                 |            |
|----------------------|-----------------------------------------|------------------------------|----------|--------------------------------|----------------------------|------------|-----------------|------------|
|                      |                                         |                              |          |                                | Standard                   | Unabridged | Number of foods | Percentage |
| V                    | Vanadium                                | V                            | µg/100 g | A                              |                            | ✓          | 332             | 11.6       |
| VAL                  | Valine                                  | VAL                          | mg/100 g | A                              |                            | ✓          | 474             | 16.6       |
| VAL_G                | Valine (g)                              | VAL_G                        | g/100 g  | C                              |                            | ✓          | 462             | 16.2       |
| VALN                 | Valine (/g N)                           | VALN                         | mg/g N   | C                              |                            | ✓          | 461             | 16.1       |
| VITA                 | Vitamin A, retinol equivalents          | VITA                         | µg/100 g | C                              | ✓                          | ✓          | 2857            | 100        |
| VITA_RAE             | Vitamin A, retinol activity equivalents | VITA_RAE                     | µg/100 g | C                              | ✓                          | ✓          | 2857            | 100        |
| VITB12               | Vitamin B <sub>12</sub>                 | VITB12                       | µg/100 g | A                              | ✓                          | ✓          | 2857            | 100        |
| VITB6A               | Vitamin B <sub>6</sub>                  | VITB6A                       | mg/100 g | A                              | ✓                          | ✓          | 2857            | 100        |
| VITC                 | Vitamin C                               | VITC                         | mg/100 g | A                              | ✓                          | ✓          | 2857            | 100        |
| VITD                 | Vitamin D; calculated by summation      | VITD                         | µg/100 g | C                              | ✓                          | ✓          | 2857            | 100        |
| VITE                 | Vitamin E, alpha-tocopherol equivalents | VITE                         | mg/100 g | C                              | ✓                          | ✓          | 2857            | 100        |
| VITK                 | Vitamin K                               | VITK                         | µg/100 g | A                              |                            | ✓          | 114             | 3.99       |
| WATER                | Water                                   | WATER                        | g/100 g  | A                              | ✓                          | ✓          | 2857            | 100        |
| ZEAX                 | Zeaxanthin                              | ZEAX                         | µg/100 g | A                              |                            | ✓          | 32              | 1.12       |
| ZN                   | Zinc                                    | ZN                           | mg/100 g | A                              | ✓                          | ✓          | 2857            | 100        |

Source: Klensin et al. (1989) and FAO/INFOODS (2012a).

FSANZ = Food Standards Australia New Zealand; TFA = Total Fatty Acid; N = Nitrogen; LC = Liquid Chromatography.

Appendix 2. Rules applied for significant numbers, decimal places and rounding

This Table describes the application of a fixed significant place for each component value, followed by a rounding rule to maximise the decimal place.

| Component                                                                                                                                                                             | Unit    | Number of places   |                 | Rounding rules |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------|-----------------|----------------|---------------|
|                                                                                                                                                                                       |         | Significant digits | Maximum decimal | Range          | Scale (limit) |
| Energy                                                                                                                                                                                | kJ/kcal | 3                  | 0               | 0–100          | 1             |
| Macronutrients                                                                                                                                                                        |         |                    |                 |                |               |
| Fat, water, dry matter, dietary fibre, and alcohol                                                                                                                                    | g       | 3                  | 1               | 0–10           | 0.1           |
| Ash, carbohydrates, protein, nitrogen total, sugars, starch, organic acids, and sorbitol                                                                                              | g       | 3                  | 2               | 0–1            | 0.01          |
| Inorganic compounds (minerals)                                                                                                                                                        |         |                    |                 |                |               |
| Calcium, potassium, magnesium, sodium, and phosphorus                                                                                                                                 | mg      | 3                  | 0               | 0–100          | 1             |
| Iron and zinc                                                                                                                                                                         | mg      | 3                  | 2               | 0–1            | 0.01          |
| Selenium and iodine                                                                                                                                                                   | µg      | 2                  | 2               | 0–1            | 0.01          |
| Copper                                                                                                                                                                                | mg      | 3                  | 3               | 0–1            | 0.001         |
| Manganese                                                                                                                                                                             | µg      | 2                  | 0               | 0–10           | 1             |
| Aluminium, arsenic, boron, cadmium, cobalt, chromium, caesium, lithium, lead, molybdenum, nickel, rubidium, sulphur, lead, silicon (acid soluble), tin, mercury, molybdenum, vanadium | µg      | 2                  | 2               | 0–1            | 0.01          |
| Vitamins                                                                                                                                                                              |         |                    |                 |                |               |
| Vitamin A, carotene (alpha- and beta-), beta-carotene equivalents, retinol, lutein, lycopene, and zeaxanthin                                                                          | µg      | 3                  | 0               | 0–100          | 1             |
| Thiamin, riboflavin, niacin (niacin from tryptophan, niacin equivalent), vitamin B6 and pantothenic acid                                                                              | mg      | 2                  | 3               | 0–1            | 0.001         |
| Vitamins B <sub>12</sub> , D (D <sub>3</sub> and D <sub>2</sub> ), K and biotin                                                                                                       | µg      | 2                  | 2               | 0–1            | 0.01          |
| Folates                                                                                                                                                                               | µg      | 2                  | 0               | 0–10           | 1             |
| Vitamin C and choline                                                                                                                                                                 | mg      | 3                  | 2               | 0–1            | 0.01          |
| Vitamin E (tocopherol)                                                                                                                                                                | mg      | 2                  | 2               | 0–1            | 0.01          |
| Others                                                                                                                                                                                |         |                    |                 |                |               |
| Caffeine                                                                                                                                                                              | mg      | 3                  | 0               | 0–100          | 1             |
| Cholesterol                                                                                                                                                                           | mg      | 3                  | 2               | 0–1            | 0.01          |
| Fatty acids total (saturated, mono-, poly- etc.)                                                                                                                                      | g       | 3                  | 2               | 0–1            | 0.01          |
| Fatty acid individual                                                                                                                                                                 | g       | 3                  | 3               | 0–1            | 0.001         |
| Amino acids                                                                                                                                                                           | mg      | 3                  | 0               | 0–100          | 1             |

Sources: Greenfield & Southgate (2003) and FAO/INFOODS (2012a).

### Appendix 3. Analytical methods

This table provides the analytical methods, including method references used to generate the values for components in the database.

| Components                                    | Method technique (primary)                                      | Method identification and references <sup>a</sup>                                                                                                                                                                                       | Lower limit of detection * |
|-----------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Alcohol                                       | Gas chromatography                                              | Palo & Ilkova (1970)                                                                                                                                                                                                                    |                            |
| Ash                                           | Gravimetric – dry ashing                                        | AOAC 900.02: Ash of sugars and syrups<br>AOAC 920.153: Ash of meat<br>AOAC 923.03: Ash of flour<br>AOAC 938.08: Ash of seafood<br>AOAC 940.26: Ash of fruits and fruit products<br>AOAC 942.05: Ash in animal feed<br>'In-house' method | 0.1 g/100 g                |
| Fat, total                                    | Acid hydrolysis – Organic solvent extraction/Soxhlet extraction | AOAC 954.02: Fat (crude) or ether extract in pet food<br>AOAC 948.22: Fat (crude) in nuts and nut products                                                                                                                              | 0.1 g/100 g                |
|                                               | Alkaline hydrolysis – Organic solvent extraction                | International Dairy Federation (2004)                                                                                                                                                                                                   |                            |
|                                               | Alkaline and acid hydrolysis – Organic solvent extraction       | 'In-house' method                                                                                                                                                                                                                       |                            |
| Nitrogen, total                               | Titrimetry (Kjeldahl method)                                    | AOAC 988.05: Protein (crude) in animal feed and pet food<br>AOAC 981.10: Crude protein in meat<br>AOAC 991.20: Nitrogen (total) in milk                                                                                                 | 0.1 g/100 g                |
|                                               | Combustion (Dumas method)                                       | AOAC 968.06: Protein (crude) in animal feed                                                                                                                                                                                             | 0.02 g/100 g               |
| Fatty acid profile                            | Methylated with acid catalyst – Gas chromatography              | Bannon et al. (1985)<br>Sukhija & Palmquist (1988)                                                                                                                                                                                      | 0.1 g/100 g                |
| Conjugated fatty acids                        |                                                                 | 'In-house' method based on Aldai et al. (2007)                                                                                                                                                                                          | 0.1 g/100 g                |
| Fibre, dietary (total, soluble and insoluble) | Enzymatic-gravimetric                                           | AOAC 991.43: Total, soluble and insoluble dietary fibre in foods<br>AOAC 985.29: Total dietary fibre in foods                                                                                                                           | 0.1 g/100 g                |
| Fibre, total dietary                          | Enzymatic-gravimetric-liquid chromatography                     | AOAC 2017.16: Total dietary fibre in foods and food ingredients: rapid integrated enzymatic-gravimetric-high-pressure liquid chromatography method                                                                                      | 0.1 g/100 g                |
| Polysaccharides, non-starch                   | Enzymatic-gravimetric                                           | Englyst et al. (1988)                                                                                                                                                                                                                   | 0.1 g/100 g                |
| Sugar profile                                 | Gas liquid chromatography                                       | 'In-house' method                                                                                                                                                                                                                       | 0.1 g/100 g                |
| Starch                                        | Enzymatic-colorimetric                                          | AOAC 996.11: Starch (total) in cereal products; (Lee et al. 1992)                                                                                                                                                                       | 0.1 g/100 g                |
| Starch, resistant                             | Enzymatic-colorimetric                                          | AOAC 2002.2: Resistant starch in starch and plant materials                                                                                                                                                                             | 0.1 g/ 100 g               |

| Components                                                               | Method technique (primary)              | Method identification and references <sup>a</sup>                                                                                                                                                                                                                                                                                                                               | Lower limit of detection *                     |
|--------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Water/Total solid                                                        | Vacuum oven/forced air-gravimetric      | AOAC 950.46: Moisture in meat<br>AOAC 930.15: Loss on drying (moisture) for feeds (at 135°C for 2 hours)/dry matter on oven drying for feeds (at 135°C for 2 hours)<br>AOAC 925.10: Solids (total) and loss on drying (moisture) in flour<br>AOAC 925.45: Loss on drying (moisture) in sugars<br>International Dairy Federation (2010)<br>International Dairy Federation (2004) | 0.1 g/100 g                                    |
|                                                                          | Titration (Karl Fisher)                 | AOAC 984.20: Loss on drying (moisture) and volatile matter in oils and fats, Karl Fisher method                                                                                                                                                                                                                                                                                 | 0.1 g/100 g                                    |
| Ca, K, Mg, Na, P, S                                                      | Biological materials digestion, ICP-OES | AOAC 984.27: Calcium, copper, iron, magnesium, manganese, phosphorous and zinc in infant formula.                                                                                                                                                                                                                                                                               | 0.001–0.005 g/100 g                            |
| Fe                                                                       |                                         | Mendham et al. (2000a)<br>Martin et al. (1994)                                                                                                                                                                                                                                                                                                                                  | 1–5 mg/kg                                      |
| Al, As, B, Co, Cr, Cs, Cu, Hg, Li, Mn, Mo, Pb, Ni, Rb, Si, Sn, V, Zn, Cd | Biological materials digestion, ICP-MS  | Martin et al. (1994)<br>Mendham et al. (2000b)                                                                                                                                                                                                                                                                                                                                  | Unknown                                        |
| I, Se, As                                                                | TMAH digestion, ICP-MS                  | Fecher et al. (1998)                                                                                                                                                                                                                                                                                                                                                            | 0.001–0.004 mg/kg                              |
| Cl                                                                       | Potentiometric                          | AOAC 971.27: Sodium chloride in canned vegetables                                                                                                                                                                                                                                                                                                                               | 10 mg/kg dry or fresh product, 1 mg/kg liquids |
| Alpha- and Beta-carotene                                                 | HPLC                                    | European Committee for Standardisation (2000a)                                                                                                                                                                                                                                                                                                                                  | 1–5 µg/100 g                                   |
|                                                                          | Ultra HPLC                              | AOAC 2016.13: Lutein, β-Carotene, and Lycopene in Infant Formula and Adult Nutritionals: Reversed – Phase Ultra – High –Performance Liquid Chromatography (UHPLC)                                                                                                                                                                                                               | 3.0 µg/100 g                                   |
| Lutein and Zeaxanthin                                                    | HPLC                                    | McGhie & Ainge (2002)<br>Barba et al. (2006)                                                                                                                                                                                                                                                                                                                                    | 5 µg/100 g                                     |
| Retinol                                                                  | Colorimetric                            | AOAC 974.29: Vitamin A in mixed feeds, premixes, and human and pet foods                                                                                                                                                                                                                                                                                                        | 10 IU/100 g                                    |
|                                                                          | HPLC                                    | European Committee for Standardisation (2000b)                                                                                                                                                                                                                                                                                                                                  |                                                |
| Folate, total<br>Folic acid                                              | Microbiological                         | DeVries et al. (2005)                                                                                                                                                                                                                                                                                                                                                           | 5 µg/100 g                                     |
| Thiamin (B <sub>1</sub> )                                                | HPLC                                    | European Committee for Standardisation (2003)                                                                                                                                                                                                                                                                                                                                   | 0.01 mg/100 g                                  |
| Riboflavin (B <sub>2</sub> )                                             |                                         | Dunbar & Stevenson (1979)                                                                                                                                                                                                                                                                                                                                                       |                                                |
| Niacin, preformed (B <sub>3</sub> )                                      |                                         | Woollard (1984)                                                                                                                                                                                                                                                                                                                                                                 |                                                |
| Vitamin B <sub>6</sub>                                                   |                                         | Bitsch & Moller (1989); Gregory (1993) and Olds et al. (1993)                                                                                                                                                                                                                                                                                                                   | 0.01 mg/100 g                                  |

| Components                                                                           | Method technique (primary)                            | Method identification and references <sup>a</sup>                                                                                                                                                           | Lower limit of detection *                                                                                                      |
|--------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Vitamins (B <sub>1</sub> , B <sub>2</sub> , B <sub>3</sub> and B <sub>6</sub> )      | Enzymatic LC–MS/MS                                    | AOAC 2015.14: Simultaneous Determination of Total Vitamins B <sub>1</sub> , B <sub>2</sub> , B <sub>3</sub> and B <sub>6</sub> in Infant Formula and Related Nutritionals: Enzymatic Digestion and LC-MS/MS | B <sub>1</sub> – 20 µg/100 g<br>B <sub>2</sub> – 30 µg/100 g<br>B <sub>3</sub> – 200 µg/100 g<br>B <sub>6</sub> – 0.01 mg/100 g |
|                                                                                      | Gas liquid chromatography                             | Davidek et al. (1985)                                                                                                                                                                                       | 1 mg/100 g                                                                                                                      |
| Pantothenic acid (Vitamin B <sub>5</sub> )                                           | LC-MS/MS                                              | AOAC 2012.16: Pantothenic Acid (Vitamin B <sub>5</sub> ) in Infant Formula and Adult/Pediatric Nutritional Formula: Ultra-High-Pressure LC-MS/MS Method                                                     |                                                                                                                                 |
| Vitamin B <sub>12</sub>                                                              | Radio isotope dilution                                | Green et al. (1974)                                                                                                                                                                                         | 0.1 µg/100 g                                                                                                                    |
|                                                                                      | Microbiological                                       | AOAC Method 952.20: Cobalamin (Vitamin B <sub>12</sub> Activity) in vitamin preparations                                                                                                                    | 0.01 µg/100 g                                                                                                                   |
|                                                                                      | HPLC                                                  | Campos-Gimenez et al. (2008)                                                                                                                                                                                | Unknown                                                                                                                         |
| Vitamin C                                                                            | LC-fluorometric detection                             | Dodson et al. (1992)<br>Giménez et al. (2017)                                                                                                                                                               |                                                                                                                                 |
|                                                                                      | LC-UV                                                 | AOAC 2012.22: Vitamin C in Infant Formula and Adult/Pediatric Nutritional Formula: Liquid Chromatography with Ultraviolet Detection (LC-UV)                                                                 | 1 mg/ 100 g                                                                                                                     |
| 25-hydroxyvitamin D <sub>3</sub>                                                     | Acetonitrile extraction followed by Radioimmune assay | Diasorin kit (Stillwater, Minnesota) from Immuno Pty Ltd                                                                                                                                                    | Unknown                                                                                                                         |
|                                                                                      | LC-MS/MS                                              | Strobel et al. (2013)                                                                                                                                                                                       | 0.5 µg/100 g                                                                                                                    |
| Cholecalciferol (vitamin D <sub>2</sub> ) & Ergocalciferol (vitamin D <sub>3</sub> ) | HPLC                                                  | AOAC 2002.05: Cholecalciferol (Vitamin D <sub>3</sub> ) in selected foods                                                                                                                                   | 0.1 µg/100 g                                                                                                                    |
|                                                                                      |                                                       | European Committee for Standardisation (2000c)                                                                                                                                                              |                                                                                                                                 |
| Vitamin K <sub>1</sub>                                                               | HPLC                                                  | Damon et al. (2005)<br>Schimpf et al. (2018)<br>AOAC 999.15: Vitamin K in milk and infant formulas                                                                                                          | 1 µg/100 g                                                                                                                      |
|                                                                                      |                                                       | AOAC 2015.09: Trans and Total ( <i>cis</i> + <i>trans</i> ) Vitamin K <sub>1</sub> in Infant, Pediatric, and Adult Nutritionals: HPLC with Fluorescence Detection                                           |                                                                                                                                 |
|                                                                                      |                                                       |                                                                                                                                                                                                             |                                                                                                                                 |
| Alpha-Beta-, Delta- and Gamma-tocopherol                                             | HPLC                                                  | Brubacher et al. (1986)<br>AOAC Official Method 971.30: α-tocopherol and α-tocopherol acetate in foods and feeds                                                                                            | 0.11 IU/100 g                                                                                                                   |
| Choline                                                                              | UHPLC-MS/MS                                           | AOAC 2015.10: Free and Total Choline and Carnitine in Infant Formula and Adult/Pediatric Nutritional Formula: Liquid Chromatography/Tandem Mass Spectrometry                                                |                                                                                                                                 |
| Biotin                                                                               | Optical biosensor-based immunoassay                   | Indyk et al. (2000)                                                                                                                                                                                         | Unknown                                                                                                                         |
|                                                                                      | HPLC-PDA                                              | AOAC 2016.02: Total Biotin in Infant Formula and Adult/Pediatric Nutritional Formulas: Liquid Chromatography Coupled with                                                                                   |                                                                                                                                 |

| Components   | Method technique (primary) | Method identification and references <sup>a</sup>                                                                                                                                                          | Lower limit of detection * |
|--------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
|              |                            | Immunoaffinity Column Cleanup Extraction                                                                                                                                                                   |                            |
| Caffeine     | HPLC                       | Woollard (1982)                                                                                                                                                                                            | 0.010 g/100 g              |
| Cholesterol  | Gas chromatography         | AOAC 933.08: Residue (unsaponifiable) of oils and fats<br>AOAC 970.50: Fat (vegetable) in butterfat<br>AOAC 970.51: Fats (animal) in vegetable fats and oils (determination of cholesterol)                | 0.5 mg/100 g               |
|              | Enzymatic HPLC method      | AOAC 2017.03 Total Tryptophan in Infant Formula and Adult/Pediatric Nutritional Formula: HPLC Following Enzymatic Hydrolysis                                                                               |                            |
| Amino acids  | HPLC                       | AOAC 988.15: Tryptophan in foods and food and feed ingredients<br>AOAC 985.28: Sulphur amino acids in food, feed ingredient, and processed foods<br>AOAC 994.12: Amino acids in feeds Landry et al. (1992) | unknown                    |
| Sorbitol     | HPLC                       | National Health and Family Planning Commission of the People's Republic of China (2016)                                                                                                                    | 0.4 g/100 g                |
| Maltodextrin |                            | Starch determination kit (Boehringer Mannheim)                                                                                                                                                             | 0.1 g/100 g                |

<sup>a</sup>AOAC methods source: Official Methods of Analysis of AOAC International (2023)

\* Lower limit of detection varies, based on the food matrix.

Note: The food components were analysed by IANZ (International Accreditation New Zealand) accredited laboratories in New Zealand and NATA (The National Association of Testing Authorities, Australia) accredited laboratory in Australia. Some of the analytical methods were not accredited for some of the food matrices. 'In-house' methods are developed or modified standard test method by the accredited laboratories.

AOAC = Association of Official Analytical Chemists; ICP-OES = Inductivity Coupled Plasma Optical Emission Spectroscopy; ICP-MS = Inductivity Coupled Plasma Optical Mass Spectrometry; HPLC = High Performance Liquid Chromatography; LC = Liquid Chromatography; PDA = Photo Diode Array; UHPLC = Ultra High Performance Liquid Chromatography.



Appendix 4. Examples for energy calculation

1. Energy calculated based on FAO/INFOODS (2012b)

| Component name: energy, total metabolisable (kJ) |                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Components Identifier                            | ENERC                                                                                                                                                                                                                                                                                                                                                                                                        |
| INFOODS tagname                                  | ENERC                                                                                                                                                                                                                                                                                                                                                                                                        |
| Formula                                          | Energy (kJ/100 g Edible Portion, EP) = (protein, total calculated from nitrogen total x 17) + (fat, total x 37) + (available carbohydrate by weight x 17) + (alcohol x 29)                                                                                                                                                                                                                                   |
| Example                                          | L1078, Kiwifruit, green-fleshed, flesh & seed, raw, Zespri™ SweetGreen Kiwifruit contains per 100 g EP: 1.27 g protein, total; calculated from nitrogen total (PROT) + 0.2 g fat, total (FAT) + 12.3 g available carbohydrate by weight (CHOAVL) + 0 g alcohol (ALC)<br><br>Calculation: [(1.27 g PROT x 17) + (0.2 g FAT x 37) + (12.3 g CHOAVL x 17) + (0.00 g ALC x 29)] = <b>ENERC = 239 kJ/100 g EP</b> |

| Component name: energy, total metabolisable (kcal) |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Components Identifier                              | ENERC_KCAL                                                                                                                                                                                                                                                                                                                                                                                                      |
| INFOODS tagname                                    | ENERC_KCAL                                                                                                                                                                                                                                                                                                                                                                                                      |
| Formula                                            | Energy (kcal/100 g EP) = (protein, total; calculated from nitrogen total x 4) + (fat, total x 9) + (available carbohydrate by weight x 4) + (alcohol x 7)                                                                                                                                                                                                                                                       |
| Example                                            | L1078, Kiwifruit, green-fleshed, flesh & seed, raw, Zespri™ SweetGreen Kiwifruit contains per 100 g EP: 1.27 g protein, total; calculated from nitrogen total (PROT) + 0.2 g fat, total (FAT) + 12.3 g available carbohydrate by weight (CHOAVL) + 0 g alcohol (ALC).<br><br>Calculation: [(1.27 g PROT x 4) + (0.2 g FAT x 9) + (12.3 g CHOAVL x 4) + (0.00 g ALC x 7)] = <b>ENERC_KCAL = 56 kcal/100 g EP</b> |

Energy calculated according to FAO/INFOODS (2012b) including dietary fibre

| Component name: energy, total metabolisable (including dietary fibre, kJ) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Components Identifier                                                     | ENERC1                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| INFOODS tagname                                                           | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Formula                                                                   | Energy (kJ/100 g EP) = (protein, total; calculated from nitrogen total x 17) + (fat, total x 37) + available carbohydrate by weight (CHOAVL) x 17) + (alcohol x 29) + (fibre, total dietary x 8)                                                                                                                                                                                                                                                                    |
| Example                                                                   | L1078, Kiwifruit, green-fleshed, flesh & seed, raw, Zespri™ SweetGreen Kiwifruit contains per 100 g EP: 1.27 g protein, total; calculated from nitrogen total (PROT) + 0.2 g fat, total (FAT) + 12.3 g available carbohydrate by weight (CHOAVL) + 0 g alcohol (ALC) + 2.4 g fibre, total dietary (FIBTG)<br><br>Calculation: [(1.27 g PROT x 17) + (0.2 g FAT x 37) + (12.3 g CHOAVL x 17) + (0 g ALC x 29) + (2.4 g FIBTG x 8)] = <b>ENERC1 = 258 kJ/100 g EP</b> |

| Component name: energy, total metabolisable (including dietary fibre, kcal) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Components Identifier                                                       | ENERC1_KCAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| INFOODS tagname                                                             | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Formula                                                                     | Energy (kJ/100 g EP) = (protein, total; calculated from nitrogen total x 4) + (fat, total x 9) + available carbohydrate by weight (CHOAVL) x 4) + (alcohol x 7) + (fibre, total dietary x 2)                                                                                                                                                                                                                                                                                     |
| Example                                                                     | <p>L1078, Kiwifruit, green-fleshed, flesh &amp; seed, raw, Zespri™ SweetGreen Kiwifruit contains per 100 g EP: 1.27 g protein, total; calculated from nitrogen total (PROT) + 0.2 g fat, total (FAT) + 12.3 g available carbohydrate by weight (CHOAVL) + 0 g alcohol (ALC) + 2.4 g fibre, total dietary (FIBTG)</p> <p>Calculation: [(1.27 g PROT x 4) + (0.2 g FAT x 9) + (12.3 g CHOAVL x 4) + (0 g ALC x 7) + (2.4 g FIBTG x 2)] = <b>ENERC1_KCAL = 61 kcal/100 g EP</b></p> |

### 3. Energy calculated according to Food Standards Australia New Zealand (2021), using carbohydrate by difference

| Component name: energy, total metabolisable, carbohydrate by difference, FSANZ (kJ) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Components Identifier                                                               | ENERC_FSANZ1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| INFOODS tagname                                                                     | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Formula                                                                             | Energy (kJ/100 g EP) = (protein, total; calculated from nitrogen total x 17) + (fat, total x 37) + (carbohydrate by difference x 17) + (alcohol x 29) + (fibre, total dietary x 8) + (organic acids, total x 13)                                                                                                                                                                                                                                                                                                                                                  |
| Example                                                                             | <p>L1078, Kiwifruit, green-fleshed, flesh &amp; seed, raw, Zespri™ SweetGreen Kiwifruit contains per 100 g EP: 1.27 g protein, total; calculated from nitrogen total (PROT) + 0.2 g fat, total (FAT) + 12.8 g carbohydrate by difference (CHOAVLDF_FSANZ) + 0.0 g alcohol (ALC) + 2.4 g fibre, total dietary (FIBTG) + 2.4 g organic acid, total (gram) (OA_G)</p> <p>Calculation: [(1.27 g PROT x 17) + (0.2 g FAT x 37) + (12.8 g CHOAVLDF_FSANZ x 17) + (0.00 g ALC x 29) + (2.4 g FIBTG x 8) + (2.4 g OA_G x 13)] = <b>ENERC_FSANZ1 = 299 kJ/100 g EP</b></p> |

| Component name: energy, total metabolisable, carbohydrate by difference, FSANZ (kcal) |                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Components Identifier                                                                 | ENERC_FSANZ1_KCAL                                                                                                                                                                                                                                                                                                       |
| INFOODS tagname                                                                       | N/A                                                                                                                                                                                                                                                                                                                     |
| Formula                                                                               | Energy (kcal/100 g EP) = Energy, total metabolisable, carbohydrate by difference FSANZ (kJ/100 g EP)/4.18                                                                                                                                                                                                               |
| Example                                                                               | <p>L1078, Kiwifruit, green-fleshed, flesh &amp; seed, raw, Zespri™ SweetGreen Kiwifruit contains 299 kJ per 100 g EP of energy, total metabolisable, carbohydrate by difference, FSANZ (kJ) (ENERC_FSANZ1)</p> <p>Calculation: [(299 kJ/100 g EP ENERC_FSANZ1)/ 4.18] = <b>ENERC_FSANZ1_KCAL = 72 kcal/100 g EP</b></p> |

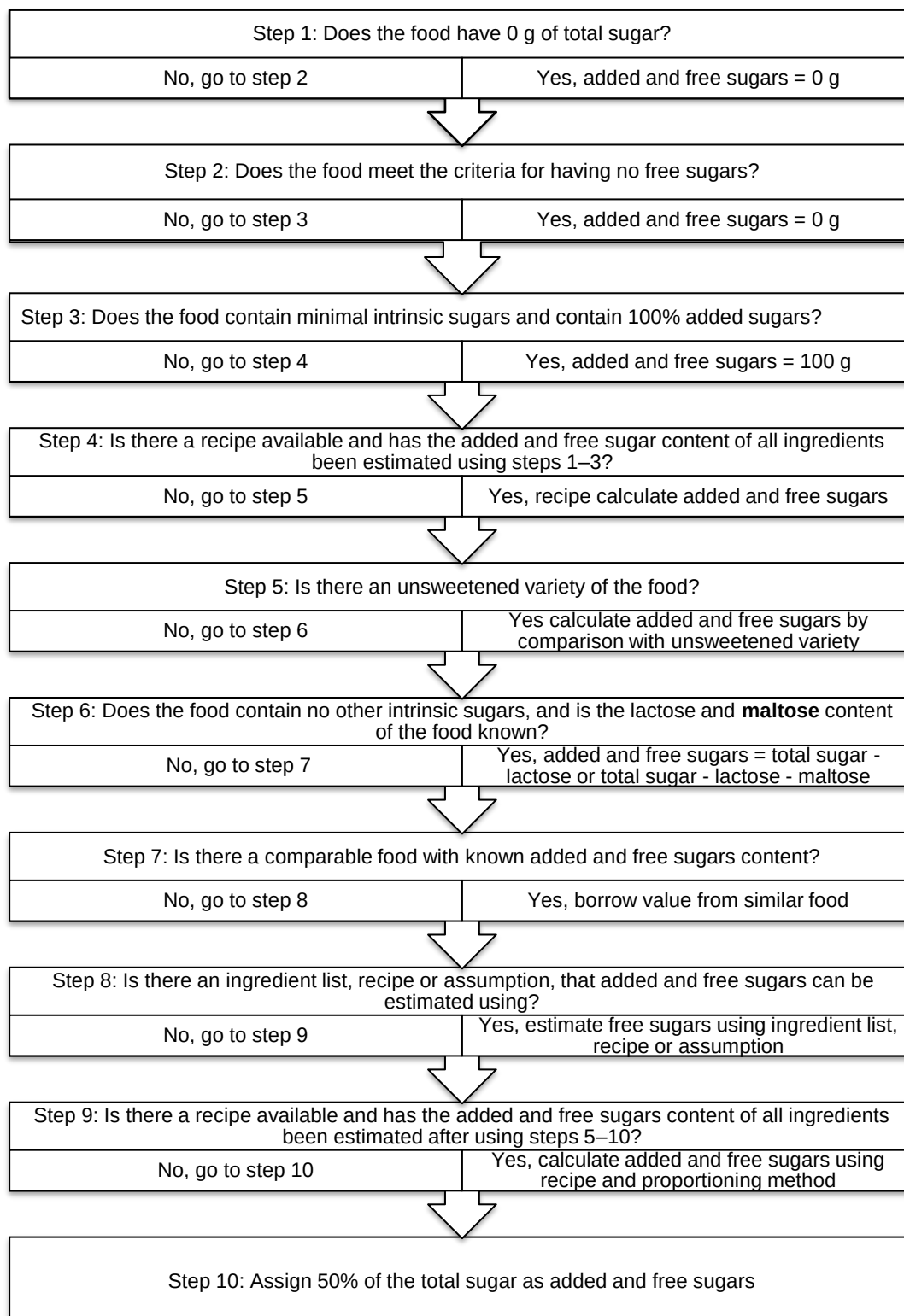
4. Energy calculated according to Food Standards Australia New Zealand (2021), using available carbohydrate

| Component name: energy, total metabolisable, available carbohydrate, FSANZ (kJ) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Components Identifier                                                           | ENERC_FSANZ2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| INFOODS tagname                                                                 | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Formula                                                                         | Energy (kJ/100 g EP) = (protein, total; calculated from nitrogen total x 17) + (fat, total x 37) + (available carbohydrate x 17) + (alcohol x 29) + (fibre, total dietary x 8) + (organic acids, total x 13)                                                                                                                                                                                                                                                                                                                           |
| Example                                                                         | L1078, Kiwifruit, green-fleshed, flesh & seed, raw, Zespri™ SweetGreen Kiwifruit contains per 100 g EP: 1.27 g protein, total; calculated from nitrogen total (PROT) + 0.2 g fat, total (FAT) + 12.3 g available carbohydrate (CHOAVL_FSANZ) + 0 g alcohol (ALC) + 2.4 g fibre, total dietary (FIBTG) + 2.4 g organic acids, total (OA_G)<br><br>Calculation: [(1.27 g PROT x 17) + (0.2 g FAT x 37) + (12.3 g CHOAVL_FSANZ x 17) + (0.00 g ALC x 29) + (2.4 g FIBTG x 8) + (2.4 g OA_G x 13)] = <b>ENERC_FSANZ2 = 289 kJ/100 g EP</b> |

| Component name: energy, total metabolisable, available carbohydrate, FSANZ (kcal) |                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Components Identifier                                                             | ENERC_FSANZ2_KCAL                                                                                                                                                                                                                                                                                     |
| INFOODS tagname                                                                   | None                                                                                                                                                                                                                                                                                                  |
| Formula                                                                           | Energy (kcal/100 g EP) = Energy, total metabolisable, available carbohydrate, FSANZ (kJ) (kJ/100 g EP)/4.18                                                                                                                                                                                           |
| Example                                                                           | L1078, Kiwifruit, green-fleshed, flesh & seed, raw, Zespri™ SweetGreen Kiwifruit contains: 289 kJ/100 g EP of energy, total metabolisable, available carbohydrate, FSANZ (kJ) (ENERC_FSANZ2)<br><br>Calculation: [(289 kJ/100 g EP ENERC_FSANZ2)/ 4.18] = <b>ENERC_FSANZ2_KCAL = 69 kcal/100 g EP</b> |

Note: To avoid additional bias, the system rounds the energy value after summation of values from the energy-producing components according to Appendix 2.

## Appendix 5. Method for estimating added and free sugars



Detailed 10-step method for estimating added and free sugars in FOODfiles 2021.

| Steps                  | Added sugar                                                                                                                                                              | Free sugar                                                                                                                                                               |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective steps</b> |                                                                                                                                                                          |                                                                                                                                                                          |
| <b>1</b>               | <b>All foods with the total sugar content of 0 g were assigned 0 g sugar</b>                                                                                             |                                                                                                                                                                          |
|                        | <b>Assigned 0 g added sugar to foods in the following groups</b>                                                                                                         | <b>Assigned 0 g free sugar to foods in the following groups</b>                                                                                                          |
|                        | (a) 100% fruit juice and vegetable juice with no added sugar (include fruit puree)                                                                                       | (a) See below step 3 (l)                                                                                                                                                 |
|                        | (b) All spices and herbs                                                                                                                                                 | (b) All spices and herbs                                                                                                                                                 |
|                        | (c) All fats and oils                                                                                                                                                    | (c) All fats and oils                                                                                                                                                    |
|                        | (d) Plain cereals including pastas, rice and flours                                                                                                                      | (d) Plain cereals including pastas, rice and flours                                                                                                                      |
|                        | (e) Eggs and egg products (excluding egg-based desserts)                                                                                                                 | (e) Eggs and egg products (excluding egg-based desserts)                                                                                                                 |
|                        | (f) Fresh fruits, vegetables (including salads with no dressing), fresh meat, fresh seafood and tofu. Includes cooked vegetables, meat and seafoods with no added sugar. | (f) Fresh fruits, vegetables (including salads with no dressing), fresh meat, fresh seafood and tofu. Includes cooked vegetables, meat and seafoods with no added sugar. |
|                        | (g) Fruit canned in 100% fruit juice or liquid sweetened with artificial sweeteners                                                                                      | (g) See below step 3 (l)                                                                                                                                                 |
|                        | (h) Beverage base (without added sugar) and prepared with water (including flavoured drink)                                                                              | (h) Beverage base (without added sugar)                                                                                                                                  |
| <b>2</b>               | (i) Legumes (fresh, dried and/or processed, except sweetened varieties)                                                                                                  | (i) Legumes (fresh, dried and/or processed, except sweetened varieties)                                                                                                  |
|                        | (j) Mixed dishes with no added sugar (decided on the basis of ingredient information; recipe)                                                                            | (j) Mixed dishes with no added sugar (decided on the basis of ingredient information; recipe)                                                                            |
|                        | (k) Non-sugar-sweetened alcoholic beverage                                                                                                                               | (k) Non-sweetened alcohol (excluding liqueurs and mixers)                                                                                                                |
|                        | (l) Non-sugar-sweetened tea and coffee                                                                                                                                   | (l) Non-sugar-sweetened tea and coffee                                                                                                                                   |
|                        | (m) Non-sugar-sweetened milk (including plant-based milk) and buttermilk; breastmilk                                                                                     | (m) Non-sugar-sweetened milk (excluding plant-based milk) and buttermilk; breastmilk                                                                                     |
|                        | (n) Non-sweetened dairy products (including those sweetened with artificial sweeteners only)                                                                             | (n) Non-sweetened dairy products (including those sweetened with artificial sweeteners only)                                                                             |
|                        | (o) All nuts and seeds (does not include sweetened varieties or nut bars)                                                                                                | (o) All nuts and seeds (does not include sweetened varieties or nut bars)                                                                                                |
|                        | (p) Oats, including porridge with no added sugar                                                                                                                         | (p) Oats, including porridge with no added sugar                                                                                                                         |
|                        | (q) Plain pastries without filling (such as chocolate, dried fruits and/or nuts)                                                                                         | (q) Plain pastries without filling (such as chocolate, dried fruits, and/or nuts)                                                                                        |
|                        | (r) Plain breads (except gluten-free), English muffin, bagels, pizza bases and naan                                                                                      | (r) Plain breads (except gluten-free), English muffin, bagels, pizza bases and naan                                                                                      |

| Steps | Added sugar                                                                                                                                                                                                    | Free sugar                                                                                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | (s) Unsweetened dried fruits                                                                                                                                                                                   | (s) Unsweetened dried fruits                                                                                                                                                                                            |
|       | <b>Assigned 100% of total sugar as added sugar to foods in the following groups</b>                                                                                                                            | <b>Assigned 100% of total sugar as free sugar to foods in the following groups</b>                                                                                                                                      |
|       | (a) All confectionery except those containing dairy products such as fudge and chocolate. Vegetable chips (including potato, cassava, soy etc.)                                                                | (a) All confectionery (excluding chocolate) and vegetable chips (including potato, cassava, soy etc.)                                                                                                                   |
|       | (b) Breakfast cereals and cereal bars without fruits, chocolate, dairy or milk                                                                                                                                 | (b) All breakfast cereals and cereal bars which do not contain fruit pieces, chocolate, or dairy or milk                                                                                                                |
|       | (c) Coffee and beverage bases with no milk solids, dry, or made up with water                                                                                                                                  | (c) Coffee and beverage bases                                                                                                                                                                                           |
|       | (d) Crumbed/battered meat and seafood                                                                                                                                                                          | (d) Crumbed/battered meat and seafood                                                                                                                                                                                   |
|       | (e) Processed meat including pies, pastries, sausages etc.                                                                                                                                                     | (e) Processed meat including pies, pastries, sausages etc.                                                                                                                                                              |
| 3     | (f) Soft drinks, sport drinks, flavoured water and non-fruit-based products that do not contain fruit, chocolate or dairy products                                                                             | (f) Sugar-sweetened soft drinks, sports drinks, flavoured waters, and energy drinks.                                                                                                                                    |
|       | (g) Savoury biscuits, sweet biscuits, cakes, and buns, donuts and batter-based products that do not contain fruit, chocolate or dairy products                                                                 | (g) All baked goods such as biscuits, cakes, buns, donuts and crackers that did not contain fruits, chocolate or dairy products. Gluten free breads.                                                                    |
|       | (h) Soy beverages and soy yoghurt without added fruits                                                                                                                                                         | (h) Soy beverages and soy yoghurt without added fruits                                                                                                                                                                  |
|       | (i) Stock powder                                                                                                                                                                                               | (i) Stock powder                                                                                                                                                                                                        |
|       | (j) Sugars and syrups                                                                                                                                                                                          | (j) All sugars and syrups                                                                                                                                                                                               |
|       | (k) Sauces and dressings, excluding pasta sauces and those that are vegetable-based such as pickles                                                                                                            | (k) Sauces and dressings, excluding pasta sauces and those that are vegetable-based such as pickles                                                                                                                     |
|       |                                                                                                                                                                                                                | (l) All fruit juices, purees, concentrates, fruit canned in fruit juice and jams (both sweetened and unsweetened varieties), including tomato pastes, sauces, and purees                                                |
|       |                                                                                                                                                                                                                | (m) Alcoholic mixed drinks, non-cream-based liqueurs                                                                                                                                                                    |
|       | <b>Calculation based on standard recipe used in the food composition database, where added sugar or free sugar contents of all ingredients were available from steps 1–3.</b>                                  |                                                                                                                                                                                                                         |
| 4     | (a) Calculation based on standard recipe used in the food composition database (by recipe calculation by FIMS)                                                                                                 |                                                                                                                                                                                                                         |
|       | (b) Added sugar content determined for undrained canned fruits in syrup:<br>Added sugar (g/100 g) = Total sugar (g/100 g) – (total sugar content of raw fruit (g/100 g) * % of raw fruit in the canned fruits. | Free sugar content determined for undrained canned fruits in fruit juice or syrup:<br>Free sugar (g/100 g) = Total sugar (g/100 g) – (total sugar content of raw fruit (g/100 g) * % of raw fruit in the canned fruits. |
| 5     | <b>Calculation based on comparison with values from the unsweetened variety</b>                                                                                                                                |                                                                                                                                                                                                                         |

| Steps | Added sugar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Free sugar |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|       | <p>Added or free sugar per 100 g is given by the formula:</p> $\text{Added or free sugar (g/100 g)} = (100 \times (S_{US} - S_T)) / (S_{US} - 100)$ <p>where <math>S_{US}</math> is the total sugar content per 100 g of the unsweetened variety of the food and <math>S_T</math> is the total sugar for the food item for which free sugars are to be estimated.</p> <p>Example: Added and free sugars calculation for sweetened (L1124) variety from the unsweetened (L1128) counterpart:</p> <p>L1124 – Peach, in light syrup, sugar-sweetened, canned, drained, composite (sugar total 14.2 g/100 g)</p> <p>L1128 – Peach, in water, artificially sweetened, canned, drained, composite (sugar total 6.6 g/100 g)</p> <p>Added or free sugar for L1124 = <math>((6.6 - 14.2) / (6.6 - 100)) \times 100 = 8.14</math> g/100 g</p> |            |
| 6     | <p><b>Decision based on analytical data of mono- and disaccharide content</b></p> <p>Lactose subtracted from total sugars for all foods</p> <p>Maltose subtracted from total sugars as intrinsic sugar</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |
| 7     | <p><b>Use “borrowed” values from similar products from steps 1–6 or from overseas databases</b></p> <p>Example H215 – Spring roll, vegetarian, deep-fried, commercial, added and free sugar estimated based on the recipe in the NZFCD (old archive Food Records, H130 – Spring roll, deep), 90% of the total sugar</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |
| 8     | <p><b>Subjective estimation of added and free sugars based on the best available information regarding ingredients and/or common recipes and/or assumption</b></p> <p>Added and free sugars estimated by searching the ingredient list of packed foods and recipes for sources.</p> <p>Example: D1055 – Breakfast cereal, muesli, ready to eat, Toasted Muesli Super Fruity, Sanitarium™, fortified vitamin E</p> <p>Ingredient list states dried fruits = 16% [raisins, cranberries]. Assumed even distribution (8.0% each). Total sugar from 8.0% raisins, 8.0% cranberries were considered as intrinsic sugars and deducted from the total sugars of D1055.</p>                                                                                                                                                                   |            |
| 9     | <p><b>Calculate based on standard recipe which includes ingredients with values assigned at steps 5–8, using the proportioning method. Step 4 is repeated here where more ingredients have their added sugar contents estimated after steps 5–8.</b></p> <p>Ingredients ratio in the standard recipe checked (step 8, % dried fruit), exclude lactose value (step 6)</p> <p>Example – A1152 – Cake, assorted fruits, rich or dark, un-iced, ready to eat, commercial</p>                                                                                                                                                                                                                                                                                                                                                             |            |
| 10    | <p><b>Assign 50% of total sugars as added or free sugars</b></p> <p>Used when it was not possible to determine free sugars using steps 1–9. Example is takeaway foods, H1058 – Salad, coleslaw with dressing, fresh, from deli, composite, assumed 50% of the total is added and free sugars</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |

Sources: Louie et al. (2015), Louie et al. (2016) and Kibblewhite et al. (2017).

Appendix 6. Individual fatty acids contributing to the total fatty acids

| Individual fatty acids | Component identifier <sup>a</sup> |       |      |      |        |          |        |       |       |       |
|------------------------|-----------------------------------|-------|------|------|--------|----------|--------|-------|-------|-------|
|                        | Undifferentiated fatty acid       | FASAT | FAMS | FAPU | FAPUN3 | FALCPUN3 | FAPUN6 | FAMST | FAPUT | FATRn |
| F4D0                   |                                   | ✓     |      |      |        |          |        |       |       |       |
| F6D0                   |                                   | ✓     |      |      |        |          |        |       |       |       |
| F8D0                   |                                   | ✓     |      |      |        |          |        |       |       |       |
| F10D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F11D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F12D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F13D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F14D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F15D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F16D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F17D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F18D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F20D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F21D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F22D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F23D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F24D0                  |                                   | ✓     |      |      |        |          |        |       |       |       |
| F10D1                  |                                   |       | ✓    |      |        |          |        |       |       |       |
| F12D1                  |                                   |       | ✓    |      |        |          |        |       |       |       |
| F14D1CN5               | F14D1                             |       | ✓    |      |        |          |        |       |       |       |
| F15D1CN5               | F15D1                             |       | ✓    |      |        |          |        |       |       |       |
| F16D1CN7               | F16D1                             |       | ✓    |      |        |          |        |       |       |       |
| F16D1T                 |                                   |       |      |      |        |          |        | ✓     |       | ✓     |
| F17D1CN7               | F17D1                             |       | ✓    |      |        |          |        |       |       |       |
| F18D1CN7               | F18D1                             |       |      |      |        |          |        |       |       |       |
| F18D1TN7               |                                   |       | ✓    |      |        |          |        | ✓     |       | ✓     |
| F18D1CN9               |                                   |       |      |      |        |          |        |       |       |       |
| F18D1TN9               |                                   |       |      |      |        |          |        | ✓     |       | ✓     |
| F20D1CN9               | F20D1                             |       | ✓    |      |        |          |        |       |       |       |
| F20D1N11               |                                   |       |      |      |        |          |        |       |       |       |
| F22D1CN9               | F22D1                             |       | ✓    |      |        |          |        |       |       |       |
| F22D1N11               |                                   |       |      |      |        |          |        |       |       |       |
| F24D1CN7               | F24D1                             |       | ✓    |      |        |          |        |       |       |       |



| Component identifier <sup>a</sup> |                             |       |      |      |        |          |        |       |       |       |
|-----------------------------------|-----------------------------|-------|------|------|--------|----------|--------|-------|-------|-------|
| Individual fatty acids            | Undifferentiated fatty acid | FASAT | FAMS | FAPU | FAPUN3 | FALCPUN3 | FAPUN6 | FAMST | FAPUT | FATRN |
| F18D2CN6                          | F18D2                       |       |      |      |        |          | ✓      |       |       |       |
| F18D2TN6                          |                             |       |      | ✓    |        |          |        |       | ✓     | ✓     |
| F18D2TCN9T11                      |                             |       |      |      |        |          |        |       | ✓     | ✓     |
| F18D3CN3                          | F18D3                       |       |      | ✓    | ✓      |          |        |       |       |       |
| F18D3CN6                          |                             |       |      |      |        |          | ✓      |       |       |       |
| F18D4N3                           | F18D4                       |       |      | ✓    | ✓      |          |        |       |       |       |
| F20D2CN6                          | F20D2                       |       |      | ✓    |        |          | ✓      |       |       |       |
| F20D3CN3                          | F20D3                       |       |      | ✓    | ✓      | ✓        |        |       |       |       |
| F20D3CN6                          |                             |       |      |      |        |          | ✓      |       |       |       |
| F20D4N3                           | F20D4                       |       |      | ✓    | ✓      | ✓        |        |       |       |       |
| F20D4CN6                          |                             |       |      |      |        |          | ✓      |       |       |       |
| F20D5CN3                          | F20D5                       |       |      | ✓    | ✓      | ✓        |        |       |       |       |
| F21D5N3                           | F21D5                       |       |      | ✓    | ✓      | ✓        |        |       |       |       |
| F22D2CN6                          | F22D2                       |       |      | ✓    |        |          | ✓      |       |       |       |
| F22D4CN6                          | F22D4                       |       |      | ✓    |        |          | ✓      |       |       |       |
| F22D5CN3                          | F22D5                       |       |      | ✓    | ✓      | ✓        |        |       |       |       |
| F22D5N6                           |                             |       |      |      |        |          | ✓      |       |       |       |
| F22D6CN3                          | F22D6                       |       |      | ✓    | ✓      | ✓        |        |       |       |       |

<sup>a</sup> Description of the component identifier can be found in Appendix 1.

Appendix 7. Recipe calculation – Mixed method

Example (R10011 - Pudding, rice, cocoa & coconut, cooked) showing nutrient calculation for a recipe food record to demonstrate the principles of the recipe calculation system 'Mixed method'.

| Ingredient                                                                                     | Amount<br>in<br>recipe<br><br>g | Weight<br>ratio (X)<br><br> | Water<br><br>g    | Ash<br><br>g | Fat<br><br>g | Starch<br><br>g | Glucose<br><br>g | Sucrose<br><br>g | Sugar<br>Total<br><br>g | Nitrogen<br>total<br><br>g | Protein<br>total<br><br>g | Fibre<br><br>g | Sodium<br><br>mg | Iron<br><br>mg | Folate,<br>total<br><br>µg |
|------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|-------------------|--------------|--------------|-----------------|------------------|------------------|-------------------------|----------------------------|---------------------------|----------------|------------------|----------------|----------------------------|
| C1036 – Baking cocoa, cocoa, powder, Cocoa, Bournville®, Cadbury® (per 100 g)                  | 37.75                           | 0.02                        |                   | 8.6          | 15.8         | 10.7            | 0                | 1.7              |                         | 3.956                      |                           | 39.1           | 102              | 9.8            | 65                         |
| E1077 – Rice, white, polished, raw (per 100 g) – (A)                                           | 206                             | 0.11                        |                   | 0.4          | 0.9          | 74.6            | 0.1              | 0.2              |                         | 1.218                      |                           | 0.7            | 0                | 0              | 0                          |
| NRF <sup>b</sup> – (B)                                                                         |                                 |                             |                   | 1            | 1            | 1               | 1                | 1                |                         | 1                          |                           | 1              | 1                | 0.95           | 0.7                        |
| E1077 – Rice, white, polished, raw, (per 100 g), nutrient values adjusted with NRF = (A x B)   |                                 |                             |                   | 0.4          | 0.9          | 74.6            | 0.1              | 0.2              |                         | 1.218                      |                           | 0.7            | 0                | 0              | 0                          |
| P10006 – Salt, from sea, non-iodised, composite (per 100 g)                                    | 5.4                             | 0.00                        |                   | 98.7         | 0            | 0               | 0                | 0                |                         | 0                          |                           | 0              | 40000            | 0              | 0                          |
| Q1015 – Coconut, milk, standard, composite (per 100 g)                                         | 452.4                           | 0.24                        |                   | 0.4          | 21.05        | 0               | 0                | 1.1              |                         | 0.2                        |                           | 0.3            | 18.7             | 0.62           | 0                          |
| C40 – Water, tap (per 100 g)                                                                   | 1000                            | 0.52                        |                   | 0.01         | 0            | 0               | 0                | 0                |                         | 0                          |                           | 0              | 1                | 0              | 0                          |
| W23 – Sugar, raw (per 100 g)                                                                   | 218                             | 0.11                        |                   | 0.05         | 0            | 0               | 0                | 99.17            |                         | 0                          |                           | 0              | 1                | 0.3            | 0                          |
| Food components in the raw recipe (per 100 g) – (weight sum of nutrient in each ingredient *X) |                                 |                             |                   | 0.598        | 5.37         | 8.22            | 0.011            | 11.6             |                         | 0.256                      |                           | 0.915          | 114              | 0.373          | 1.28                       |
| Total weight of the ingredients (Y)                                                            | 1920                            |                             |                   |              |              |                 |                  |                  |                         |                            |                           |                |                  |                |                            |
| Weight Yield factor (per g) (z)                                                                |                                 |                             |                   |              |              |                 |                  |                  |                         | 0.97                       |                           |                |                  |                |                            |
| Food components in the cooked dish (per 100 g, R10011) (X*Z) <sup>a</sup>                      |                                 |                             | 70.8 <sup>b</sup> | 0.61         | 5.53         | 8.47            | 0.01             | 11.9             | 11.9 <sup>c</sup>       | 0.26                       | 1.65 <sup>d</sup>         | 0.9            | 128              | 0.38           | 1                          |

<sup>a</sup> To avoid introducing additional bias, all the values were rounded after the calculation (summation, multiplication etc.), Appendix 2. <sup>b</sup> Water value 70.8 g/100 g is calculated by the difference from 100 minus other proximate components: ash, fat, protein, fibre, starch, and sugar total Calculated using formulas: sugar value <sup>c</sup>11.9 g/ 100 g is the sum of individual sugars (fructose, lactose, and maltose values zero for all the ingredients) and <sup>d</sup>protein, total value is 1.65 g/100 g from nitrogen total (0.26 g/ 100 g) multiplied by nitrogen to protein conversion factor 6.25.

Appendix 8. Units of measurement and New Zealand metric standards

| Units           | Definition                                                                        |
|-----------------|-----------------------------------------------------------------------------------|
| 1 teaspoon      | 5 mL                                                                              |
| 1 tablespoon    | 15 mL                                                                             |
| 1 cup           | 250 mL                                                                            |
| mL              | millilitres                                                                       |
| L               | litre                                                                             |
| cm <sup>3</sup> | cubic centimetre                                                                  |
| µg              | microgram                                                                         |
| mg              | milligram                                                                         |
| g               | grams                                                                             |
| kg              | kilogram                                                                          |
| cm              | centimetre                                                                        |
| kcal            | kilocalories                                                                      |
| kJ              | kilojoules                                                                        |
| g/100 g TFA     | grams per 100 g of total fatty acids or grams per percentage of total fatty acids |
| mg/g N          | milligrams of amino acid per gram of nitrogen present in food                     |
| °C              | degrees Celsius                                                                   |
| %ABV            | percentage alcohol by volume                                                      |

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